A Comparative Study of Modern Inference Techniques for Structured Discrete Energy Minimization Problems : Supplementary Material 2

Jörg H. Kappes Bjoern Andres Fred A. Hamprecht Christoph Schnörr Sebastian Nowozin Dhruv Batra Sungwoong Kim Bernhard X. Kausler Thorben Kröger Jan Lellmann Nikos Komodakis Bogdan Savchynskyy Carsten Rother

1 Evaluation per Instance

Table 1: brain-3mm

brain-3mm		o-Exp-pct	o-Exp-VIEW	FastPD	FastPD-pct	opti-FastPD-LP1	opti-FastPD-LP2	opti-ICM	opti-LP-1	opti-LP-2	opti-TRWS-LP1	opti-TRWS-LP2	o3-SwapVirW	opti-BPS	opti-LBP-0.5	opti-LBP-0.95	opti-TRBP-0.5	opti-TRBP-0.95	opti-ADSAL	opti-BUNDLE-A	opti-BUNDLE-H	opti-SG-A	opti-SG-AS	opti-BUNDLE-A-	opti-SG-A-	opti-BUNDLE-A-	opti-SG-A-	TRWS-TL	TRWS-pct	MCI-pct	opti-CombiLP
brain-0	value	30411507.00	30411444.00	30411921.00	30411683.00	30411921.00	30410746.00	30758587.00	30758985.00	30510720.00	30409134.00	30409070.00	30411991.00	30413911.00	30426547.00	30414154.00	30414461.00	30414414.00	30414203.00	30425902.00	30409179.00	30414785.00	30411712.00	30414561.00	30414205.00	30414225.00	30925223.00	30410729.00	30520678.00	30409315.00	30409353.00	**30409020.00**	30409152.00
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30409069.00	30409069.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30409065.54	30409065.54	30408529.09	30312727.96	30409454.71	30409501.70	30409469.09	30409469.09	30409469.09	**30409469.00**
	runtime	15.72	132.47	2.16	7.56	17.65	19.92	11.42	36.98	29.23	1433.63	909.02	192.30	625.72	3601.31	3606.00	3605.99	3628.04	3611.52	3603.21	3601.69	3600.44	3606.47	3603.71	3601.33	3606.27	3607.33	318.10	27.37	36.04	5094.84	6.83 GiS
	memory (GB)	2.55 GB	3.04 GB	5.15 GB	2.55 GB	7.04 GB	7.85 GB	2.19 GB	2.86 GB	3.68 GB	3.76 GB	4.56 GB	3.04 GB	1.80 GB	12.17 GB	12.17 GB	12.17 GB	12.60 GB	12.60 GB	5.57 GB	9.59 GB	9.59 GB	6.15 GB	9.58 GB	6.15 GB	1.79 GB	2.55 GB	2.77 GB	6.83 GB		
brain-1	value	30411579.00	30411149.00	30411838.00	30411716.00	30411838.00	30410882.00	30762052.00	30762052.00	30510008.00	30409504.00	30409321.00	30412522.00	30413484.00	30413803.00	30413504.00	30413504.00	30413504.00	30413568.00	30413907.00	30413379.00	30411444.00	30413907.00	30413504.00	30413504.00	30413504.00	30413504.00	30413504.00	30413504.00	30409409.00	30409409.00	30409503.00
	bound	-Inf	-Inf	27357673.00	-Inf	27357673.00	27357673.00	-Inf	-Inf	-Inf	30409469.00	30409469.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30409468.83	30409469.55	30409362.34	30409654.54	30408529.09	30312727.96	30409454.71	30409501.70	30409469.09	30409469.09	**30409469.00**
	runtime	10.44	169.46	2.16	8.81	11.64	79.64	19.38	31.21	3.68 GiS	156.73	237.05	185.72	657.17	3605.31	3604.83	3606.14	3623.79	3611.02	3602.81	3600.00	3602.73	3606.21	3605.17	3605.20	3605.17	178.89	26.64	34.48	4963.84	
	memory (GB)	2.55 GB	3.04 GB	5.16 GB	2.55 GB	7.04 GB	7.85 GB	2.18 GB	2.86 GB	3.68 GB	3.75 GB	4.56 GB	3.04 GB	1.80 GB	12.18 GB	12.18 GB	12.17 GB	12.60 GB	12.60 GB	5.57 GB	9.59 GB	9.59 GB	6.16 GB	9.58 GB	6.16 GB	1.79 GB	2.55 GB	2.75 GB	6.84 GB		
brain-2	value	30427100.00	30397535.00	30407834.00	30407427.00	30407834.00	30406626.00	30753875.00	30754259.00	30505105.00	30409113.00	30405069.00	30407692.00	30409338.00	30425507.00	30413960.00	30413960.00	30413962.00	30413961.00	30423159.00	30405222.00	30410374.00	30407921.00	30417489.00	30411987.00	30409837.00	30407219.00	30405227.00	30405308.00	30405170.00	**30405503.00**	30405170.00
	bound	-Inf	-Inf	27352852.60	-Inf	27352852.60	27352852.60	-Inf	-Inf	-Inf	30405069.00	30405043.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30405870.41	30404464.63	30404393.81	30146261.86	30404902.01	30307470.34	30405083.00	30405083.00	30405083.00	**30405503.00**	30405002.99
	runtime	7.41	147.07	2.17	6.97	7.41	7.45	2.18	2.47	3.68 GiS	156.73	236.98	142.41	624.61	3607.14	3607.14	3607.14	3607.14	3607.14	3602.81	3600.00	3603.01	3606.21	3605.17	3605.20	3605.17	178.89	26.64	34.48	4963.84	
	memory (GB)	2.55 GB	3.04 GB	5.15 GB	2.55 GB	7.04 GB	7.85 GB	2.18 GB	2.86 GB	3.68 GB	3.75 GB	4.56 GB	3.04 GB	1.80 GB	12.17 GB	12.17 GB	12.17 GB	12.60 GB	12.60 GB	5.57 GB	9.59 GB	9.59 GB	6.16 GB	9.58 GB	6.16 GB	1.79 GB	2.55 GB	2.73 GB	6.84 GB		
brain-3	value	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**	**34420480.00**
	bound	34420480.00	34420480.00	-Inf	34420480.00	-Inf	-Inf	34420480.00	34420480.00	34420480.00	34420480.00	34420480.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	4.96	28.73	1.38	4.88	7.02	6.87	7.72	3.73	68.86	1.70	1.98	25.64	671.69	21.71	219.55	63.37	285.43	82.51	27.17	6.67	12.94	12.44	12.19	12.33	11.65	12.59	12.47	1.11	5.16	5.39	6.77
	memory (GB)	2.57 GB	3.04 GB	5.16 GB	2.55 GB	6.85 GB	6.84 GB	2.57 GB	2.57 GB	3.68 GB	3.56 GB	3.50 GB	3.04 GB	1.80 GB	12.18 GB	12.17 GB	12.17 GB	12.60 GB	12.60 GB	5.57 GB	8.84 GB	8.84 GB	6.14 GB	8.84 GB	6.13 GB	1.79 GB	2.55 GB	2.55 GB	6.83 GB		
mean energy		216441.82.75	216437.00	216443.18.25	216443.26	216443.18.25	215436583.50	254253309.50	254253594.00	253238328.25	215622960.25	215623535.00	216469471.25	215665594.00	215670740.00	215668724.25	215668731.75	215668637.00	215668636.25	215710731.75	215452633.25	215699059.25	215417413.75	215452633.25	215453335.75	215263795.00	215244347.75	215262703.75	215262848.50	215262493.00	215262505.25
mean bound		-Inf	-Inf	229738001.00	-Inf	229738001.00	229738001.00	-Inf	-Inf	-Inf	215622960.25	215623535.00	216469471.25	215665594.00	215670740.00	215668724.25	215668731.75	215668637.00	215668636.25	215710731.75	215452633.25	215699059.25	215417413.75	215452633.25	215453335.75	215263795.00	215244347.75	215262703.75	215262848.50	215262493.00	215262505.25
mean runtime		9.57	126.94	2.02	7.56	10.51	61.27	16.60	9.48	83.39	285.81	343.41	136.43	644.82	2707.16	1930.36	2718.32	2774.78	2723.08	2710.81	2720.64	2708.90	2705.77	2704.54	2708.06	2709.06	2707.39	2707.93	282.73	40.16	27.35	3874.65
best value		25.00	0.00	25.00	25.00	25.00	25.00	0.00	0.00	0.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	100.00	100.00	100.00	25.00
best bound		25.00	0.00	0.00	25.00	25.00	25.00	0.00	0.00	0.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	100.00	100.00	100.00	25.00
verified opt		25.00	0.00	0.00	25.00	25.00	25.00	0.00	0.00	0.00	25.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00

Table 2: brain-5mm

brain-5mm		o-Exp-pct	o-Exp-VIEW	FastPD	FastPD-pct	opti-FastPD-LP1	opti-FastPD-LP2	opti-ICM	opti-LP-1	opti-LP-2	opti-TRWS-LP1	opti-TRWS-LP2	o3-SwapVirW	opti-BPS	opti-LBP-0.5	opti-LBP-0.95	opti-TRBP-0.5	opti-TRBP-0.95	opti-ADSAL	opti-BUNDLE-A	opti-BUNDLE-H	opti-SG-A	opti-SG-AS	opti-BUNDLE-A-	opti-SG-A-	opti-BUNDLE-A-	opti-SG-A-	TRWS-TL	TRWS-pct	MCI-pct	opti-CombiLP
brain-0	value	19080483.00	19080581.00	19080917.00	19080646.00	19080917.00	19080215.00	19205232.00	19205605.00	19131474.00	19079536.00	19079982.00	19081991.00	19082058.00	19091688.00	19083020.00	19082765.00	19082986.00	19082745.00	19091173.00	19079037.00	19079563.00	19081395.00	19080273.00	19297661.00	19079433.00	19111785.00	19079102.00	19079800.00	**19079858.00**	19079515.00
	bound	-Inf	-Inf	17039344.00	-Inf	17039344.00	17039344.00	-Inf	-Inf	-Inf	19079469.00	19079560.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	19079058.04	19079837.49	19079763.46	19056581.44	19079053.04	19079437.99	19079654.00	19079654.00	19079654.00	**19079858.00**
	runtime	6.98	82.55	1.32	4.67	6.98	46.79	13.17	6.43	51.66	85.67	125.11	97.79	486.57	3600.09	1094.78	1904.77	2258.06	3604.57	3603.03	3609.82	3606.33	3603.44	3601.81	3600.57	3602.99	3601.98	3604.99	83.33	24.39	25.93	3603.14
	memory (GB)	1.53 GB	1.82 GB	3.11 GB	1.53 GB	4.21 GB	4.59 GB	1.31 GB	1.69 GB	2.09 GB	2.24 GB	2.62 GB	1.83 GB	1.08 GB	7.28 GB	7.28 GB	7.28 GB	7.54 GB	7.54 GB	3.36 GB	5.76 GB	5.77 GB	3.70 GB	5.77 GB	3.70 GB	1.08 GB	1.53 GB	1.66 GB	4.84 GB		
brain-1	value	19093714.00	19093808.00	19094221.00	19094014.00	19094221.00	19093523.00	19274841.00	19274985.00	19134003.00	19092431.00	19092398.00	19094409.00	19095435.00	19109751.00	19095859.00	19095934.00	19095781.00	19095832.00	19102197.00	19092508.00	19094227.00	19093355.00	19093765.00	19094839.00	19093728.00	19093341.00	19093597.00	19092532.00	19092598.00	**19092383.00**	**19092383.00**
	bound	-Inf	-Inf	17058725.00	-Inf	17058725.00	17058725.00	-Inf	-Inf	-Inf	19092431.00	19092398.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	19093343.42	19093113.07	19092129.21	19075.36	19092377.43	19086267.20	19092383.00	19092383.00	19092383.00	**19092383.00**
	runtime	7.28	114.25	1.33	4.69	7.28	56.44	11.40	6.46	52.69	232.04	253.46	97.54	611.51	3600.73	1122.22	3601.63	3602.40	3601.49	3603.19	3612.05	3606.21	3601.29	3602.73	3607.99	3604.89	1577.72	23.26	4.79	1398.66	
	memory (GB)	1.53 GB	1.82 GB	3.11 GB	1.53 GB	4.21 GB	4.59 GB	1.31 GB	1.69 GB	2.09 GB	2.24 GB	2.62 GB	1.83 GB	1.08 GB	7.28 GB	7.28 GB	7.28 GB	7.54 GB	7.54 GB	3.36 GB	5.76 GB	5.77 GB	3.69 GB	5.77 GB	3.69 GB	1.08 GB	1.53 GB	1.53 GB	4.84 GB		
brain-2	value	19093722.00	19093794.00	19094221.00	19094014.00	19094221.00	19093523.00	19274841.00	19274985.00	19134003.00	19092431.00	19092398.00	19094409.00	19095435.00	19109751.00	19095859.00	19095934.00	19095781.00	19095832.00	19102197.00	19092508.00	19094227.00	19093355.00	19093765.00	19094839.00	19093728.00	19093341.00	19093597.00	19092532.00	19092598.00	**19092383.00**	**19092383.00**
	bound	-Inf	-Inf	17037151.00	-Inf	17037151.00	17037151.00	-Inf	-Inf	-Inf	19058753.00	19058753.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	19058753.00	19058753.00	19058728.32	19058718.48	19058655.88	19058660.20	18945166.88	19058750.12	19058729.16	19058753.00	**19058753.00**	**19058753.00**
	runtime	6.44	102.94	1.32	4.64	6.98	46.48	11.28	6.55	52.15	127.24	193.55	88.51	486.62	3601.38	1111.67	3601.76	3602.21	3601.47	3605.77	3616.39	3603.78	3606.24	3602.33	3603.62	3600.91	3601.80	125.74	26.70	25.45	635.07	
	memory (GB)	1.53 GB	1.82 GB	3.11 GB	1.53 GB	4.21 GB	4.59 GB	1.31 GB	1.69 GB	2.09 GB	2.24 GB	2.62 GB	1.83 GB	1.08 GB	7.28 GB	7.28 GB	7.28 GB	7.54 GB	7.54 GB	3.36 GB	5.76 GB	5.77 GB	3.70 GB	5.77 GB	3.69 GB	1.08 GB	1.53 GB	1.95 GB	4.84 GB		
brain-3	value	19121553.00	19121607.00	19121553.00	19121553.00	19121553.00	19121553.00	19245.00	19245.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	19121553.00	**19121556.00**	**19121556.00**
	bound	-Inf	-Inf	17075137.00	-Inf	17075137.00	17075137.00	-Inf	-Inf	-Inf	19120356.00	19120356.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	19120356.00	19120344.38	19120310.39	19120243.47	19120081.31	18999933.35	19120337.49	19119920.33	19120356.00	**19120356.00**	**19120356.00**
	runtime	7.28	141.65	241365.85	241370.70	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85	241365.85
	memory (GB)	1.53 GB	1.82 GB	3.11 GB	1.53 GB	4.29 GB	4.59 GB	1.31 GB	1.69 GB	2.09 GB	2.24 GB	2.62 GB	1.83 GB	1.08 GB	7.28 GB	7.28 GB	7.28 GB	7.54 GB	7.54 GB	3.36 GB	5.76 GB	5.77 GB	3.70 GB	5.77 GB	3.70 GB	1.08 GB	1.53 GB	1.95 GB	4.84 GB		
mean energy		19089989.75	19089980.00	19089984.75	19089984.25	19089984.75	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25
mean bound		-Inf	-Inf	17052669.25	-Inf	17052669.25	17052669.25	-Inf	-Inf	-Inf	19089984.25	19089984.25	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25	19089984.25
mean runtime		7.02	100.66	1.32	4.66	7.19	48.51	11.30	6.49	51.81	125.92	184.22	91.84	450.43	3601.41	1116.27	1574.35	2948.01	3483.22	3602.69	3610.13	3605.55	3603.13	3602.62	3602.10	3602.34	3601.02	3603.66	120.29	21.93	25.63	2022.37
best value		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.00
best bound		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	100.00	75.00
verified opt		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.00

Table 3: brain-9mm

brain-9mm		o-Exp-pct	o-Exp-VIEW	FastPD	FastPD-pct	opti-FastPD-LP1	opti-FastPD-LP2	opti-ICM	opti-LP-1	opti-LP-2	opti-TRWS-LP1	opti-TRWS-LP2	o3-SwapVirW	opti-BPS	opti-LBP-0.5	opti-LBP-0.95	opti-TRBP-0.5	opti-TRBP-0.95	opti-ADSAL	opti-BUNDLE-A	opti-BUNDLE-H	opti-SG-A	opti-SG-AS	opti-BUNDLE-A-	opti-SG-A-	opti-BUNDLE-A-	opti-SG-A-	TRWS-TL	TRWS-pct	MCI-pct	opti-CombiLP
brain-0	value	11816550.00	11816530.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	11816720.00	**11816945.00**	**11816945.00**
	bound	-Inf	-Inf	9951215.00	-Inf	9951215.00	9951215.00	-Inf	-Inf	-Inf	11779945.00	11779945.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	11779944.97	11779941.73	11779931.02	11779853.22	11779898.13	11779719.56	11779945.00	11779945.00	11779945.00	**11816945.00**
	runtime	6.87	81.30	0.83	3.91	6.87	45.31	10.73	6.03	51.13	1.75 GiS	1.92 GiS	1.01 GiS	660.13	4.02 GiS	4.02 GiS	4.02 GiS	4.16 GiS	4.16 GiS	1.90 GiS	2.09 GiS	3.22 GiS	2.09 GiS	3.22 GiS	2.09 GiS	0.69 GiS	0.85 GiS	1.09 GiS	2.71 GiS		
brain-1	value	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	11816836.00	**11816945.00**	**11816945.00**
	bound	-Inf	-Inf	9951215.00	-Inf	9951215.00	9951215.00	-Inf	-Inf	-Inf	11779945.00	11779945.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	11779944.97	11779941.73	11779931.02	11779853.22	11779898.13	11779719.56	11779945.00	11779945.00	11779945.00	**11816945.00**
	runtime	6.87	81.30	0.83	3.91	6.87	45.31	10.73	6.03	51.13	1.75 GiS	1.92 GiS	1.01 GiS	660.13	4.02 GiS	4.02 GiS	4.02 GiS	4.16 GiS	4.16 GiS	1.90 GiS	2.09 GiS	3.22 GiS	2.09 GiS	3.22 GiS	2.09 GiS	0.69 GiS	0.85 GiS	1.09 GiS	2.71 GiS		
brain-2	value	11239900.00	11239965.00	11240264.00	11240120.00	11240264.00	11239927.00	11342450.00	11342467.00	11290818.00	11239428.00	11239361.00	11240722.00	11241092.00	11244700.00	11241077.00	11240992.00	11240994.00	11240941.00	11240928.00	11239407.00	11239756.00	11239774.00	11242250.00	11240151.00	11234902.00	11239941.00	11239219.00	11239749.00	11239472.00	**11239339.00**	**11239339.00**
	bound	-Inf	-Inf	9997375.00	-Inf	9997375.00	9997375.00	-Inf	-Inf	-Inf	11239339.00	11239339.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	11239339.00	11239336.22	11239198.12	11239175.26	11239333.54	11238445.99	11239339.00	11239339.00	11239339.00	**11239383.00**
	runtime	4.23	61.67	0.83	2.91	4.16	27.81	6.86	3.92	30.87	76.70	80.63	54.22	229.52	228.62	579.95	437.81	1151.35	695.88	2831.79	2698.11	3066.10	3601.46	2814.82	3600.94	3556.93	3603.51	3345.00	57.92	7.67	10.95	344.19
	memory (GB)	0.85 GB	1.01 GB	1.71 GB	0.85 GB	2.26 GB	2.29 GB	0.67 GB	0.93 GB	1.10 GB	1.16 GB	1.16 GB	1.01 GB	0.60 GB	4.03 GB	4.02 GB	4.02 GB	4.16 GB	4.16 GB	1.86 GB	2.94 GB	2.94 GB	2.09 GB	2.96 GB	2.08 GB	2.94 GB	2.09 GB	0.85 GB	0.85 GB	2.29 GB		
brain-3	value	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**	**31421600.00**
	bound	31421600.00	31421600.00	-Inf	31421600.00	-Inf	-Inf	31421600.00	31421600.00	31421600.00	31421600.00	31421600.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	1.79	10.31	0.76	2.63	3.71	21.41	5.76	3.24	25.95	80.76	55.10	14.51	203.30	1608.94	1078.96	1167.33	1078.96	1161.51	1020.13	930.94	2566.53	2566.53	2292.99	2295.56	2581.48	2355.54	66.40	14.15	8.32	895.33	
	memory (GB)	0.85 GB	1.01 GB	1.71 GB	0.85 GB	2.26 GB	2.29 GB	0.67 GB	0.93 GB	1.10 GB	1.16 GB	1.16 GB	1.01 GB	0.60 GB	4.03 GB	4.02 GB	4.02 GB	4.16 GB	4.16 GB	1.86 GB	2.94 GB	2.94 GB	2.09 GB	2.96 GB	2.08 GB	2.94 GB	2.09 GB	0.85 GB	0.85 GB	2.29 GB		
mean energy		31836841.25	31836805.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25	31836841.25
mean bound		-Inf	-Inf	8260110.75	-Inf	8260110.75	8260110.75	-Inf	-Inf	-Inf	31815280.75	31815280.75	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	31815279.54	31815278.59	31815277.04	31815276.04	31815277.04	31815276.04	31815277.04	31815277.04	31815277.04	31815277.04
mean runtime		3.51	37.19	0.75	2.62	3.71	21.41	5.76	3.24	25.95	80.76	55.10	14.51	203.30	1608.94	408.82	1167.33	1078.96	1161.51	1852.20	2102.64	2193.95	2566.53	2292.99	2295.56	2581.48	2355.54	66.40	14.15	8.32	895.33	
best value		25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	100.00
best bound		25.00	0.00	0.00	25.00	25.00	25.00	0.00	0.00	0.00	100.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	100.00	100.00	100.00
verified opt		25.00	0.00	0.00	25.00	25.00	25.00	0.00	0.00	0.00	25.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	25.00	100.00	100.00	100.00

Table 4: cell-tracking

cell-tracking		opti-ICM	opti-LBP-1.2	opti-LP-1	opti-LP-2	opti-LP-3	opti-LBP-0.95	opti-TRBP-0.95	opti-TRBP	ADDD	MPLP	opti-SG-A	opti-BUNDLE-H	opti-SG-A-	opti-BUNDLE-A-	opti-SG-A-	opti-LP-CP	ADDD-BB	opti-ILP-pct	opti-ILP
opti_model	value	46359947.14	75155751.61	46359947.14	140757443.46	84616934.24	46715126.01	30751387.54	30751387.54	34006664296.29	107514359.61	76966513.52	74785433.52	76966513.52	23012687.07	75019965.37	75165359.61	75165359.61	**75144121.29**	**75144121.29**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	6059951.32	75144121.29	75144121.29	75144121.29	75144121.29	75144121.29	75144121.29	75144121.29

correlation=clustering		ogm-ICM	ogm-LP-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCL-CC1	MCL-CCIPD
00006097	value	-537.91	-537.91	-599.51	-599.51	-599.51	**-600.68**	**-600.68**	**-600.68**
	bound	-Inf	-Inf	-602.18	-602.18	-602.18	**-600.68**	**-600.68**	**-600.68**
	runtime	0.44	0.20	0.09	0.05	0.06	0.54	1.05	1.17
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00006759	value	-441.82	-441.82	-499.68	-499.68	-499.68	**-500.43**	**-500.43**	**-500.43**
	bound	-Inf	-Inf	-502.03	-502.03	-502.03	**-500.43**	**-500.43**	**-500.43**
	runtime	0.56	0.24	0.05	0.03	0.03	0.48	0.28	0.34
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00006794	value	-284.59	-284.59	-320.00	-320.00	-320.00	**-321.74**	**-321.74**	**-321.74**
	bound	-Inf	-Inf	-325.04	-325.04	-325.04	**-321.74**	**-321.74**	**-321.74**
	runtime	2.20	1.05	0.06	0.03	0.04	1.37	1.02	1.61
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
00006952	value	-709.38	-709.38	-761.55	-761.55	-761.55	**-762.66**	**-762.66**	**-762.66**
	bound	-Inf	-Inf	-764.25	-764.25	-764.25	**-762.66**	**-762.66**	**-762.66**
	runtime	2.03	0.94	0.08	0.04	0.05	0.71	1.46	1.65
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
00010177	value	-548.64	-548.64	-591.43	-591.43	-591.43	**-593.05**	**-593.05**	**-593.05**
	bound	-Inf	-Inf	-593.14	-593.14	-593.14	**-593.05**	**-593.05**	**-593.05**
	runtime	0.82	0.39	0.07	0.04	0.05	0.15	0.49	0.59
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.03 GB
00022136	value	-718.99	-718.99	-751.63	-751.63	-751.63	**-752.29**	**-752.29**	**-752.29**
	bound	-Inf	-Inf	-754.33	-754.33	-754.33	**-752.29**	**-752.29**	**-752.29**
	runtime	0.64	0.28	0.33	0.20	0.23	0.78	1.43	2.12
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
00023395	value	-386.78	-386.78	-474.04	-474.04	-474.04	**-474.71**	**-474.71**	**-474.71**
	bound	-Inf	-Inf	-475.27	-475.27	-475.27	**-474.71**	**-474.71**	**-474.71**
	runtime	0.32	0.14	0.03	0.02	0.02	2.16	2.93	2.73
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00023755	value	-872.34	-872.34	-721.56	-721.56	-721.56	**-724.99**	**-724.99**	**-724.99**
	bound	-Inf	-Inf	-725.88	-725.88	-725.88	**-724.99**	**-724.99**	**-724.99**
	runtime	0.41	0.18	0.07	0.05	0.06	0.25	0.90	0.60
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
00031178	value	-900.49	-900.49	-951.03	-951.03	-951.03	**-952.34**	**-952.34**	**-952.34**
	bound	-Inf	-Inf	-952.37	-952.37	-952.37	**-952.34**	**-952.34**	**-952.34**
	runtime	1.20	0.55	0.15	0.13	0.14	0.30	0.85	1.99
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.04 GB	0.05 GB
00034223	value	-558.02	-558.02	-569.13	-569.13	-569.13	**-570.74**	**-570.74**	**-570.74**
	bound	-Inf	-Inf	-570.92	-570.92	-570.92	**-570.74**	**-570.74**	**-570.74**
	runtime	0.36	0.16	0.03	0.03	0.04	0.17	0.40	0.41
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
00034463	value	-543.84	-543.84	-582.69	-582.69	-582.69	**-582.80**	**-582.80**	**-582.80**
	bound	-Inf	-Inf	-583.09	-583.09	-583.09	**-582.80**	**-582.80**	**-582.80**
	runtime	0.41	0.18	0.07	0.05	0.06	0.25	0.90	0.60
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
00037728	value	-571.78	-571.78	-630.48	-630.48	-630.48	**-634.37**	**-634.37**	**-634.37**
	bound	-Inf	-Inf	-634.56	-634.56	-634.56	**-634.37**	**-634.37**	**-634.37**
	runtime	0.41	0.18	0.07	0.05	0.06	0.25	0.90	0.60
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.05 GB	0.04 GB
00037793	value	-678.07	-678.07	-724.97	-724.97	-724.97	**-725.87**	**-725.87**	**-725.87**
	bound	-Inf	-Inf	-726.72	-726.72	-726.72	**-725.87**	**-725.87**	**-725.87**
	runtime	0.18	0.08	0.01	0.01	0.01	0.08	0.24	0.27
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
00039357	value	-830.46	-830.46	**-882.14**	**-882.14**	**-882.14**	**-882.14**	**-882.14**	**-882.14**
	bound	-Inf	-Inf	**-882.14**	**-882.14**	**-882.14**	**-882.14**	**-882.14**	**-882.14**
	runtime	3.34	1.16	4.04	0.05	0.04	0.05	0.35	0.29
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00040038	value	-683.37	-683.37	**-710.58**	**-710.58**	**-710.58**	**-710.58**	**-710.58**	**-710.58**
	bound	-Inf	-Inf	**-710.58**	**-710.58**	**-710.58**	**-710.58**	-710.59	**-710.58**
	runtime	0.30	0.14	0.08	0.06	0.06	0.06	0.54	0.33
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
00040069	value	-624.77	-624.77	-672.67	-672.67	-672.67	**-674.60**	**-674.60**	**-674.60**
	bound	-Inf	-Inf	-674.97	-674.97	-674.97	**-674.60**	**-674.60**	**-674.60**
	runtime	0.58	0.26	0.09	0.05	0.05	0.11	1.27	0.68
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00040294	value	-381.98	-381.98	-375.25	-375.25	-375.25	**-375.10**	**-375.10**	**-375.10**
	bound	-Inf	-Inf	-375.25	-375.25	-375.25	**-375.10**	**-375.10**	**-375.10**
	runtime	0.28	0.13	0.03	0.02	0.02	0.06	0.54	0.38
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
00041498	value	-470.12	-470.12	**-523.70**	**-523.70**	**-523.70**	**-523.70**	**-523.70**	**-523.70**
	bound	-Inf	-Inf	-525.04	-525.04	-525.04	**-523.70**	**-523.70**	**-523.70**
	runtime	0.44	0.20	0.03	0.02	0.03	0.06	0.21	0.28
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
00041774	value	-797.59	-797.59	-827.37	-827.37	-827.37	**-829.46**	**-829.46**	**-829.46**
	bound	-Inf	-Inf	-829.47	-829.47	-829.47	**-829.46**	-829.47	**-829.46**
	runtime	0.36	0.18	0.02	0.01	0.02	0.26	0.29	0.43
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00042074	value	-338.95	-338.95	-388.75	-388.75	-388.75	**-388.85**	**-388.85**	**-388.85**
	bound	-Inf	-Inf	-389.03	-389.03	-389.03	**-388.85**	-388.98	**-388.85**
	runtime	1.21	0.60	0.05	0.04	0.05	0.14	1.80	2.51
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
00042079	value	-941.03	-941.03	-942.09	-942.09	-942.09	**-942.05**	**-942.05**	**-942.05**
	bound	-Inf	-Inf	-942.10	-942.10	-942.10	-942.06	**-942.05**	**-942.05**
	runtime	0.46	0.20	0.22	0.17	0.17	0.57	1.09	0.96
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
00045633	value	-484.20	-484.20	-524.96	-524.96	-524.96	**-525.71**	**-525.71**	**-525.71**
	bound	-Inf	-Inf	-526.11	-526.11	-526.11	**-525.71**	**-525.71**	**-525.71**
	runtime	0.32	0.15	0.03	0.03	0.03	0.07	0.22	0.21
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
00066575	value	-886.89	-886.89	-926.12	-926.12	-926.12	**-932.65**	**-932.65**	**-932.65**
	bound	-Inf	-Inf	-932.65	-932.65	-932.65	**-932.65**	-932.66	**-932.65**
	runtime	6.39	3.39	0.45	0.30	0.32	0.72	3.03	4.09
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB	0.06 GB	0.06 GB
00072323	value	-606.87	-606.87	-640.69	-640.69	-640.69	**-643.50**	**-643.50**	**-643.50**
	bound	-Inf	-Inf	-643.76	-643.76	-643.76	**-643.50**	**-643.50**	**-643.50**
	runtime	0.66	0.28	0.04	0.03	0.03	0.11	0.44	0.60
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
00073545	value	-560.91	-560.91	-537.99	-537.99	-537.99	**-541.51**	**-541.51**	**-541.51**
	bound	-Inf	-Inf	-542.56	-542.56	-542.56	**-541.51**	**-541.51**	**-541.51**
	runtime	0.42	0.18	0.02	0.01	0.02	0.07	0.29	0.42
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
00076932	value	-387.32	-387.32	-428.31	-428.31	-428.31	**-429.63**	**-429.63**	**-429.63**
	bound	-Inf	-Inf	-429.64	-429.64	-429.64	**-429.63**	**-429.63**	**-429.63**
	runtime	0.39	0.16	0.03	0.02	0.02	0.04	0.19	0.22
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB
00099212	value	-328.14	-328.14	-369.07	-369.07	-369.07	**-373.77**	**-373.77**	**-373.77**
	bound	-Inf	-Inf	-373.23	-373.23	-373.23	**-373.77**	**-373.77**	**-373.77**
	runtime	1.07	0.51	0.29	0.15	0.19	0.58	1.79	1.87
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
00010830	value	-443.85	-443.85	-498.53	-498.53	-498.53	**-499.14**	**-499.14**	**-499.14**
	bound	-Inf	-Inf	-499.23	-499.23	-499.23	**-499.14**	**-499.14**	**-499.14**
	runtime	0.14	0.06	0.01	0.01	0.01	0.03	0.07	0.11
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
00010950	value	-627.02	-627.02	-667.88	-667.88	-667.88	**-670.87**	**-670.87**	**-670.87**
	bound	-Inf	-Inf	-671.61	-671.61	-671.61	**-670.87**	**-670.87**	**-670.87**
	runtime	0.47	0.21	0.03	0.02	0.02	0.08	0.24	0.34
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
00011003	value	-432.49	-432.49	**-495.38**	**-495.38**	**-495.38**	**-495.38**	**-495.38**	**-495.38**
	bound	-Inf	-Inf	**-495.38**	**-495.38**	**-495.38**	**-495.38**	**-495.38**	**-495.38**
	runtime	0.62	0.29	0.02	0.02	0.02	0.02	0.12	0.15
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
00011033	value	-407.77	-407.77	-438.43	-438.43	-438.43	**-439.17**	**-439.17**	**-439.17**
	bound	-Inf	-Inf	-440.08	-440.08	-440.08	**-439.17**	**-439.17**	**-439.17**
	runtime	0.71	0.33	0.17	0.10	0.12	0.34	0.74	0.79
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
00011073	value	-935.76	-935.76	-968.75	-968.75	-968.75	**-969.55**	**-969.55**	**-969.55**
	bound	-Inf	-Inf	-970.28	-970.28	-970.28	**-969.55**	**-969.55**	**-969.55**
	runtime	1.61	0.74	0.14	0.10	0.10	0.75	1.06	1.06
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
00011088	value	-674.52	-674.52	-738.79	-738.79	-738.79	**-739.16**	**-739.16**	**-739.16**
	bound	-Inf	-Inf	-740.05	-740.05	-740.05	**-739.16**	**-739.16**	**-739.16**
	runtime	0.61	0.28	0.04	0.03	0.04	0.11	0.44	0.60
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
00011116	value	-713.28	-713.28	-731.04	-731.04	-731.04	**-732.86**	**-732.86**	**-732.86**
	bound	-Inf	-Inf	-734.43	-734.43	-734.43	**-732.86**	**-732.86**	**-732.86**
	runtime	1.91	0.91	0.27	0.12	0.13	1.78	2.61	2.50
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
00011154	value	-276.00	-276.00	**-311.32**	**-311.32**	**-311.32**	**-311.32**	**-311.32**	**-311.32**
	bound	-Inf	-Inf	**-311.32**	**-311.32**	**-311.32**	**-311.32**	**-311.32**	**-311.32**
	runtime	0.66	0.29	0.02	0.01	0.02	0.02	0.21	0.27
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01000026	value	-833.31	-833.31	-696.13	-696.13	-696.13	**-696.67**	**-696.67**	**-696.67**
	bound	-Inf	-Inf	-697.02	-697.02	-697.02	**-696.67**	**-696.67**	**-696.67**
	runtime	0.42	0.20	0.03	0.02	0.02	0.06	0.19	0.21
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01000030	value	-538.00	-538.00	**-587.09**	**-587.09**	**-587.09**	**-587.09**	**-587.09**	**-587.09**
	bound	-Inf	-Inf	-587.56	-587.56	-587.56	**-587.09**	**-587.09**	**-587.09**
	runtime	0.31	0.14	0.02	0.02	0.02	0.25	0.29	0.27
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01000061	value	-239.04	-239.04	-266.18	-266.18	-266.18	**-267.58**	**-267.58**	**-267.58**
	bound	-Inf	-Inf	-266.17	-266.17	-266.17	**-267.58**	**-267.58**	**-267.58**
	runtime	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
01000091	value	-506.76	-506.76	**-551.73**	**-551.73**	**-551.73**	**-551.73**	**-551.73**	**-551.73**
	bound	-Inf	-Inf	**-551.73**	**-551.73**	**-551.73**	**-551.73**	**-551.73**	**-551.73**
	runtime	1.74	0.81	0.10	0.07	0.07	0.07	0.35	0.36
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB
01000113	value	-717.76	-717.76	**-728.21**	**-728.21**	**-728.21**	**-728.21**	**-728.21**	**-728.21**
	bound	-Inf	-Inf	**-728.21**	**-728.21**	**-728.21**	**-728.21**	**-728.21**	**-728.21**
	runtime	0.61	0.28	0.04	0.03	0.03	0.07	0.29	0.34
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
01000132	value	-489.99	-489.99	-546.80	-546.80	-546.80	**-548.21**	**-548.21**	**-548.21**
	bound	-Inf	-Inf	-550.60	-550.60	-550.60	**-548.21**	**-548.21**	**-548.21**
	runtime	0.16	0.07	0.04	0.03	0.03	0.93	1.25	1.39
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB
01000164	value	-238.52	-238.52	-261.29	-261.29	-261.29	**-261.58**	**-261.58**	**-261.58**
	bound	-Inf	-Inf	-262.05	-262.05	-262.05	**-261.58**	**-261.58**	**-261.58**
	runtime	4.49	0.21	0.02	0.01	0.01	0.26	0.48	0.48
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01002241	value	-623.54	-623.54	-648.12	-648.12	-648.12	**-651.09**	**-651.09**	**-651.09**
	bound	-Inf	-Inf	-651.17	-651.17	-651.17	**-651.09**	**-651.09**	**-651.09**
	runtime	0.28	0.13	0.03	0.04	0.04	0.11	0.42	0.64
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
01004477	value	-597.07	-597.07	-562.46	-562.46	-562.46	**-566.80**	**-566.80**	**-566.80**
	bound	-Inf	-Inf	-569.36	-569.36	-569.36	**-566.80**	**-566.80**	**-566.80**
	runtime	1.31	0.58	0.09	0.06	0.07	0.13	1.01	1.41
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01010570	value	-599.03	-599.03	**-604.76**	**-604.76**	**-604.76**	**-604.76**	**-604.76**	**-604.76**
	bound	-Inf	-Inf	**-604.76**	**-604.76**	**-604.76**	**-604.76**	**-604.76**	**-604.76**
	runtime	0.46	0.20	0.03	0.02	0.02	0.02	0.33	0.35
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
01010628	value	-799.03	-799.03	-868.89	-868.89	-868.89	**-870.21**	**-870.21**	**-870.21**
	bound	-Inf	-Inf	-871.15	-871.15	-871.15	**-870.21**	**-870.21**	**-870.21**
	runtime	1.69	2.21	0.09	0.06	0.07	1.19	1.82	1.79
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
01010740	value	-583.24	-583.24	-632.98	-632.98	-632.98	**-632.98**	**-632.98**	**-632.98**
	bound	-Inf	-Inf	-634.23	-634.23	-634.23	**-632.98**	**-632.98**	**-632.98**
	runtime	0.45	0.25	0.02	0.02	0.02	0.05	0.15	0.13
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB
01010757	value	-768.38	-768.38	-811.92	-811.92	-811.92	**-813.76**	**-813.76**	**-813.76**
	bound	-Inf	-Inf	-815.82	-815.82	-815.82	**-813.76**	**-813.76**	**-813.76**
	runtime	1.21	0.56	0.10	0.06	0.06	3.08	4.23	4.25
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
01010922	value	-566.54	-566.54	-622.38	-622.38	-622.38	**-621.37**	**-621.37**	**-621.37**
	bound	-Inf	-Inf	-622.38	-622.38	-622.38	**-621.37**	**-621.37**	**-621.37**
	runtime	1.16	0.52	0.13	0.09	0.09	0.44	2.46	2.58
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.05 GB
01010851	value	-531.26	-531.26	-561.27	-561.27	-561.27	**-564.52**	**-564.52**	**-564.52**
	bound	-Inf	-Inf	-569.39	-569.39	-569.35	**-564.52**	**-564.52**	**-564.52**
	runtime	0.28	0.12	0.13	0.08	0.09	2.03	2.94	2.26
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01010855	value	-277.33	-277.33	-329.01	-329.01	-329.01	**-329.67**	**-329.67**	**-329.67**
	bound	-Inf	-Inf	-329.67	-329.67	-329.67	**-329.67**	-329.68	**-329.67**
	runtime	0.69	0.31	0.04	0.02	0.02	0.04	0.55	0.53
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01010892	value	-659.52	-659.52	-727.51	-727.51	-727.51	**-727.96**	**-727.96**	**-727.96**
	bound	-Inf	-Inf	-727.99	-727.99	-727.99	**-727.96**	**-727.96**	**-727.96**
	runtime	0.42	0.21	0.03	0.02	0.02	0.10	0.33	0.38
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
01010935	value	-365.88	-365.88	-440.74	-440.74	-440.74	**-442.12**	**-442.12**	**-442.12**
	bound	-Inf	-Inf	-443.43	-443.43	-443.43	**-442.12**	**-442.12**	**-442.12**
	runtime	0.54	0.24	0.03	0.02	0.02	0.06	0.54	0.38
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
01010960	value	-451.09	-451.09	-492.30	-492.30	-492.30	**-483.35**	**-483.35**	**-483.35**
	bound	-Inf	-Inf	-484.29	-484.29	-484.29	**-483.35**	**-483.35**	**-483.35**
	runtime	0.63	0.26	0.14	0.11	0.11	1.57	2.65	1.91
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01011121	value	-276.37	-276.37	**-327.82**	**-327.82**	**-327.82**	**-327.82**	**-327.82**	**-327.82**
	bound	-Inf	-Inf	**-327.82**	**-327.82**	**-327.82**	**-327.82**	**-327.82**	**-327.82**
	runtime	0.07	0.03	0.02	0.02	0.02	0.02	0.07	0.10
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
01011434	value	-332.97	-332.97	-415.48	-415.48	-415.48	**-418.68**	**-418.68**	**-418.68**
	bound	-Inf	-Inf	-420.02	-420.02	-420.02	**-418.68**	-418.68	**-418.68**
	runtime	0.52	0.23	0.07	0.04	0.05	0.12	0.55	0.54
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01011463	value	-653.76	-653.76	**-707.85**	**-707.85**	**-707.85**	**-707.85**	**-707.85**	**-707.85**
	bound	-Inf	-Inf	**-707.85**	**-707.85**	**-707.85**	**-707.85**	-707.85	**-707.85**
	runtime	0.47	0.21	0.03	0.02	0.02	0.07	0.24	0.34
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
01011488	value	-284.85	-284.85	-331.20	-331.20	-331.20	**-331.61**	**-331.61**	**-331.61**
	bound	-Inf	-Inf	-331.83	-331.83	-331.83	**-331.61**	-331.61	**-331.61**
	runtime	0.08	0.04	0.02	0.01	0.01	0.23	0.33	0.45
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
01011492	value	-523.20	-523.20	-563.76	-563.76	-563.76	**-565.50**	**-565.50**	**-565.50**
	bound	-Inf	-Inf	-566.20	-566.20	-566.20	**-565.50**	**-565.50**	**-565.50**
	runtime	0.61	0.28	0.06	0.03	0.02	0.07	0.02	0.23
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
01011801	value	-676.44	-676.44	-717.31	-717.31	-719.29	**-718.31**	**-718.31**	**-718.31**
	bound	-Inf	-Inf	-719.29	-719.29	-719.29	**-718.31**	**-718.31**	**-718.31**
	runtime	1.83	0.80	0.10	0.05	0.05	0.75	1.06	1.04
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
01010939	value	-720.23	-720.23	**-773.58**	**-773.58**	**-773.58**	**-773.58**	**-773.58**	**-773.58**
	bound	-Inf	-Inf	**-773.58**	**-773.58**	**-773.58**	**-773.58**	-773.58	**-773.58**
	runtime	0.64	0.28	0.04	0.03	0.03	0.07	0.29	0.34
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
01012170	value	-509.01	-509.03	**-530.12**	**-530.12**	**-530.12**	**-530.12**	**-530.12**	**-530.12**
	bound	-Inf	-Inf	**-530.12**	**-530.12**	**-530.12**	**-530.12**	**-530.12**	**-530.12**
	runtime	0.79	0.36	0.09	0.06	0.05	0.06	0.32	0.31
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
01012234	value	-697.81	-697.81	**-758.50**	**-758.50**	**-758.50**	**-758.50**	**-758.50**	**-758.50**
	bound	-Inf	-Inf	-759.45	-759.45	-759.45	**-758.50**	**-758.50**	**-758.50**
	runtime	1.85	0.45	0.04	0.04	0.04	0.28	0.33	0.45
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
01024003	value	-780.60	-780.60	-798.03	-798.03	-798.03	**-803.76**	**-803.76**	**-803.76**
	bound	-Inf	-Inf	-804.45	-804.45	-804.45	**-803.76**	**-803.76**	**-803.76**
	runtime	0.50	0.30	0.08	0.06	0.06	0.28	0.76	0.71
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
01024435	value	-1064.74	-1064.74	-1078.90	-1078.90	-1078.90	**-1079.66**	**-1079.66**	**-1079.66**
	bound	-Inf	-Inf	-1080.33	-1080.33	-1080.33	**-1079.66**	**-1079.66**	**-1079.66**
	runtime	0

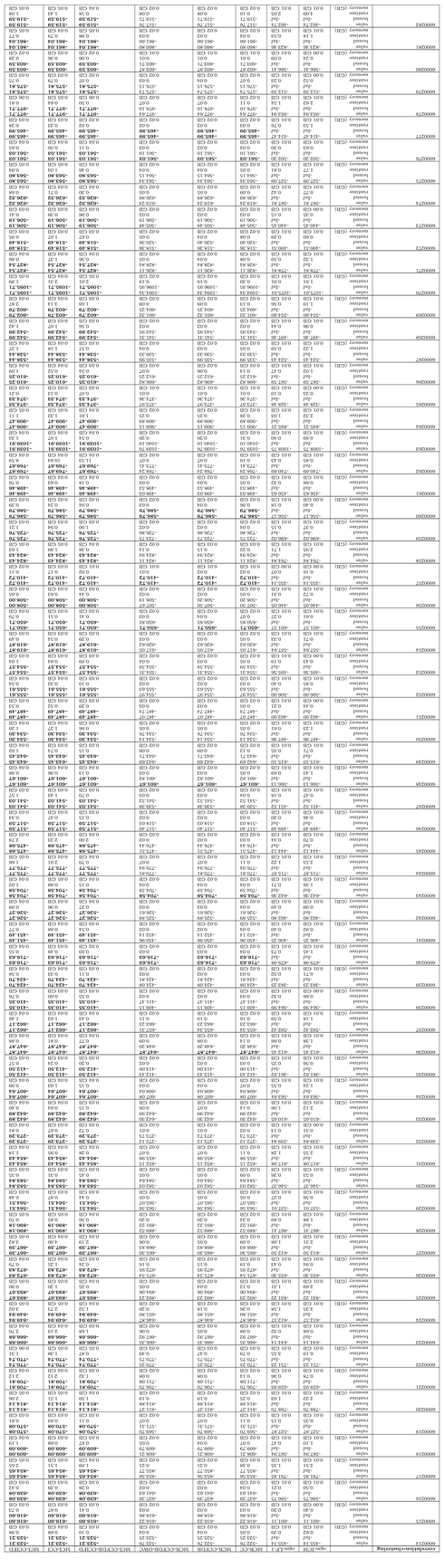
| overwhelm-interesting | quiz-like | sickness | sickness | sickness

correlation=clustering		ogm-ILCM	ogm-LP-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCLCC†	MCLCCPD
4000086	value	-512.83	-512.83	-550.54	-550.54	-550.54	**-550.96**	**-550.96**	**-550.96**
	bound	-Inf	-Inf	-552.97	-552.67	-552.67	**-550.96**	**-550.96**	**-550.96**
	runtime	0.83	0.40	0.08	0.07	0.07	0.25	0.80	0.97
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
4100066	value	-329.28	-320.28	-556.79	-556.79	-556.79	**-557.66**	**-557.66**	**-557.66**
	bound	-Inf	-Inf	-557.66	-557.66	-557.66	**-557.66**	**-557.66**	**-557.66**
	runtime	0.56	0.28	0.08	0.05	0.05	0.20	0.54	0.43
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
4100246	value	-595.44	-595.44	-634.05	-634.05	-634.05	**-635.17**	**-635.17**	**-635.17**
	bound	-Inf	-Inf	-635.33	-635.33	-635.33	**-635.17**	**-635.17**	**-635.17**
	runtime	1.43	0.63	0.13	0.07	0.07	0.14	0.57	0.61
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
4100280	value	-796.90	-796.90	**-877.45**	**-877.45**	**-877.45**	**-877.45**	**-877.45**	**-877.45**
	bound	-Inf	-Inf	-877.45	-877.45	-877.45	**-877.45**	**-877.45**	**-877.45**
	runtime	0.92	0.42	0.11	0.07	0.07	0.06	0.61	0.66
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB	0.04 GB	0.06 GB
5000016	value	-753.97	-753.97	**-783.18**	**-783.18**	**-783.18**	**-783.18**	**-783.18**	**-783.18**
	bound	-Inf	-Inf	-783.37	-783.37	-783.37	**-783.18**	**-783.18**	**-783.18**
	runtime	0.94	0.43	0.11	0.17	0.16	0.36	0.36	0.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000119	value	-516.98	-516.98	**-634.49**	**-634.49**	**-634.49**	**-634.49**	**-634.49**	**-634.49**
	bound	-Inf	-Inf	**-634.49**	**-634.49**	**-634.49**	**-634.49**	**-634.49**	**-634.49**
	runtime	0.33	0.13	0.07	0.02	0.02	0.02	0.22	0.19
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
5000120	value	-398.38	-398.38	-453.00	-453.00	-453.00	**-454.80**	**-454.80**	**-454.80**
	bound	-Inf	-Inf	-455.54	-455.54	-455.54	**-454.80**	-454.92	**-454.80**
	runtime	0.75	0.37	0.03	0.02	0.02	0.83	0.83	0.89
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
5000121	value	-457.20	-457.20	-486.38	-486.38	-486.38	**-495.05**	**-495.05**	**-495.05**
	bound	-Inf	-Inf	-495.38	-495.38	-495.38	**-495.05**	**-495.05**	**-495.05**
	runtime	0.84	0.38	0.04	0.02	0.02	0.03	0.14	0.06
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000122	value	-380.99	-380.99	-421.47	-421.47	-421.47	**-425.47**	**-425.47**	**-425.47**
	bound	-Inf	-Inf	-427.49	-427.49	-427.49	**-425.47**	**-425.47**	**-425.47**
	runtime	0.84	0.38	0.04	0.02	0.02	0.75	1.81	1.82
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000123	value	-394.56	-394.56	-429.84	-429.84	-429.84	**-429.95**	**-429.95**	**-429.95**
	bound	-Inf	-Inf	-429.95	-429.95	-429.95	**-429.95**	**-429.95**	**-429.95**
	runtime	0.86	0.39	0.04	0.03	0.03	0.03	0.64	0.67
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
5000124	value	-556.19	-556.19	-592.98	-592.98	-592.98	**-587.95**	**-587.95**	**-587.95**
	bound	-Inf	-Inf	-588.97	-588.97	-588.97	**-587.95**	**-587.95**	**-587.95**
	runtime	0.68	0.30	0.11	0.09	0.08	1.34	3.39	2.71
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000125	value	-581.51	-581.51	-637.64	-637.64	-637.64	**-639.51**	**-639.51**	**-639.51**
	bound	-Inf	-Inf	-640.68	-640.68	-640.68	**-639.51**	**-639.51**	**-639.51**
	runtime	0.84	0.38	0.04	0.02	0.02	0.03	0.14	0.16
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
5000126	value	-1827.77	-1827.77	**-1843.63**	**-1843.63**	**-1843.63**	**-1843.63**	**-1843.63**	**-1843.63**
	bound	-Inf	-Inf	-1843.86	-1843.86	-1843.86	**-1843.63**	**-1843.63**	**-1843.63**
	runtime	4.88	2.85	0.09	0.09	0.09	0.42	0.82	1.16
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.06 GB	0.06 GB	0.06 GB
5000127	value	-290.81	-290.81	-363.48	-363.48	-363.48	**-364.53**	**-364.53**	**-364.53**
	bound	-Inf	-Inf	-364.91	-364.91	-364.91	**-364.53**	**-364.53**	**-364.53**
	runtime	0.72	0.35	0.02	0.02	0.02	0.76	1.10	1.48
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
5000128	value	-1164.27	-1164.27	-1220.92	-1220.92	-1220.92	**-1221.02**	**-1221.02**	**-1221.02**
	bound	-Inf	-Inf	-1221.76	-1221.76	-1221.76	**-1221.02**	**-1221.02**	**-1221.02**
	runtime	1.65	0.75	0.17	0.13	0.13	0.49	1.19	1.54
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
5000129	value	-637.66	-637.66	**-673.62**	**-673.62**	**-673.62**	**-673.62**	**-673.62**	**-673.62**
	bound	-Inf	-Inf	**-673.62**	**-673.62**	**-673.62**	**-673.62**	**-673.62**	**-673.62**
	runtime	0.23	0.10	0.01	0.01	0.01	0.02	0.09	0.12
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB	0.02 GB
5000130	value	-1316.07	-1316.07	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**
	bound	-Inf	-Inf	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**	**-1351.90**
	runtime	7.21	3.51	0.53	0.34	0.34	0.35	1.19	1.19
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB
5000131	value	-579.50	-579.50	-600.93	-600.93	-600.93	**-602.56**	**-602.56**	**-602.56**
	bound	-Inf	-Inf	-603.98	-603.98	-603.98	**-602.56**	**-602.56**	**-602.56**
	runtime	1.59	0.82	0.04	0.03	0.03	1.00	1.10	1.20
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000132	value	-496.54	-496.54	-584.28	-584.28	-584.28	**-586.21**	**-586.21**	**-586.21**
	bound	-Inf	-Inf	-586.21	-586.21	-586.21	**-586.21**	**-586.21**	**-586.21**
	runtime	0.53	0.27	0.06	0.03	0.03	0.23	0.48	0.69
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000133	value	-344.64	-344.64	-391.06	-391.06	-391.06	**-391.28**	**-391.28**	**-391.28**
	bound	-Inf	-Inf	-391.41	-391.41	-391.41	**-391.28**	**-391.28**	**-391.28**
	runtime	0.19	0.08	0.01	0.01	0.01	0.08	0.22	0.35
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB
5000137	value	-833.68	-833.68	**-834.68**	**-834.68**	**-834.68**	**-834.68**	**-834.68**	**-834.68**
	bound	-Inf	-Inf	**-834.68**	**-834.68**	**-834.68**	**-834.68**	**-834.68**	**-834.68**
	runtime	1.16	0.56	0.03	0.03	0.03	0.03	0.12	0.12
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
5000144	value	-665.20	-665.20	**-725.12**	**-725.12**	**-725.12**	**-725.12**	**-725.12**	**-725.12**
	bound	-Inf	-Inf	**-725.12**	**-725.12**	**-725.12**	**-725.12**	**-725.12**	**-725.12**
	runtime	0.54	0.24	0.03	0.03	0.03	0.03	0.10	0.11
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
5000147	value	-497.10	-497.10	-519.03	-519.03	-519.03	**-520.82**	**-520.82**	**-520.82**
	bound	-Inf	-Inf	-520.82	-520.82	-520.82	**-520.82**	**-520.82**	**-520.82**
	runtime	0.16	0.07	0.05	0.07	0.07	0.02	1.57	1.04
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.02 GB
5000149	value	-808.24	-808.24	-813.19	-813.19	-813.19	**-813.81**	**-813.81**	**-813.81**
	bound	-Inf	-Inf	-814.56	-814.56	-814.56	**-813.81**	**-813.81**	**-813.81**
	runtime	1.54	0.75	0.07	0.05	0.05	0.54	1.10	0.85
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.06 GB	0.06 GB	0.05 GB
5000150	value	-452.36	-452.36	-496.42	-496.42	-496.42	**-496.80**	**-496.80**	**-496.80**
	bound	-Inf	-Inf	-496.80	-496.80	-496.80	**-496.80**	**-496.80**	**-496.80**
	runtime	0.64	0.29	0.04	0.02	0.02	0.03	0.14	0.04
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000162	value	-523.11	-523.11	-574.24	-574.24	-574.24	**-575.46**	**-575.46**	**-575.46**
	bound	-Inf	-Inf	-575.67	-575.67	-575.67	**-575.46**	**-575.46**	**-575.46**
	runtime	1.92	1.09	0.07	0.04	0.05	0.25	0.61	0.70
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
5000163	value	-832.29	-832.29	-876.12	-876.12	-876.12	**-876.40**	**-876.40**	**-876.40**
	bound	-Inf	-Inf	-877.01	-877.01	-877.01	**-876.40**	**-876.40**	**-876.40**
	runtime	0.24	0.10	0.02	0.01	0.01	0.03	0.10	0.10
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000164	value	-756.15	-756.15	-825.72	-825.72	-825.72	**-829.15**	**-829.15**	**-829.15**
	bound	-Inf	-Inf	-830.38	-830.38	-830.38	**-829.15**	**-829.15**	**-829.15**
	runtime	0.66	0.30	0.13	0.12	0.12	1.43	2.23	2.40
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000165	value	-714.51	-714.51	**-731.05**	**-731.05**	**-731.05**	**-731.05**	**-731.05**	**-731.05**
	bound	-Inf	-Inf	**-731.05**	**-731.05**	**-731.05**	**-731.05**	**-731.05**	**-731.05**
	runtime	0.41	0.18	0.03	0.02	0.02	0.02	0.03	0.04
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
5000166	value	-1011.33	-1011.33	-1030.36	-1030.36	-1030.36	**-1031.79**	**-1031.79**	**-1031.79**
	bound	-Inf	-Inf	-1031.91	-1031.91	-1031.91	**-1031.79**	**-1031.79**	**-1031.79**
	runtime	4.41	2.42	0.56	0.30	0.29	0.55	1.53	2.00
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.06 GB	0.06 GB
5000168	value	-728.53	-728.53	-765.33	-765.33	-765.33	**-770.89**	**-770.89**	**-770.89**
	bound	-Inf	-Inf	-771.57	-771.57	-771.57	**-770.89**	**-770.89**	**-770.89**
	runtime	1.82	0.91	0.26	0.13	0.13	0.61	2.01	2.27
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000172	value	-890.40	-890.40	**-915.17**	**-915.17**	**-915.17**	**-915.17**	**-915.17**	**-915.17**
	bound	-Inf	-Inf	**-915.17**	**-915.17**	**-915.17**	**-915.17**	**-915.17**	**-915.17**
	runtime	1.41	0.69	0.15	0.11	0.10	0.11	1.32	0.84
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB
5000173	value	-907.50	-907.50	-942.17	-942.17	-942.17	**-942.40**	**-942.40**	**-942.40**
	bound	-Inf	-Inf	-942.67	-942.67	-942.67	**-942.40**	**-942.40**	**-942.40**
	runtime	0.64	0.29	0.03	0.02	0.02	0.03	0.13	0.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000174	value	-609.34	-609.34	**-651.23**	**-651.23**	**-651.23**	**-651.23**	**-651.23**	**-651.23**
	bound	-Inf	-Inf	**-651.23**	**-651.23**	**-651.23**	**-651.23**	**-651.23**	**-651.23**
	runtime	0.51	0.27	0.03	0.02	0.02	0.02	0.24	0.25
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
5000175	value	-923.33	-923.33	-943.96	-943.96	-943.96	**-944.86**	**-944.86**	**-944.86**
	bound	-Inf	-Inf	-945.37	-945.37	-945.37	**-944.86**	**-944.86**	**-944.86**
	runtime	1.84	0.93	0.25	0.17	0.16	1.01	1.09	1.20
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
5000176	value	-985.40	-985.40	-1018.31	-1024.31	-1018.31	**-1022.83**	**-1022.83**	**-1022.83**
	bound	-Inf	-Inf	-1027.32	-1027.32	-1027.32	**-1022.83**	**-1022.83**	**-1022.83**
	runtime	1.97	1.82	4.78	2.53	2.52	15.68	25.82	12.40
	memory (GB)	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.03 GB	0.08 GB	0.08 GB	0.07 GB
5000180	value	-698.24	-698.24	**-719.00**	**-719.00**	**-719.00**	**-719.00**	**-719.00**	**-719.00**
	bound	-Inf	-Inf	**-719.00**	**-719.00**	**-719.00**	**-719.00**	**-719.00**	**-719.00**
	runtime	3.62	1.79	0.84	0.33	0.32	0.33	0.92	1.07
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB
5000181	value	-321.99	-321.99	**-350.75**	**-350.75**	**-350.75**	**-350.75**	**-350.75**	**-350.75**
	bound	-Inf	-Inf	-350.86	-350.86	-350.86	**-350.75**	**-350.75**	**-350.75**
	runtime	0.84	0.40	0.07	0.02	0.02	0.03	0.04	0.04
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000182	value	-1009.59	-1009.59	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**
	bound	-Inf	-Inf	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**	**-1053.66**
	runtime	1.20	0.57	0.05	0.04	0.04	0.05	0.39	0.36
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
5000183	value	-675.85	-675.85	**-731.41**	**-731.41**	**-731.41**	**-731.41**	**-731.41**	**-731.41**
	bound	-Inf	-Inf	**-731.41**	**-731.41**	**-731.41**	**-731.41**	**-731.41**	**-731.41**
	runtime	0.51	0.23	0.03	0.02	0.02	0.02	0.03	0.04
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
5000184	value	-873.89	-873.89	**-908.78**	**-908.78**	**-908.78**	**-908.78**	**-908.78**	**-908.78**
	bound	-Inf	-Inf	-909.81	-909.81	-909.81	**-908.78**	**-908.78**	**-908.78**
	runtime	1.06	0.47	0.25	0.13	0.13	0.24	0.76	0.70
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
5000188	value	-619.05	-619.05	-646.21	-646.21	-646.21	**-646.70**	**-646.70**	**-646.70**
	bound	-Inf	-Inf	-646.97	-646.97	-646.97	**-646.70**	**-646.70**	**-646.70**
	runtime	1.09	0.50	0.11	0.26	0.26	0.67	1.81	1.56
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000190	value	-661.35	-661.35	**-694.76**	**-694.76**	**-694.76**	**-694.76**	**-694.76**	**-694.76**
	bound	-Inf	-Inf	-697.06	-697.06	-697.06	**-694.76**	**-694.76**	**-694.76**
	runtime	0.42	0.20	0.15	0.10	0.10	0.16	0.24	0.24
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000191	value	-658.31	-658.31	**-699.02**	**-699.02**	**-699.02**	**-699.02**	**-699.02**	**-699.02**
	bound	-Inf	-Inf	**-699.02**	**-699.02**	**-699.02**	**-699.02**	**-699.02**	**-699.02**
	runtime	0.44	0.19	0.03	0.02	0.02	0.02	0.11	0.16
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000192	value	-513.49	-513.49	-513.77	-513.77	-513.77	**-513.49**	**-513.49**	**-513.49**
	bound	-Inf	-Inf	-513.77	-513.77	-513.77	**-513.49**	**-513.49**	**-513.49**
	runtime	0.45	0.20	0.03	0.02	0.02	0.18	0.34	0.29
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000194	value	-138.34	-138.34	**-201.84**	**-201.84**	**-201.84**	**-201.84**	**-201.84**	**-201.84**
	bound	-Inf	-Inf	**-201.84**	**-201.84**	**-201.84**	**-201.84**	**-201.84**	**-201.84**
	runtime	0.05	0.02	0.00	0.00	0.00	0.00	0.04	0.06
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
5000195	value	-278.77	-278.77	**-321.84**	**-321.84**	**-321.84**	**-321.84**	**-321.84**	**-321.84**
	bound	-Inf	-Inf	**-321.84**	**-321.84**	**-321.84**	**-321.84**	**-321.84**	**-321.84**
	runtime	0.25	0.11	0.02	0.01	0.01	0.01	0.09	0.09
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
5000196	value	-443.35	-443.35	**-471.86**	**-471.86**	**-471.86**	**-471.86**	**-471.86**	**-471.86**
	bound	-Inf	-Inf	**-471.86**	**-471.86**	**-471.86**	**-471.86**	**-471.86**	**-471.86**
	runtime	0.37	0.15	0.01	0.01	0.01	0.01	0.10	0.10
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
5000197	value	-420.64	-420.64	-487.37	-487.37	-487.37	**-485.98**	**-485.98**	**-485.98**
	bound	-Inf	-Inf	-487.37	-487.37	-487.37	**-485.98**	**-485.98**	**-485.98**
	runtime	0.72	0.36	0.05	0.03	0.04	0.58	0.96	1.63
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
5000198	value	-301.06	-301.06	**-336.76**	**-336.76**	**-336.76**	**-336.76**	**-336.76**	**-336.76**
	bound	-Inf	-Inf	**-336.76**	**-336.76**	**-336.76**	**-336.76**	**-336.76**	**-336.76**
	runtime	0.20	0.09	0.01	0.01	0.01	0.01	0.08	0.06
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
5000199	value	-564.20	-564.20	-597.38	-597.38	-597.38	**-598.26**	**-598.26**	**-598.26**
	bound	-Inf	-Inf	-598.26	-598.26	-598.26	**-598.26**	**-598.26**	**-598.26**
	runtime	1.32	0.65	0.02	0.02	0.03	0.31	1.19	1.47
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
5000200	value	-204.87	-204.87	-237.65	-237.65	-237.65	**-238.42**	**-238.42**	**-238.42**
	bound	-Inf	-Inf	-239.74	-239.74	-239.74	**-238.42**	**-238.42**	**-238.42**
	runtime	0.23	0.10	0.02	0.02	0.01	0.33	0.43	0.80
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
5000201	value	-543.62	-543.62	-606.07	-606.07	-606.07	**-609.80**	**-609.80**	**-609.80**
	bound	-Inf	-Inf	-610.58	-610.58	-610.58	**-609.80**	**-609.80**	**-609.80**
	runtime	2.91	1.51	0.36	0.34	0.34	0.35	1.19	1.19
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
5000202	value	-546.81	-546.81	-628.46	-628.46	-628.46	**-629.27**	**-629.27**	**-629.27**
	bound	-Inf	-Inf	-630.08	-630.08	-630.08	**-629.27**	**-629.27**	**-629.27**
	runtime	0.59	0.26	0.02	0.01	0.01	0.05	0.14	0.18
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
5000203	value	-318.79	-318.79	-410.82	-410.82	-410.82	**-411.27**	**-411.27**	**-411.27**
	bound	-Inf	-Inf	-411.27	-411.27	-411.27	**-411.27**	**-411.27**	**-411.27**
	runtime	0.90	0.50	0.07	0.04	0.04	0.14	0.77	0.73
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
5000204	value	-997.11	-997.11	**-1025.32**	**-1025.32**	**-1025.32**	**-1025.32**	**-1025.32**	**-1025.32**
	bound	-Inf	-Inf	-1025.91	-1025.91	-1025.91	-1025.51	**-1025.32**	**-1025.32**
	runtime	4.76	2.93	0.19	0.22	0.22	0.97	3.61	2.41
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.06 GB	0.06 GB	0.06 GB
5000205	value	-478.51	-478.51	-544.56	-544.56	-544.56	**-545.13**	**-545.13**	**-545.13**
	bound	-Inf	-Inf	-545.26	-545.26	-545.26	**-545.13**	**-545.13**	**-545.13**
	runtime	0.21	0.09	0.02	0.02	0.02	0.02	0.11	0.11
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
5000226	value	-491.20	-491.20	-511.91	-511.91	-511.91	**-511.99**	**-511.99**	**-511.99**
	bound	-Inf	-Inf	-512.15	-512.15	-512.15	**-511.99**	**-511.99**	**-511.99**
	runtime	0.16	0.07	0.01	0.01	0.01	0.04	0.22	0.17
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
5000234	value	-819.35	-819.35	-857.52	-857.52	-857.52	**-858.86**	**-858.86**	**-858.86**
	bound	-Inf	-Inf	-860.79	-860.79	-860.79	**-858.86**	**-858.86**	**-858.86**
	runtime	0.55	0.25	0.08	0.05	0.05	0.03	0.13	0.23
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000257	value	-628.55	-628.55	-664.52	-664.52	-664.52	**-665.10**	**-665.10**	**-665.10**
	bound	-Inf	-Inf	-667.27	-667.27	-667.27	**-665.10**	**-665.10**	**-665.10**
	runtime	0.55	0.25	0.08	0.05	0.06	0.28	0.57	0.68
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
5000264	value	-492.68	-492.68	-518.91	-518.91	-518.91	**-520.68**	**-520.68**	**-520.68**
	bound	-Inf	-Inf	-524.05	-524.05	-524.05	**-520.68**	**-520.68**	**-520.68**
	runtime	0.64	0.29	0.04	0.02	0.02	0.03	0.14	0.04
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
5000265	value	-395.35	-395.35	-451.81	-451.81	-451.81	**-452.13**	**-452.13**	**-452.13**
	bound	-Inf	-Inf	-452.28	-452.28	-452.28	**-452.13**	**-452.13**	**-452.13**
	runtime	0.38	0.19	0.03	0.01	0.02	0.15	0.33	0.41
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000000	value	-551.26	-551.26	-575.92	-575.92	-575.92	**-590.23**	**-590.23**	**-590.23**
	bound	-Inf	-Inf	-592.84	-592.84	-592.84	**-590.23**	**-590.25**	**-590.23**
	runtime	1.88	0.86	4.13	0.09	0.10	0.16	1.21	0.97
	memory (GB)	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000001	value	-754.59	-754.59	-788.72	-788.72	-788.72	**-789.77**	**-789.77**	**-789.77**
	bound	-Inf	-Inf	-790.00	-790.00	-790.00	**-789.77**	**-789.77**	**-789.77**
	runtime	0.57	0.27	0.07	0.05	0.05	0.39	0.69	0.79
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 G

correlation=clustering		ogm-ILCM	ogm-LP-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCL-CC1	MCL-CCIPD
6000007	value	-590.16	-590.16	-638.31	-638.31	-638.31	**-641.39**	**-641.39**	**-641.39**
	bound	-Inf	-Inf	-643.18	-643.18	-643.18	**-641.39**	**-641.39**	**-641.39**
	runtime	0.97	0.55	0.21	0.22	0.23	0.76	1.41	2.13
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000008	value	-542.85	-542.85	**-586.78**	**-586.78**	**-586.78**	**-586.78**	**-586.78**	**-586.78**
	bound	-Inf	-Inf	**-586.78**	**-586.78**	**-586.78**	**-586.78**	**-586.78**	**-586.78**
	runtime	0.37	0.16	0.03	0.02	0.02	0.02	0.25	0.30
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
6000009	value	-659.28	-659.28	-680.40	-680.40	-680.40	**-682.12**	**-682.12**	**-682.12**
	bound	-Inf	-Inf	-682.46	-682.46	-682.46	**-682.12**	**-682.12**	**-682.12**
	runtime	2.68	1.40	0.18	0.10	0.10	0.79	1.09	1.41
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000010	value	-870.82	-870.82	-902.66	-902.56	-902.66	**-903.38**	**-903.38**	**-903.38**
	bound	-Inf	-Inf	-903.77	-903.77	-903.77	**-903.38**	**-903.38**	**-903.38**
	runtime	3.61	2.04	0.21	0.11	0.11	0.85	1.20	1.56
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.05 GB	0.06 GB	0.06 GB
6000011	value	-546.27	-546.27	-611.65	-611.65	-611.65	**-616.57**	**-616.57**	**-616.57**
	bound	-Inf	-Inf	-617.81	-617.81	-617.81	**-616.57**	**-616.57**	**-616.57**
	runtime	0.39	0.18	0.04	0.02	0.02	0.68	0.89	0.87
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000012	value	-567.79	-567.79	-599.26	-599.26	-599.26	**-600.13**	**-600.13**	**-600.13**
	bound	-Inf	-Inf	-601.14	-601.14	-601.14	**-600.13**	**-600.13**	**-600.13**
	runtime	0.99	0.46	0.14	0.08	0.09	0.22	0.22	2.03
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000013	value	-678.24	-678.24	-733.25	-733.25	-733.25	**-733.64**	**-733.64**	**-733.64**
	bound	-Inf	-Inf	-736.69	-736.69	-736.69	**-733.64**	**-733.64**	**-733.64**
	runtime	1.88	0.95	0.10	0.05	0.06	3.96	5.16	4.74
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000014	value	-920.70	-920.70	-938.55	-938.55	-938.55	**-940.94**	**-940.94**	**-940.94**
	bound	-Inf	-Inf	-943.31	-943.31	-943.31	**-940.94**	**-940.94**	**-940.94**
	runtime	2.64	1.31	0.21	0.11	0.11	0.84	1.20	1.56
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
6000015	value	-672.53	-672.53	-715.37	-715.37	-715.37	**-716.32**	**-716.32**	**-716.32**
	bound	-Inf	-Inf	-716.81	-716.81	-716.81	**-716.32**	**-716.32**	**-716.32**
	runtime	0.43	0.19	0.06	0.03	0.04	0.22	0.59	0.64
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000016	value	-581.65	-581.65	-612.35	-612.35	-612.35	**-612.70**	**-612.70**	**-612.70**
	bound	-Inf	-Inf	-613.58	-613.58	-613.58	**-612.70**	**-612.70**	**-612.70**
	runtime	0.93	0.46	0.09	0.05	0.06	0.23	0.89	0.93
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
6000017	value	-576.58	-576.58	-597.24	-597.24	-597.24	**-599.46**	**-599.46**	**-599.46**
	bound	-Inf	-Inf	-600.58	-601.02	-601.02	**-599.46**	**-599.46**	**-599.46**
	runtime	1.73	0.83	0.28	0.16	0.16	0.92	2.03	1.76
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000018	value	-788.45	-788.45	-795.20	-795.20	-795.20	**-796.75**	**-796.75**	**-796.75**
	bound	-Inf	-Inf	-797.88	-797.88	-797.88	**-796.75**	**-796.75**	**-796.75**
	runtime	1.74	0.81	0.14	0.06	0.06	0.21	1.50	0.85
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000019	value	-716.38	-716.58	-743.09	-743.09	-743.09	**-743.99**	**-743.99**	**-743.99**
	bound	-Inf	-Inf	-744.13	-744.13	-744.13	**-743.99**	**-743.99**	**-743.99**
	runtime	1.74	0.81	0.14	0.06	0.06	0.21	1.50	0.85
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000020	value	-621.97	-621.97	-649.59	-649.59	-649.59	**-653.26**	**-653.26**	**-653.26**
	bound	-Inf	-Inf	-653.69	-653.09	-653.69	**-653.26**	**-653.26**	**-653.26**
	runtime	0.97	0.48	0.15	0.10	0.11	0.41	0.85	0.89
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000021	value	-757.66	-757.66	-757.60	-757.60	-757.60	**-758.47**	**-758.47**	**-758.47**
	bound	-Inf	-Inf	-761.53	-761.53	-761.53	**-758.47**	**-758.47**	**-758.47**
	runtime	0.92	0.40	0.08	0.05	0.05	0.53	0.87	1.14
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000022	value	-361.84	-361.84	-361.83	-361.83	-361.83	**-364.28**	**-364.28**	**-364.28**
	bound	-Inf	-Inf	-365.26	-365.26	-365.26	**-364.28**	**-364.28**	**-364.28**
	runtime	0.09	0.05	0.01	0.01	0.01	0.06	0.18	0.21
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
6000023	value	-497.68	-497.68	-530.94	-530.94	-530.94	**-530.94**	**-530.94**	**-530.94**
	bound	-Inf	-Inf	-533.42	-533.42	-533.42	**-530.94**	**-530.94**	**-530.94**
	runtime	0.17	0.07	0.05	0.05	0.05	0.60	1.39	1.36
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000024	value	-505.63	-505.63	-536.01	-536.01	-536.01	**-536.14**	**-536.14**	**-536.14**
	bound	-Inf	-Inf	-537.64	-537.64	-537.64	**-536.14**	**-536.14**	**-536.14**
	runtime	1.17	0.50	0.12	0.08	0.09	7.25	3.26	2.63
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000025	value	-465.84	-465.84	-534.52	-534.52	-534.52	**-535.57**	**-535.57**	**-535.57**
	bound	-Inf	-Inf	-535.57	-535.57	-535.57	**-535.57**	**-535.57**	**-535.57**
	runtime	0.68	0.29	0.04	0.03	0.03	0.53	1.24	1.22
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000026	value	-480.54	-480.54	-520.62	-520.62	-520.62	**-521.61**	**-521.61**	**-521.61**
	bound	-Inf	-Inf	-523.41	-523.41	-523.41	**-521.61**	**-521.61**	**-521.61**
	runtime	0.74	0.37	0.06	0.03	0.04	0.72	0.99	1.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000027	value	-529.54	-529.54	-577.64	-577.64	-577.64	**-576.54**	**-576.54**	**-576.54**
	bound	-Inf	-Inf	-577.64	-577.64	-577.64	**-576.54**	**-576.54**	**-576.54**
	runtime	0.29	0.12	0.05	0.03	0.03	0.23	0.83	0.62
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000028	value	-829.69	-829.69	-873.74	-873.74	-873.74	**-874.15**	**-874.15**	**-874.15**
	bound	-Inf	-Inf	-874.70	-874.70	-874.70	**-874.15**	**-874.15**	**-874.15**
	runtime	1.59	0.75	0.20	0.13	0.14	1.61	1.80	3.25
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
6000029	value	-1082.41	-1082.41	**-1106.25**	**-1106.25**	**-1106.25**	**-1106.25**	**-1106.25**	**-1106.25**
	bound	-Inf	-Inf	-1107.52	-1107.52	-1107.52	**-1106.25**	**-1106.25**	**-1106.25**
	runtime	0.73	0.35	0.69	0.51	0.61	0.64	0.61	0.69
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000030	value	-754.30	-754.30	**-794.99**	**-794.99**	**-794.99**	**-794.99**	**-794.99**	**-794.99**
	bound	-Inf	-Inf	-795.01	-795.01	-795.01	**-794.99**	**-794.99**	**-794.99**
	runtime	0.57	0.27	0.09	0.06	0.06	0.16	0.46	0.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000031	value	-589.54	-589.54	-596.74	-596.74	-596.74	**-589.23**	**-589.23**	**-589.23**
	bound	-Inf	-Inf	-589.27	-589.27	-589.27	**-589.23**	**-589.23**	-589.51
	runtime	0.29	0.12	0.05	0.03	0.03	0.23	0.83	0.62
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
6000032	value	-785.25	-785.25	-826.57	-826.57	-826.57	**-826.81**	**-826.81**	**-826.81**
	bound	-Inf	-Inf	-827.15	-827.15	-827.15	**-826.81**	**-826.81**	**-826.81**
	runtime	1.35	0.62	0.20	0.16	0.16	0.86	1.21	1.56
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000033	value	-554.85	-554.85	**-596.64**	**-596.64**	**-596.64**	**-596.64**	**-596.64**	**-596.64**
	bound	-Inf	-Inf	**-596.64**	**-596.64**	**-596.64**	**-596.64**	**-596.64**	**-596.64**
	runtime	2.60	1.16	0.09	0.04	0.04	0.04	0.64	0.94
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB
6000034	value	-856.72	-856.72	-886.51	-886.51	-886.51	**-886.70**	**-886.70**	**-886.70**
	bound	-Inf	-Inf	-887.24	-887.24	-887.24	**-886.70**	**-886.70**	**-886.70**
	runtime	2.95	1.53	0.34	0.32	0.34	0.95	1.42	2.07
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.06 GB	0.06 GB
6000035	value	-532.05	-532.05	-550.95	-550.95	-550.95	**-551.69**	**-551.69**	**-551.69**
	bound	-Inf	-Inf	-552.02	-552.02	-552.02	**-551.69**	**-551.69**	**-551.69**
	runtime	0.39	0.18	0.04	0.02	0.02	0.68	0.89	0.87
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.03 GB
6000036	value	-582.81	-582.81	-619.67	-619.67	-619.67	**-620.93**	**-620.93**	**-620.93**
	bound	-Inf	-Inf	-621.70	-621.70	-621.70	**-620.93**	**-620.93**	**-620.93**
	runtime	0.44	0.17	0.02	0.02	0.02	0.69	0.57	0.77
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000037	value	-409.39	-409.39	-440.97	-440.97	-440.97	**-441.63**	**-441.63**	**-441.63**
	bound	-Inf	-Inf	-442.59	-442.59	-442.59	**-441.63**	**-441.63**	**-441.63**
	runtime	1.23	0.61	0.09	0.03	0.03	0.45	0.85	0.82
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
6000038	value	-752.74	-752.74	-805.84	-805.84	-805.84	**-807.76**	**-807.76**	**-807.76**
	bound	-Inf	-Inf	-810.37	-810.37	-810.37	**-807.76**	**-807.76**	**-807.76**
	runtime	0.39	0.16	0.06	0.02	0.03	0.44	0.74	0.69
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000039	value	-506.07	-506.07	**-573.89**	**-573.89**	**-573.89**	**-573.89**	**-573.89**	**-573.89**
	bound	-Inf	-Inf	**-573.89**	**-573.89**	**-573.89**	**-573.89**	**-573.89**	**-573.89**
	runtime	0.29	0.14	0.02	0.01	0.01	0.02	0.15	0.02
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
6000040	value	-636.29	-636.68	-694.16	-694.16	-694.16	**-695.56**	**-695.56**	**-695.56**
	bound	-Inf	-Inf	-695.62	-695.62	-695.62	**-695.56**	**-695.56**	**-695.56**
	runtime	0.66	0.32	0.12	0.10	0.10	0.24	0.56	1.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000041	value	-495.09	-495.09	-519.27	-519.27	-519.27	**-525.29**	**-525.29**	**-525.29**
	bound	-Inf	-Inf	-526.33	-526.33	-526.33	**-525.29**	**-525.29**	**-525.29**
	runtime	1.09	0.54	0.05	0.04	0.04	0.26	0.64	1.02
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.05 GB
6000042	value	-552.80	-552.80	-576.08	-576.08	-576.08	**-576.94**	**-576.94**	**-576.94**
	bound	-Inf	-Inf	-577.16	-577.16	-577.16	**-576.94**	**-576.94**	**-576.94**
	runtime	0.62	0.27	0.05	0.03	0.04	0.35	0.48	0.51
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000043	value	-459.59	-459.59	-502.29	-502.29	-502.29	**-503.03**	**-503.03**	**-503.03**
	bound	-Inf	-Inf	-503.81	-503.81	-503.81	**-503.03**	**-503.03**	**-503.03**
	runtime	0.58	0.28	0.05	0.03	0.03	1.04	1.30	1.61
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000044	value	-471.42	-471.42	**-507.70**	**-507.70**	**-507.70**	**-507.70**	**-507.70**	**-507.70**
	bound	-Inf	-Inf	-509.36	-509.36	-509.36	**-507.70**	**-507.70**	**-507.70**
	runtime	0.97	0.46	0.09	0.05	0.05	0.68	0.89	0.87
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000045	value	-548.93	-548.93	-580.73	-580.73	-580.73	**-584.43**	**-584.43**	**-584.43**
	bound	-Inf	-Inf	-586.40	-586.40	-586.40	**-584.43**	**-584.43**	**-584.43**
	runtime	1.29	0.62	0.06	0.04	0.04	0.72	0.98	1.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000046	value	-468.14	-468.14	**-510.98**	**-510.98**	**-510.98**	**-510.98**	**-510.98**	**-510.98**
	bound	-Inf	-Inf	**-510.98**	**-510.98**	**-510.98**	**-510.98**	**-510.98**	**-510.98**
	runtime	0.44	0.21	0.03	0.02	0.02	0.68	0.89	0.87
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
6000047	value	-515.54	-515.54	-563.58	-563.58	-563.58	**-564.16**	**-564.16**	**-564.16**
	bound	-Inf	-Inf	-565.81	-565.81	-565.81	**-564.16**	**-564.16**	**-564.16**
	runtime	0.99	0.41	0.05	0.04	0.05	0.88	1.48	1.93
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000048	value	-624.00	-624.00	**-690.03**	**-690.03**	**-690.03**	**-690.03**	**-690.03**	**-690.03**
	bound	-Inf	-Inf	**-690.03**	**-690.03**	**-690.03**	**-690.03**	**-690.03**	**-690.03**
	runtime	4.76	0.38	0.07	0.04	0.04	0.05	0.31	0.29
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000049	value	-474.35	-474.35	-510.49	-510.49	-510.49	**-511.07**	**-511.07**	**-511.07**
	bound	-Inf	-Inf	-511.48	-511.48	-511.48	**-511.07**	**-511.07**	**-511.07**
	runtime	1.16	0.49	0.10	0.07	0.07	0.27	1.10	0.85
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000050	value	-752.96	-722.96	-742.03	-742.03	-742.03	**-752.27**	**-752.27**	**-752.27**
	bound	-Inf	-Inf	-753.94	-753.94	-753.94	**-752.27**	**-752.27**	**-752.27**
	runtime	1.32	0.62	0.19	0.12	0.12	1.06	2.38	3.00
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000051	value	-446.53	-446.53	-471.62	-471.62	-471.62	**-471.62**	**-471.62**	**-471.62**
	bound	-Inf	-Inf	-472.56	-472.56	-472.56	**-471.62**	**-471.62**	**-471.62**
	runtime	1.08	0.52	0.04	0.03	0.03	0.83	1.33	1.73
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000052	value	-413.20	-413.20	-447.32	-447.32	-447.32	**-447.62**	**-447.62**	**-447.62**
	bound	-Inf	-Inf	-448.27	-448.27	-448.27	**-447.62**	**-447.62**	**-447.62**
	runtime	0.73	0.32	0.10	0.06	0.06	0.33	0.83	1.33
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000053	value	-593.18	-593.18	-636.83	-636.83	-636.83	**-636.83**	**-636.83**	**-636.83**
	bound	-Inf	-Inf	-637.25	-637.25	-637.25	**-636.83**	**-636.83**	**-636.83**
	runtime	1.16	0.53	0.08	0.04	0.04	0.39	0.64	0.77
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000054	value	-622.08	-622.08	-671.20	-671.20	-671.20	**-674.72**	**-674.72**	**-674.72**
	bound	-Inf	-Inf	-675.22	-675.22	-675.16	**-674.72**	**-674.72**	**-674.72**
	runtime	5.86	3.17	0.29	0.24	0.24	1.24	1.93	2.86
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.05 GB	0.06 GB
6000055	value	-958.00	-958.00	-991.12	-991.12	-991.12	**-992.44**	**-992.44**	**-992.44**
	bound	-Inf	-Inf	-993.12	-993.12	-993.12	**-992.44**	**-992.44**	**-992.44**
	runtime	4.98	2.45	0.27	0.13	0.13	1.09	2.39	2.51
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.05 GB	0.06 GB
6000056	value	-615.59	-615.59	-667.80	-667.80	-667.80	**-669.62**	**-669.62**	**-669.62**
	bound	-Inf	-Inf	-669.94	-669.94	-669.94	**-669.62**	**-669.62**	**-669.62**
	runtime	1.22	0.62	0.04	0.03	0.03	0.52	1.07	1.10
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000057	value	-452.82	-452.82	-481.61	-481.61	-481.61	**-481.90**	**-481.90**	**-481.90**
	bound	-Inf	-Inf	-482.02	-482.02	-482.02	**-481.90**	**-481.90**	**-481.90**
	runtime	0.60	0.28	0.03	0.02	0.02	0.30	0.57	0.59
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000058	value	-321.53	-321.53	-351.43	-351.43	-351.43	**-351.72**	**-351.72**	**-351.72**
	bound	-Inf	-Inf	-352.55	-352.55	-352.55	**-351.72**	**-351.72**	**-351.72**
	runtime	0.97	0.44	0.03	0.02	0.02	1.14	0.83	1.18
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000059	value	-587.27	-587.27	-630.22	-630.22	-630.22	**-631.11**	**-631.11**	**-631.11**
	bound	-Inf	-Inf	-631.69	-631.69	-631.69	**-631.11**	**-631.11**	**-631.11**
	runtime	0.92	0.41	0.10	0.07	0.07	0.27	0.83	0.82
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000060	value	-496.84	-496.84	-546.36	-546.36	-546.36	**-547.57**	**-547.57**	**-547.57**
	bound	-Inf	-Inf	-547.75	-547.75	-547.75	**-547.57**	**-547.57**	**-547.57**
	runtime	0.46	0.23	0.03	0.02	0.02	0.10	0.21	0.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000061	value	-831.45	-831.45	-861.64	-861.64	-861.64	**-862.95**	**-862.95**	**-862.95**
	bound	-Inf	-Inf	-864.15	-864.15	-864.15	**-862.95**	**-862.95**	**-862.95**
	runtime	0.85	0.38	0.11	0.13	0.13	0.72	2.06	2.93
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000062	value	-485.81	-485.81	**-546.36**	**-546.36**	**-546.36**	**-546.36**	**-546.36**	**-546.36**
	bound	-Inf	-Inf	-546.62	-546.62	-546.62	**-546.36**	**-546.36**	**-546.36**
	runtime	0.43	0.20	0.04	0.02	0.02	0.49	0.51	0.64
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000063	value	-544.35	-544.35	-600.10	-600.10	-600.10	**-603.33**	**-603.33**	**-603.33**
	bound	-Inf	-Inf	-604.48	-604.48	-604.48	**-603.33**	**-603.33**	**-603.33**
	runtime	0.81	0.41	0.09	0.05	0.05	0.23	0.83	0.62
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
6000064	value	-696.30	-696.30	-736.13	-736.13	-736.13	**-739.15**	**-739.15**	**-739.15**
	bound	-Inf	-Inf	-740.55	-740.55	-740.55	**-739.15**	**-739.15**	**-739.15**
	runtime	0.65	0.29	0.05	0.03	0.03	0.42	1.03	0.89
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000065	value	-513.17	-513.17	-559.09	-559.09	-559.09	**-559.67**	**-559.67**	**-559.67**
	bound	-Inf	-Inf	-559.68	-559.68	-559.68	**-559.67**	**-559.67**	**-559.67**
	runtime	0.62	0.28	0.04	0.03	0.03	0.23	0.83	0.62
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000066	value	-427.51	-427.51	-479.90	-479.90	-479.90	**-480.39**	**-480.39**	**-480.39**
	bound	-Inf	-Inf	-480.63	-480.63	-480.63	**-480.39**	**-480.39**	**-480.39**
	runtime	0.92	0.43	0.04	0.02	0.03	0.61	0.72	0.67
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000067	value	-611.96	-611.96	-650.68	-650.68	-650.68	**-652.52**	**-652.52**	**-652.52**
	bound	-Inf	-Inf	-653.02	-653.02	-653.02	**-652.52**	**-652.52**	**-652.52**
	runtime	0.97	0.46	0.09	0.05	0.05	0.68	0.89	0.87
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000068	value	-595.08	-595.08	**-627.24**	**-627.24**	**-627.24**	**-627.24**	**-627.24**	**-627.24**
	bound	-Inf	-Inf	-628.61	-628.61	-628.61	**-627.24**	**-627.24**	**-627.24**
	runtime	1.43	0.73	0.37	0.23	0.22	3.65	4.12	4.14
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000069	value	-387.10	-387.10	-403.31	-403.31	-403.31	**-407.65**	**-407.65**	**-407.65**
	bound	-Inf	-Inf	-407.88	-407.88	-407.88	**-407.65**	**-407.65**	**-407.65**
	runtime	0.25	0.11	0.02	0.01	0.01	0.17	0.34	0.29
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
6000070	value	-747.02	-747.02	-806.29	-806.29	-806.29	**-807.44**	**-807.44**	**-807.44**
	bound	-Inf	-Inf	-807.65	-807.65	-807.65	**-807.44**	**-807.44**	**-807.44**
	runtime	1.43	0.78	0.10	0.07	0.06	0.50	1.15	1.22
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000071	value	-442.41	-442.41	-499.54	-499.54	-499.54	**-500.20**	**-500.20**	**-500.20**
	bound	-Inf	-Inf	-500.30	-500.30	-500.30	**-500.20**	**-500.20**	**-500.20**
	runtime	1.23	0.60	0.05	0.03

| overwhelm-universe | open-the | MECHANISM | MECHANISM | MECHANISM

correlation=clustering		ogm-ICM	ogm-LF-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCL-CC1	MCL-CCIPD
6000145	value	-470.18	-470.18	-504.10	-504.10	-504.10	**-504.40**	**-504.40**	**-504.40**
	bound	-Inf	-Inf	-504.41	-504.41	-504.41	**-504.40**	**-504.40**	**-504.40**
	runtime	0.27	0.11	0.04	0.02	0.03	0.07	0.28	0.34
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000146	value	-312.84	-312.84	-355.16	-355.16	-355.16	**-355.28**	**-355.28**	**-355.28**
	bound	-Inf	-Inf	-355.17	-355.17	-355.17	**-355.28**	**-355.28**	**-355.28**
	runtime	0.39	0.16	0.02	0.02	0.02	0.13	0.60	0.44
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000147	value	-668.64	-668.64	-702.56	-702.56	-702.56	**-706.17**	**-706.17**	**-706.17**
	bound	-Inf	-Inf	-707.32	-707.32	-707.32	**-706.17**	**-706.17**	**-706.17**
	runtime	2.96	1.46	0.18	0.15	0.13	1.04	3.38	1.84
	memory (GB)	0.01 GB	0.03 GB	0.03 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
6000148	value	-884.49	-884.49	**-928.50**	**-928.50**	**-928.50**	**-928.50**	**-928.50**	**-928.50**
	bound	-Inf	-Inf	-928.71	-928.71	-928.71	**-928.50**	**-928.50**	**-928.50**
	runtime	0.63	0.24	0.04	0.02	0.02	0.09	0.34	0.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.04 GB	0.05 GB
6000149	value	-676.59	-676.59	-716.41	-716.41	-716.41	**-716.99**	**-716.99**	**-716.99**
	bound	-Inf	-Inf	-717.34	-717.34	-717.34	**-716.99**	**-716.99**	**-716.99**
	runtime	1.17	0.52	0.05	0.03	0.04	0.43	0.66	0.93
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000150	value	-693.07	-693.07	-739.38	-739.38	-739.38	**-741.33**	**-741.33**	**-741.33**
	bound	-Inf	-Inf	-741.43	-741.43	-741.43	**-741.33**	**-741.33**	**-741.33**
	runtime	7.89	3.82	0.43	0.24	0.25	1.93	7.37	3.08
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB	0.06 GB	0.07 GB
6000151	value	-562.98	-562.98	-580.96	-580.96	-580.96	**-585.19**	**-585.19**	**-585.19**
	bound	-Inf	-Inf	-585.85	-585.85	-585.85	**-585.19**	**-585.19**	**-585.19**
	runtime	0.75	0.32	0.07	0.05	0.05	0.72	1.56	1.58
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
6000152	value	-447.41	-447.41	-468.26	-468.26	-468.26	**-473.27**	**-473.27**	**-473.27**
	bound	-Inf	-Inf	-473.79	-473.79	-473.79	**-473.27**	**-473.27**	**-473.27**
	runtime	0.54	0.24	0.04	0.02	0.02	0.08	0.29	0.34
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000153	value	-428.51	-428.33	-497.12	-467.12	-467.12	**-467.50**	**-467.50**	**-467.50**
	bound	-Inf	-Inf	-467.92	-467.92	-467.92	**-467.50**	**-467.50**	**-467.50**
	runtime	0.54	0.24	0.07	0.04	0.04	0.29	0.89	1.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000154	value	-488.33	-488.33	-543.00	-543.00	-543.00	**-544.82**	**-544.82**	**-544.82**
	bound	-Inf	-Inf	-545.30	-545.30	-545.30	**-544.82**	**-544.82**	**-544.82**
	runtime	0.69	0.25	0.03	0.03	0.02	0.08	0.10	0.53
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000155	value	-580.96	-580.96	-631.16	-621.16	-631.16	**-634.56**	**-634.56**	**-634.56**
	bound	-Inf	-Inf	-635.71	-635.71	-635.71	**-634.56**	**-634.56**	**-634.56**
	runtime	1.18	0.61	0.16	0.14	0.14	0.76	2.03	2.24
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000156	value	-1006.12	-1006.12	-1031.01	-1031.01	-1031.01	**-1032.75**	**-1032.75**	**-1032.75**
	bound	-Inf	-Inf	-1033.59	-1033.59	-1033.59	**-1032.75**	**-1032.75**	**-1032.75**
	runtime	1.63	0.81	0.09	0.08	0.08	0.65	2.03	2.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB	0.05 GB	0.06 GB	0.07 GB
6000157	value	-743.91	-743.91	-783.19	-783.19	-783.19	**-785.69**	**-785.69**	**-785.69**
	bound	-Inf	-Inf	-785.79	-785.79	-785.79	**-785.69**	**-785.69**	**-785.69**
	runtime	1.36	0.64	0.26	0.19	0.20	0.34	0.51	1.00
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
6000158	value	-408.55	-408.55	-491.35	-491.35	-491.35	**-492.50**	**-492.50**	**-492.50**
	bound	-Inf	-Inf	-493.19	-493.19	-493.19	**-492.50**	**-492.50**	**-492.50**
	runtime	1.33	0.65	0.04	0.02	0.02	1.14	1.56	1.37
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000159	value	-606.05	-606.05	-638.10	-638.10	-638.10	**-633.84**	**-633.84**	**-633.84**
	bound	-Inf	-Inf	-636.18	-636.18	-636.18	**-633.84**	**-633.84**	**-633.84**
	runtime	0.43	0.18	0.11	0.08	0.08	5.48	15.45	16.01
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000160	value	-456.70	-456.70	-510.18	-510.18	-510.18	**-510.64**	**-510.64**	**-510.64**
	bound	-Inf	-Inf	-512.45	-512.45	-512.45	**-510.64**	**-510.64**	**-510.64**
	runtime	0.30	0.13	0.03	0.02	0.02	0.09	0.33	0.59
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
6000161	value	-281.00	-281.00	-294.58	-294.58	-294.58	**-293.91**	**-293.91**	**-293.91**
	bound	-Inf	-Inf	-294.58	-294.58	-294.58	**-293.91**	**-293.91**	**-293.91**
	runtime	0.36	0.18	0.01	0.01	0.01	0.53	0.87	0.96
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000162	value	-528.44	-528.44	-594.79	-594.79	-594.79	**-594.89**	**-594.89**	**-594.89**
	bound	-Inf	-Inf	-596.54	-596.54	-596.54	**-594.89**	**-594.89**	**-594.89**
	runtime	0.84	0.37	0.06	0.04	0.05	1.51	1.52	1.70
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000163	value	-866.17	-866.17	-907.54	-907.54	-907.54	**-900.91**	**-900.91**	**-900.91**
	bound	-Inf	-Inf	-907.54	-907.54	-907.54	**-900.91**	**-900.91**	**-900.91**
	runtime	2.71	1.32	0.17	0.09	0.09	0.56	3.84	2.13
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
6000164	value	-512.21	-512.21	**-550.01**	**-550.01**	**-550.01**	**-550.01**	**-550.01**	**-550.01**
	bound	-Inf	-Inf	-550.70	-550.70	-550.70	**-550.01**	**-550.01**	**-550.01**
	runtime	1.59	0.74	0.03	0.02	0.02	0.54	0.68	0.64
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000165	value	-538.01	-538.01	-555.89	-555.89	-555.89	**-555.52**	**-555.52**	**-555.52**
	bound	-Inf	-Inf	-555.89	-555.89	-555.89	**-555.52**	**-555.52**	**-555.52**
	runtime	0.26	0.11	0.05	0.04	0.04	0.34	0.51	0.54
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000166	value	-661.11	-661.11	-686.18	-686.18	-686.18	**-686.56**	**-686.56**	**-686.56**
	bound	-Inf	-Inf	-686.57	-686.57	-686.57	**-686.56**	**-686.56**	**-686.56**
	runtime	1.08	0.50	0.05	0.05	0.05	0.12	0.69	0.74
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000167	value	-204.06	-204.06	-235.57	-235.57	-235.57	**-237.81**	**-237.81**	**-237.81**
	bound	-Inf	-Inf	-237.81	-237.81	-237.81	**-237.81**	**-237.81**	**-237.81**
	runtime	0.64	0.28	0.02	0.01	0.01	0.48	0.74	0.72
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000168	value	-504.80	-504.80	-547.94	-547.94	-547.94	**-548.07**	**-548.07**	**-548.07**
	bound	-Inf	-Inf	-548.20	-548.20	-548.20	**-548.07**	**-548.07**	**-548.07**
	runtime	2.52	1.19	0.08	0.05	0.06	0.16	0.40	0.59
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.05 GB
6000169	value	-683.41	-683.41	-740.93	-740.93	-740.93	**-742.32**	**-742.32**	**-742.32**
	bound	-Inf	-Inf	-743.32	-743.32	-743.32	**-742.32**	**-742.32**	**-742.32**
	runtime	1.94	0.91	0.08	0.05	0.05	0.21	0.65	0.72
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
6000170	value	-404.39	-404.39	**-427.11**	**-427.11**	**-427.11**	**-427.11**	**-427.11**	**-427.11**
	bound	-Inf	-Inf	-427.13	-427.13	-427.13	**-427.11**	**-427.11**	**-427.11**
	runtime	1.04	0.53	0.07	0.03	0.04	0.14	0.68	0.95
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000171	value	-574.91	-574.91	-580.07	-580.07	-580.07	**-583.91**	**-583.91**	**-583.91**
	bound	-Inf	-Inf	-584.08	-584.08	-584.08	**-583.91**	**-583.91**	**-583.91**
	runtime	0.75	0.32	0.06	0.03	0.03	0.51	1.44	1.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000172	value	-380.31	-380.31	-415.47	-415.47	-415.47	**-416.10**	**-416.10**	**-416.10**
	bound	-Inf	-Inf	-416.17	-416.17	-416.17	**-416.10**	**-416.10**	**-416.10**
	runtime	0.64	0.31	0.05	0.03	0.03	0.20	0.65	0.72
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000173	value	-726.48	-726.48	**-773.76**	**-773.76**	**-773.76**	**-773.76**	**-773.76**	**-773.76**
	bound	-Inf	-Inf	**-773.76**	**-773.76**	**-773.76**	**-773.76**	**-773.76**	**-773.76**
	runtime	2.62	1.27	0.09	0.04	0.04	0.14	0.58	1.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000174	value	-960.89	-960.89	**-983.44**	**-983.44**	**-983.44**	**-983.44**	**-983.44**	**-983.44**
	bound	-Inf	-Inf	-984.80	-984.80	-984.80	**-983.44**	**-983.44**	**-983.44**
	runtime	6.80	3.35	0.53	0.30	0.31	0.49	1.47	1.13
	memory (GB)	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.05 GB	0.05 GB
6000175	value	-875.00	-875.00	-900.91	-900.91	-900.91	**-904.61**	**-904.61**	**-904.61**
	bound	-Inf	-Inf	-906.37	-906.37	-906.37	**-904.61**	**-904.61**	**-904.61**
	runtime	4.36	2.26	0.43	0.53	0.54	1.91	4.91	2.56
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB	0.06 GB	0.07 GB
6000176	value	-512.73	-512.73	-593.27	-593.27	-593.27	**-593.28**	**-593.28**	**-593.28**
	bound	-Inf	-Inf	-593.28	-593.28	-593.28	**-593.28**	**-593.28**	**-593.28**
	runtime	2.62	1.27	0.09	0.04	0.04	0.13	0.58	1.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000177	value	-397.95	-397.95	-440.84	-440.84	-440.84	**-442.22**	**-442.22**	**-442.22**
	bound	-Inf	-Inf	-443.54	-443.54	-443.54	**-442.22**	**-442.22**	**-442.22**
	runtime	0.63	0.30	0.04	0.02	0.02	0.08	0.31	0.42
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000178	value	-588.38	-588.68	**-632.23**	**-632.23**	**-632.23**	**-632.23**	**-632.23**	**-632.23**
	bound	-Inf	-Inf	-633.04	-633.04	-633.04	**-632.23**	**-632.23**	**-632.23**
	runtime	1.69	0.86	0.06	0.04	0.05	0.48	0.82	1.53
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
6000179	value	-496.53	-496.53	-540.82	-540.82	-540.82	**-541.14**	**-541.14**	**-541.14**
	bound	-Inf	-Inf	-541.29	-541.29	-541.29	**-541.14**	**-541.14**	**-541.14**
	runtime	1.98	0.91	0.04	0.03	0.03	0.40	1.28	1.43
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000180	value	-458.96	-458.96	**-506.95**	**-506.95**	**-506.95**	**-506.95**	**-506.95**	**-506.95**
	bound	-Inf	-Inf	-507.09	-507.09	-507.09	**-506.95**	**-506.95**	**-506.95**
	runtime	0.69	0.30	0.04	0.02	0.02	0.09	0.34	0.44
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000181	value	-431.22	-431.22	-465.67	-465.67	-465.67	**-466.31**	**-466.31**	**-466.31**
	bound	-Inf	-Inf	-466.49	-466.49	-466.49	**-466.31**	**-466.31**	**-466.31**
	runtime	0.79	0.34	0.04	0.02	0.02	0.14	0.63	0.96
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000182	value	-468.31	-468.31	-533.21	-533.21	-533.21	**-534.06**	**-534.06**	**-534.06**
	bound	-Inf	-Inf	-534.66	-534.66	-534.66	**-534.06**	**-534.06**	**-534.06**
	runtime	0.69	0.25	0.03	0.03	0.02	0.08	0.10	0.53
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000183	value	-608.31	-608.31	-636.95	-636.95	-636.95	**-637.04**	**-637.04**	**-637.04**
	bound	-Inf	-Inf	-637.10	-637.10	-637.10	**-637.04**	**-637.04**	**-637.04**
	runtime	1.29	0.56	0.05	0.03	0.04	0.24	0.65	0.39
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000184	value	-721.36	-721.36	-788.94	-788.94	-788.94	**-791.26**	**-791.26**	**-791.26**
	bound	-Inf	-Inf	-791.99	-791.99	-791.99	**-791.26**	**-791.26**	**-791.26**
	runtime	1.63	0.81	0.09	0.08	0.08	0.65	2.03	2.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000185	value	-562.23	-562.23	-585.35	-585.35	-585.35	**-587.05**	**-587.05**	**-587.05**
	bound	-Inf	-Inf	-587.28	-587.28	-587.28	**-587.05**	**-587.05**	**-587.05**
	runtime	1.70	0.85	0.08	0.04	0.04	0.33	0.87	0.62
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000186	value	-505.67	-505.67	-536.22	-536.22	-536.22	**-538.48**	**-538.48**	**-538.48**
	bound	-Inf	-Inf	-539.26	-539.26	-539.26	**-538.48**	**-538.48**	**-538.48**
	runtime	1.88	1.34	0.08	0.05	0.05	0.31	2.80	1.68
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
6000187	value	-508.29	-508.29	-532.69	-532.69	-532.69	**-533.84**	**-533.84**	**-533.84**
	bound	-Inf	-Inf	-534.16	-534.16	-534.16	**-533.84**	**-533.84**	**-533.84**
	runtime	1.99	0.96	0.07	0.04	0.04	0.36	0.43	0.74
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000188	value	-589.26	-589.26	-622.19	-622.19	-622.19	**-624.74**	**-624.74**	**-624.74**
	bound	-Inf	-Inf	-624.90	-624.90	-624.90	**-624.74**	**-624.74**	**-624.74**
	runtime	7.38	3.53	0.20	0.12	0.11	0.66	1.19	1.37
	memory (GB)	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.05 GB	0.05 GB
6000189	value	-518.34	-518.34	-552.43	-552.43	-552.43	**-552.34**	**-552.34**	**-552.34**
	bound	-Inf	-Inf	-552.43	-552.43	-552.43	**-552.34**	**-552.34**	**-552.34**
	runtime	1.66	0.83	0.12	0.07	0.07	0.21	1.37	1.12
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
6000190	value	-452.09	-452.09	**-481.77**	**-481.77**	**-481.77**	**-481.77**	**-481.77**	**-481.77**
	bound	-Inf	-Inf	**-481.77**	**-481.77**	**-481.77**	**-481.77**	**-481.77**	**-481.77**
	runtime	0.49	0.21	0.03	0.02	0.02	0.02	0.18	0.33
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000191	value	-348.64	-348.64	**-415.54**	**-415.54**	**-415.54**	**-415.54**	**-415.54**	**-415.54**
	bound	-Inf	-Inf	**-415.54**	**-415.54**	**-415.54**	**-415.54**	**-415.54**	**-415.54**
	runtime	0.32	0.16	0.04	0.02	0.02	0.02	0.35	0.51
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
6000192	value	-316.60	-316.60	-349.89	-349.89	-349.89	**-350.16**	**-350.16**	**-350.16**
	bound	-Inf	-Inf	-350.22	-350.22	-350.22	**-350.16**	**-350.16**	**-350.16**
	runtime	0.67	0.31	0.02	0.01	0.01	0.07	0.25	0.28
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000193	value	-412.68	-412.68	-456.05	-456.05	-456.05	**-455.59**	**-455.59**	**-455.59**
	bound	-Inf	-Inf	-456.05	-456.05	-456.05	**-455.59**	**-455.59**	**-455.59**
	runtime	0.57	0.29	0.08	0.06	0.06	0.29	0.61	0.52
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000194	value	-422.97	-422.97	**-485.00**	**-485.00**	**-485.00**	**-485.00**	**-485.00**	**-485.00**
	bound	-Inf	-Inf	-485.53	-485.53	-485.53	**-485.00**	**-485.00**	**-485.00**
	runtime	0.93	0.44	0.04	0.03	0.03	0.15	0.80	0.62
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
6000195	value	-557.41	-557.41	**-607.04**	**-607.04**	**-607.04**	**-607.04**	**-607.04**	**-607.04**
	bound	-Inf	-Inf	**-607.04**	**-607.04**	**-607.04**	**-607.04**	**-607.04**	**-607.04**
	runtime	0.70	0.34	0.05	0.05	0.04	0.05	0.32	0.54
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB
6000196	value	-429.88	-429.88	-487.20	-487.20	-487.20	**-487.73**	**-487.73**	**-487.73**
	bound	-Inf	-Inf	-488.10	-488.10	-488.10	**-487.73**	**-487.73**	**-487.73**
	runtime	1.75	0.79	0.06	0.03	0.04	0.34	0.60	1.32
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
6000197	value	-533.82	-533.82	-584.90	-584.90	-584.90	**-585.40**	**-585.40**	**-585.40**
	bound	-Inf	-Inf	-585.40	-585.40	-585.40	**-585.40**	**-585.40**	**-585.40**
	runtime	0.42	0.20	0.05	0.04	0.04	0.53	0.87	0.93
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000198	value	-401.00	-401.00	**-443.19**	**-443.19**	**-443.19**	**-443.19**	**-443.19**	**-443.19**
	bound	-Inf	-Inf	-443.19	-443.19	-443.19	**-443.19**	**-443.19**	**-443.19**
	runtime	0.50	0.22	0.02	0.01	0.01	0.07	0.21	0.37
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000199	value	-417.32	-417.32	**-452.62**	**-452.62**	**-452.62**	**-452.62**	**-452.62**	**-452.62**
	bound	-Inf	-Inf	**-452.62**	**-452.62**	**-452.62**	**-452.62**	**-452.62**	**-452.62**
	runtime	0.37	0.15	0.02	0.01	0.01	0.03	0.09	0.12
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
6000200	value	-437.97	-437.97	-464.76	-464.76	-464.76	**-465.76**	**-465.76**	**-465.76**
	bound	-Inf	-Inf	-466.72	-466.72	-466.72	**-465.76**	**-465.76**	**-465.76**
	runtime	0.86	0.46	0.08	0.06	0.06	0.52	0.65	1.02
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
6000201	value	-524.87	-524.87	-598.25	-598.25	-598.25	**-599.92**	**-599.92**	**-599.92**
	bound	-Inf	-Inf	-601.11	-601.11	-601.11	**-599.92**	**-599.92**	**-599.92**
	runtime	0.42	0.20	0.04	0.03	0.03	0.13	0.44	0.53
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000202	value	-791.27	-791.27	**-843.50**	**-843.50**	**-843.50**	**-843.50**	**-843.50**	**-843.50**
	bound	-Inf	-Inf	**-843.50**	**-843.50**	**-843.50**	**-843.50**	**-843.50**	**-843.50**
	runtime	2.24	1.15	0.16	0.10	0.10	0.11	1.24	0.79
	memory (GB)	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB
6000203	value	-629.96	-629.96	-651.90	-651.90	-651.78	**-655.51**	**-655.51**	**-655.51**
	bound	-Inf	-Inf	-655.69	-655.69	-655.99	**-655.51**	**-655.51**	**-655.51**
	runtime	1.34	0.68	0.05	0.03	0.03	0.14	0.71	0.72
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
6000204	value	-484.84	-484.84	**-522.53**	**-522.53**	**-522.53**	**-522.53**	**-522.53**	**-522.53**
	bound	-Inf	-Inf	**-522.53**	**-522.53**	**-522.53**	**-522.53**	**-522.53**	**-522.53**
	runtime	1.40	0.75	0.04	0.03	0.02	0.03	0.40	0.40
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
6000205	value	-751.98	-751.98	-796.49	-796.49	-796.49	**-796.78**	**-796.78**	**-796.78**
	bound	-Inf	-Inf	-797.23	-797.23	-797.23	**-796.78**	**-796.78**	**-796.78**
	runtime	1.63	0.81	0.09	0.07	0.07	0.21	0.65	0.72
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.03 GB	0.05 GB	0.05 GB	0.05 GB
6000206	value	-664.03	-664.03	-705.41	-705.41	-705.41	**-706.88**	**-706.88**	**-706.88**
	bound	-Inf	-Inf	-708.55	-708.55	-708.55	**-706.88**	**-706.88**	**-706.88**
	runtime	1.02	0.47	0.08	0.06	0.06	0.46	2.05	1.17
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000207	value	-418.89	-418.89	-439.13	-439.13	-439.13	**-440.05**	**-440.05**	**-440.05**
	bound	-Inf	-Inf	-440.08	-440.08	-440.08	**-440.05**	**-440.05**	**-440.05**
	runtime	0.43	0.21	0.04	0.03	0.03	0.17	0.54	0.45
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000208	value	-451.17	-451.17	**-487.76**	**-487.76**	**-487.76**	**-487.76**	**-487.76**	**-487.76**
	bound	-Inf	-Inf	**-487.76**	**-487.76**	**-487.76**	**-487.76**	**-487.76**	**-487.76**
	runtime	1.21	0.59	0.04	0.03	0.02	0.02	0.41	0.30
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
6000209	value	-530.30	-530.30	-575.02	-575.02	-575.02	**-576.55**	**-576.55**	**-576.55**
	bound	-Inf	-Inf	-576.84	-576.84	-576.84	**-576.55**	**-576.55**	**-576.55**



correlation=clustering		ogm-ICM	ogm-LP-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCL-CC1	MCL-CCIPD
60002283	value	-572.90	-572.90	-611.21	-611.21	-611.21	**-611.78**	**-611.78**	**-611.78**
	bound	-Inf	-Inf	-611.86	-611.86	-611.86	**-611.78**	**-611.78**	**-611.78**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.04 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002284	value	-671.54	-671.54	-710.52	-710.52	-710.52	**-711.28**	**-711.28**	**-711.28**
	bound	-Inf	-Inf	-711.28	-711.28	-711.28	**-711.28**	**-711.28**	**-711.28**
	runtime	2.42	1.18	0.10	0.08	0.07	1.39	2.33	2.06
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002285	value	-484.31	-484.33	**-507.32**	**-507.32**	**-507.32**	**-507.32**	**-507.32**	**-507.32**
	bound	-Inf	-Inf	-507.70	-507.70	-507.70	**-507.32**	**-507.32**	**-507.32**
	runtime	0.73	0.35	0.03	0.02	0.02	0.34	0.67	0.61
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
60002286	value	-560.16	-560.16	-607.34	-607.34	-607.34	**-607.87**	**-607.87**	**-607.87**
	bound	-Inf	-Inf	-608.59	-608.59	-608.59	**-607.87**	**-607.87**	**-607.87**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002287	value	-725.12	-725.12	-757.22	-757.22	-757.22	**-760.19**	**-760.19**	**-760.19**
	bound	-Inf	-Inf	-760.70	-760.70	-760.70	**-760.19**	**-760.19**	**-760.19**
	runtime	0.27	0.31	0.90	0.58	0.57	2.70	5.92	4.20
	memory (GB)	0.01 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB	0.06 GB	0.06 GB	0.05 GB
60002288	value	-554.84	-554.84	**-565.34**	**-565.34**	**-565.34**	**-565.34**	**-565.34**	**-565.34**
	bound	-Inf	-Inf	**-565.34**	**-565.34**	**-565.34**	**-565.34**	**-565.34**	**-565.34**
	runtime	1.39	0.81	0.11	0.10	0.10	0.31	0.61	0.46
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB
60002289	value	-520.33	-520.33	-552.77	-552.77	-552.90	**-554.32**	**-554.32**	**-554.32**
	bound	-Inf	-Inf	-554.74	-554.74	-554.73	**-554.32**	**-554.32**	**-554.32**
	runtime	2.36	1.35	0.14	0.09	0.10	0.27	2.47	2.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
60002290	value	-575.88	-572.90	**-603.54**	**-603.54**	**-603.54**	**-603.54**	**-603.54**	**-603.54**
	bound	-Inf	-Inf	-603.62	-603.62	-603.62	**-603.54**	**-603.54**	**-603.54**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002291	value	-531.15	-531.15	-577.57	-577.57	-577.57	**-578.45**	**-578.45**	**-578.45**
	bound	-Inf	-Inf	-579.21	-579.21	-579.21	**-578.45**	**-578.45**	**-578.45**
	runtime	2.92	1.51	0.16	0.11	0.11	0.88	1.82	1.58
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002292	value	-1089.41	-1089.41	-1140.24	-1140.24	-1140.24	**-1143.01**	**-1143.01**	**-1143.01**
	bound	-Inf	-Inf	-1143.87	-1143.87	-1143.87	**-1143.01**	**-1143.01**	**-1143.01**
	runtime	4.08	2.03	0.23	0.24	0.24	0.40	2.40	2.28
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002293	value	-339.60	-339.60	-365.13	-365.13	-365.13	**-365.44**	**-365.44**	**-365.44**
	bound	-Inf	-Inf	-366.36	-366.36	-366.36	**-365.44**	**-365.44**	**-365.44**
	runtime	0.82	0.36	0.02	0.01	0.01	0.10	0.27	0.28
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002294	value	-439.60	-439.60	-476.67	-476.67	-476.67	**-481.54**	**-481.54**	**-481.54**
	bound	-Inf	-Inf	-482.17	-482.17	-482.17	**-481.54**	**-481.54**	**-481.54**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002295	value	-604.78	-604.78	-648.16	-648.16	-648.16	**-648.67**	**-648.67**	**-648.67**
	bound	-Inf	-Inf	-649.44	-649.44	-649.44	**-648.67**	**-648.67**	**-648.67**
	runtime	2.16	1.07	0.21	0.14	0.14	1.29	2.30	2.06
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002296	value	-705.70	-705.70	-741.91	-741.91	-741.91	**-742.81**	**-742.81**	**-742.81**
	bound	-Inf	-Inf	-743.85	-743.85	-743.85	**-742.81**	**-742.81**	**-742.81**
	runtime	3.29	1.20	0.15	0.08	0.08	0.54	1.49	1.69
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002297	value	-633.46	-633.46	-664.15	-664.15	-664.15	**-665.13**	**-665.13**	**-665.13**
	bound	-Inf	-Inf	-665.38	-665.38	-665.38	**-665.13**	**-665.13**	**-665.13**
	runtime	1.24	0.65	0.17	0.13	0.13	0.52	1.43	1.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002298	value	-531.96	-521.96	-536.53	-536.53	-536.53	**-538.14**	**-538.14**	**-538.14**
	bound	-Inf	-Inf	-538.50	-538.50	-538.50	**-538.14**	**-538.14**	**-538.14**
	runtime	1.87	0.85	0.12	0.08	0.09	0.21	0.79	0.71
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002299	value	-742.81	-742.81	-791.65	-791.65	-791.65	**-791.65**	**-791.65**	**-791.65**
	bound	-Inf	-Inf	-793.12	-793.12	-793.11	**-791.65**	**-791.65**	**-791.65**
	runtime	4.82	2.24	0.13	0.10	0.11	0.49	1.95	1.83
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.06 GB
60002300	value	-469.50	-469.50	-516.02	-516.02	-516.02	**-518.73**	**-518.73**	**-518.73**
	bound	-Inf	-Inf	-519.82	-519.82	-519.82	**-518.73**	**-518.73**	**-518.73**
	runtime	1.53	0.70	0.23	0.17	0.17	1.00	4.08	3.21
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002301	value	-550.97	-550.97	-612.17	-612.17	-612.17	**-613.49**	**-613.49**	**-613.49**
	bound	-Inf	-Inf	-613.49	-613.49	-614.16	**-613.49**	**-613.49**	**-613.49**
	runtime	4.46	2.12	0.16	0.13	0.14	0.80	1.23	1.32
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.04 GB
60002302	value	-504.07	-504.07	-559.37	-559.37	-559.37	**-560.45**	**-560.45**	**-560.45**
	bound	-Inf	-Inf	-560.70	-560.70	-560.70	**-560.45**	**-560.45**	**-560.45**
	runtime	0.54	0.22	0.04	0.03	0.03	0.10	0.31	0.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002303	value	-602.64	-602.64	-651.26	-651.26	-651.26	**-651.10**	**-651.10**	**-651.10**
	bound	-Inf	-Inf	-651.26	-651.26	-651.26	**-651.10**	**-651.10**	**-651.10**
	runtime	2.35	1.16	0.08	0.04	0.05	0.19	0.78	1.32
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
60002304	value	-503.16	-503.16	-528.51	-528.51	-528.51	**-528.86**	**-528.86**	**-528.86**
	bound	-Inf	-Inf	-528.90	-528.90	-528.90	**-528.86**	**-528.86**	**-528.86**
	runtime	1.79	0.90	0.12	0.07	0.08	0.14	1.07	0.85
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002305	value	-549.62	-549.62	-593.90	-593.90	-593.90	**-595.68**	**-595.68**	**-595.68**
	bound	-Inf	-Inf	-595.68	-595.68	-595.68	**-595.68**	**-595.68**	**-595.68**
	runtime	2.22	1.18	0.12	0.08	0.09	0.29	0.50	0.55
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
60002306	value	-599.08	-599.08	-650.93	-650.93	-650.93	**-651.90**	**-651.90**	**-651.90**
	bound	-Inf	-Inf	-652.32	-652.32	-652.32	**-651.90**	**-651.90**	**-651.90**
	runtime	1.56	0.78	0.07	0.07	0.07	0.44	1.18	1.30
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002307	value	-658.98	-658.98	**-692.78**	**-692.78**	**-692.78**	**-692.78**	**-692.78**	**-692.78**
	bound	-Inf	-Inf	**-692.78**	**-692.78**	**-692.78**	**-692.78**	**-692.78**	**-692.78**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.05 GB
60002308	value	-712.74	-712.74	-739.96	-739.96	-739.96	**-740.57**	**-740.57**	**-740.57**
	bound	-Inf	-Inf	-741.77	-741.77	-741.77	**-740.57**	**-740.57**	**-740.57**
	runtime	1.18	0.58	0.25	0.14	0.15	2.74	3.17	2.84
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002309	value	-542.82	-542.82	-591.00	-591.00	-591.00	**-593.16**	**-593.16**	**-593.16**
	bound	-Inf	-Inf	-594.05	-594.05	-594.05	**-593.16**	**-593.16**	**-593.16**
	runtime	0.55	0.28	0.06	0.04	0.04	0.14	0.38	0.34
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002310	value	-607.30	-607.30	-641.21	-641.21	-641.21	**-641.77**	**-641.77**	**-641.77**
	bound	-Inf	-Inf	-642.07	-642.07	-642.07	**-641.77**	**-641.77**	**-641.77**
	runtime	0.63	0.30	0.07	0.05	0.05	0.27	1.09	1.06
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002311	value	-632.22	-632.22	-661.15	-661.15	-661.15	**-667.11**	**-667.11**	**-667.11**
	bound	-Inf	-Inf	-668.67	-668.67	-668.67	**-667.11**	**-667.11**	**-667.11**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002312	value	-562.86	-562.86	-580.00	-580.00	-580.00	**-582.30**	**-582.30**	**-582.30**
	bound	-Inf	-Inf	-582.57	-582.57	-582.57	**-582.30**	**-582.30**	**-582.30**
	runtime	0.97	0.43	0.07	0.05	0.05	0.20	0.61	0.53
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002313	value	-395.60	-395.60	-453.23	-453.23	-453.23	**-454.95**	**-454.95**	**-454.95**
	bound	-Inf	-Inf	-455.22	-455.22	-455.22	**-454.95**	**-454.95**	**-454.95**
	runtime	0.50	0.24	0.06	0.04	0.04	0.14	0.34	0.66
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002314	value	-500.92	-500.92	**-550.60**	**-550.60**	**-550.60**	**-550.60**	**-550.60**	**-550.60**
	bound	-Inf	-Inf	**-550.60**	**-550.60**	**-550.60**	**-550.60**	**-550.60**	**-550.60**
	runtime	0.81	0.39	0.02	0.02	0.02	0.02	0.14	0.21
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
60002315	value	-544.20	-544.20	-594.23	-594.23	-594.23	**-594.80**	**-594.80**	**-594.80**
	bound	-Inf	-Inf	-595.54	-595.54	-595.54	**-594.80**	**-594.80**	**-594.80**
	runtime	0.58	0.28	0.06	0.04	0.04	0.14	0.34	0.39
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
60002316	value	-433.29	-433.29	**-484.05**	**-484.05**	**-484.05**	**-484.05**	**-484.05**	**-484.05**
	bound	-Inf	-Inf	**-484.05**	**-484.05**	**-484.05**	**-484.05**	**-484.05**	**-484.05**
	runtime	0.57	0.25	0.03	0.02	0.01	0.02	0.19	0.32
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB
60002317	value	-797.68	-797.68	-821.55	-821.55	-822.52	**-824.20**	**-824.20**	**-824.20**
	bound	-Inf	-Inf	-825.17	-825.17	-825.14	**-824.20**	**-824.20**	**-824.20**
	runtime	1.22	0.56	0.11	0.09	0.10	0.36	0.54	0.48
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.03 GB
60002318	value	-624.12	-624.12	-656.02	-656.02	-656.02	**-658.52**	**-658.52**	**-658.52**
	bound	-Inf	-Inf	-661.41	-661.41	-661.41	**-658.52**	**-658.52**	**-658.52**
	runtime	1.54	0.74	0.29	0.17	0.17	1.13	2.04	1.39
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002319	value	-318.33	-318.33	-354.87	-354.87	-354.87	**-354.92**	**-354.92**	**-354.92**
	bound	-Inf	-Inf	-354.93	-354.93	-354.93	**-354.92**	**-354.92**	**-354.92**
	runtime	0.97	0.44	0.03	0.02	0.02	0.10	0.33	0.45
	memory (GB)	0.03 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002320	value	-370.07	-370.07	-415.61	-415.61	-415.61	**-416.34**	**-416.34**	**-416.34**
	bound	-Inf	-Inf	-416.81	-416.81	-416.81	**-416.34**	**-416.34**	**-416.34**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002321	value	-543.05	-543.05	-586.67	-586.67	-586.67	**-587.50**	**-587.50**	**-587.50**
	bound	-Inf	-Inf	-587.96	-587.96	-587.96	**-587.50**	**-587.50**	**-587.50**
	runtime	0.62	0.26	0.07	0.04	0.04	0.17	0.74	0.47
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002322	value	-452.99	-452.99	-499.35	-499.35	-499.35	**-499.71**	**-499.71**	**-499.71**
	bound	-Inf	-Inf	-500.47	-500.47	-500.47	**-499.71**	**-499.71**	**-499.71**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002323	value	-592.93	-592.93	**-633.16**	**-633.16**	**-633.16**	**-633.16**	**-633.16**	**-633.16**
	bound	-Inf	-Inf	-633.23	-633.23	-633.23	**-633.16**	**-633.16**	**-633.16**
	runtime	0.52	0.22	0.07	0.05	0.05	0.14	0.40	0.54
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
60002324	value	-383.20	-383.20	-413.48	-413.48	-413.48	**-415.08**	**-415.08**	**-415.08**
	bound	-Inf	-Inf	-415.52	-415.52	-415.52	**-415.08**	**-415.08**	**-415.08**
	runtime	0.37	0.16	0.04	0.02	0.03	0.22	0.71	0.69
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002325	value	-331.28	-331.28	-376.06	-376.06	-376.06	**-377.44**	**-377.44**	**-377.44**
	bound	-Inf	-Inf	-376.33	-378.33	-378.33	**-377.44**	**-377.44**	**-377.44**
	runtime	0.39	0.19	0.02	0.01	0.02	0.28	0.33	0.35
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002326	value	-468.90	-468.90	-483.17	-483.17	-483.17	**-487.08**	**-487.08**	**-487.08**
	bound	-Inf	-Inf	-488.04	-488.04	-488.04	**-487.08**	**-487.08**	**-487.08**
	runtime	0.58	0.27	1.03	0.02	0.02	0.14	0.34	0.46
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002327	value	-544.18	-544.18	-580.18	**-580.18**	**-580.18**	**-580.18**	**-580.18**	**-580.18**
	bound	-Inf	-Inf	-580.51	-580.51	-580.51	**-580.18**	**-580.18**	**-580.18**
	runtime	1.03	0.46	0.18	0.07	0.08	0.97	1.55	2.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002328	value	-558.34	-558.34	**-586.12**	**-586.12**	**-586.12**	**-586.12**	**-586.12**	**-586.12**
	bound	-Inf	-Inf	**-586.12**	**-586.12**	**-586.12**	**-586.12**	**-586.12**	**-586.12**
	runtime	0.75	0.34	0.03	0.02	0.03	0.02	0.25	0.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002329	value	-568.03	-568.03	-627.07	-627.07	-627.07	**-627.67**	**-627.67**	**-627.67**
	bound	-Inf	-Inf	-628.24	-628.24	-628.20	**-627.67**	**-627.67**	**-627.67**
	runtime	0.77	0.33	0.07	0.06	0.07	0.14	0.81	0.91
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
60002330	value	-508.86	-508.86	-544.81	-544.81	-544.81	**-545.65**	**-545.65**	**-545.65**
	bound	-Inf	-Inf	-546.16	-546.16	-546.16	**-545.65**	**-545.65**	**-545.65**
	runtime	1.11	0.50	0.09	0.05	0.06	0.56	1.25	1.38
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002331	value	-798.10	-798.10	**-827.10**	**-827.10**	**-827.10**	**-827.10**	**-827.10**	**-827.10**
	bound	-Inf	-Inf	**-827.10**	**-827.10**	**-827.10**	**-827.10**	**-827.10**	**-827.10**
	runtime	1.58	0.78	0.40	0.18	0.18	0.18	0.95	0.93
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB
60002332	value	-579.89	-579.89	**-623.34**	**-623.34**	**-623.34**	**-623.34**	**-623.34**	**-623.34**
	bound	-Inf	-Inf	-624.01	-624.01	-624.01	**-623.34**	**-623.34**	**-623.34**
	runtime	0.99	0.45	0.15	0.09	0.10	0.14	0.69	0.55
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002333	value	-574.70	-574.70	-622.88	-622.88	-622.88	**-624.09**	**-624.09**	**-624.09**
	bound	-Inf	-Inf	-624.09	-624.09	-624.09	**-624.09**	**-624.09**	**-624.09**
	runtime	0.63	0.28	0.12	0.07	0.07	0.06	0.49	0.42
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002334	value	-741.96	-741.96	-785.20	-785.20	-785.20	**-786.21**	**-786.21**	**-786.21**
	bound	-Inf	-Inf	-786.96	-786.96	-786.96	**-786.21**	**-786.21**	**-786.21**
	runtime	0.98	0.48	0.08	0.05	0.05	0.34	1.46	2.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002335	value	-470.12	-470.12	-500.95	-500.95	-500.95	**-501.74**	**-501.74**	**-501.74**
	bound	-Inf	-Inf	-501.98	-501.98	-501.98	**-501.74**	**-501.74**	**-501.74**
	runtime	1.43	0.55	0.05	0.04	0.04	0.14	0.34	0.41
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002336	value	-527.62	-527.62	-549.80	-549.80	-549.80	**-552.31**	**-552.31**	**-552.31**
	bound	-Inf	-Inf	-554.09	-554.09	-554.09	**-552.31**	**-552.31**	**-552.31**
	runtime	1.99	1.01	0.11	0.06	0.07	6.92	8.92	8.30
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
60002337	value	-412.58	-412.58	-429.51	-429.51	-429.51	**-431.05**	**-431.05**	**-431.05**
	bound	-Inf	-Inf	-431.64	-431.64	-431.64	**-431.05**	**-431.05**	**-431.05**
	runtime	0.60	0.26	0.05	0.04	0.04	0.15	0.79	0.65
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002338	value	-640.70	-640.70	-686.76	-686.76	-686.76	**-687.22**	**-687.22**	**-687.22**
	bound	-Inf	-Inf	-687.23	-687.23	**-687.22**	**-687.22**	**-687.22**	**-687.22**
	runtime	1.05	0.46	0.08	0.05	0.05	0.11	0.53	0.70
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
60002339	value	-550.17	-550.17	**-604.94**	**-604.94**	**-604.94**	**-604.94**	**-604.94**	**-604.94**
	bound	-Inf	-Inf	**-604.94**	**-604.94**	**-604.94**	**-604.94**	**-604.94**	**-604.94**
	runtime	0.54	0.22	0.04	0.03	0.03	0.10	0.31	0.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
60002340	value	-594.05	-594.05	-633.53	-633.53	-633.53	**-635.78**	**-635.78**	**-635.78**
	bound	-Inf	-Inf	-636.56	-636.56	-636.56	**-635.78**	**-635.78**	**-635.78**
	runtime	0.34	0.14	0.10	0.07	0.06	0.25	0.31	0.50
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002341	value	-645.56	-645.56	-695.56	-695.56	-695.56	**-696.50**	**-696.50**	**-696.50**
	bound	-Inf	-Inf	-697.63	-697.63	-697.63	**-696.50**	**-696.50**	**-696.50**
	runtime	0.64	0.27	0.08	0.03	0.04	0.14	0.34	0.66
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
60002342	value	-820.89	-820.89	-872.68	-872.68	-872.68	**-873.53**	**-873.53**	**-873.53**
	bound	-Inf	-Inf	-873.73	-873.73	-873.73	**-873.53**	**-873.53**	**-873.53**
	runtime	1.99	0.89	0.09	0.05	0.05	0.53	0.73	0.61
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
60002343	value	-671.90	-671.90	-717.75	-717.75	-717.75	**-719.87**	**-719.87**	**-719.87**
	bound	-Inf	-Inf	-722.14	-722.14	-722.14	**-719.87**	**-719.87**	**-719.87**
	runtime	1.37	0.66	0.18	0.12	0.11	0.21	1.35	1.28
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
60002344	value	-550.13	-550.13	-603.12	-603.12	-603.12	**-604.99**	**-604.99**	**-604.99**
	bound	-Inf	-Inf	-605.68	-605.68	-605.68	**-604.99**	**-604.99**	**-604.99**
	runtime	1.46	0.67	0.11	0.06	0.07	0.24	1.07	1.49
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
60002345	value	-534.22	-534.22	-611.54	-611.54	-612.51	**-616.82**	**-616.82**	**-616.82**
	bound	-Inf	-Inf	-617.67	-617.67	-617.66	**-616.82**	**-616.82**	**-616.82**
	runtime	0.71	0.29	0.06	0.04	0.05	0.28	0.77	1.12
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
60002346	value	-417.98	-417.98	-447.61	-447.61	-447.61	**-448.79**	**-448.79**	**-448.79**
	bound	-Inf	-Inf	-448.95	-448.95	-448.95	**-448.79**	**-448.79**	**-448.79**
	runtime	0.64	0.26	0.03	0.02	0.02	0.21	0.34	0.63
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
60002347	value	-666.25	-666.25	-709.15	-709.15	-709.15	**-709.64**	**-709.64**	**-709.64**
	bound	-Inf	-Inf	-710.45	-710.45	-710.45	**-709.64**	**-709.64**	**-709.64**
	runtime	0.98	0.47

correlation=clustering		ogm-ILCM	ogm-LP-1	MCR-CC	MCR-CCPDB	MCR-CCPDB-OWC	MCL-CCPDB-CCIPD	MCL-CC1	MCL-CCIPD
6000352	value	-355.69	-355.69	-383.99	-383.99	-383.99	**-385.43**	**-385.43**	**-385.43**
	bound	-Inf	-Inf	-385.69	-385.69	-385.69	**-385.43**	**-385.43**	**-385.43**
	runtime	1.10	0.49	0.06	0.03	0.03	0.22	0.75	0.67
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000353	value	-566.74	-566.74	-618.37	-618.37	-619.17	**-619.62**	**-619.62**	**-619.62**
	bound	-Inf	-Inf	-619.62	-619.62	-619.62	**-619.62**	**-619.62**	**-619.62**
	runtime	1.27	0.59	0.04	0.02	0.03	0.22	0.39	0.42
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
6000354	value	-362.65	-362.65	-402.68	-402.68	-402.68	**-402.91**	**-402.91**	**-402.91**
	bound	-Inf	-Inf	-403.27	-403.27	-403.27	**-402.91**	**-402.91**	**-402.91**
	runtime	0.48	0.24	0.01	0.01	0.01	0.25	0.47	0.42
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
6000355	value	-508.28	-508.28	-544.93	-544.93	-544.93	**-546.79**	**-546.79**	**-546.79**
	bound	-Inf	-Inf	-548.44	-548.44	-548.44	**-546.79**	**-546.79**	**-546.79**
	runtime	0.61	0.30	0.03	0.02	0.02	0.22	0.43	0.44
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
8000811	value	-524.20	-524.20	**-562.98**	**-562.98**	**-562.98**	**-562.98**	**-562.98**	**-562.98**
	bound	-Inf	-Inf	**-562.98**	**-562.98**	**-562.98**	**-562.98**	**-562.98**	**-562.98**
	runtime	0.12	0.05	0.03	0.03	0.03	0.03	0.16	0.47
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.03 GB
8001155	value	-336.33	-336.33	-366.76	-366.76	-366.76	**-367.82**	**-367.82**	**-367.82**
	bound	-Inf	-Inf	-367.82	-367.82	-367.82	**-367.82**	**-367.82**	**-367.82**
	runtime	0.22	0.11	0.03	0.02	0.02	0.17	0.53	0.58
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB
8001974	value	-1212.80	-1212.80	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**
	bound	-Inf	-Inf	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**	**-1243.94**
	runtime	4.23	2.14	0.27	0.22	0.21	0.23	1.81	1.47
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.02 GB	0.05 GB	0.06 GB
8002274	value	-822.23	-822.23	**-848.73**	**-848.73**	**-848.73**	**-848.73**	**-848.73**	**-848.73**
	bound	-Inf	-Inf	**-848.73**	**-848.73**	**-848.73**	**-848.73**	**-848.73**	**-848.73**
	runtime	0.29	0.13	0.01	0.01	0.01	0.02	0.07	0.11
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
8002764	value	-312.42	-312.42	-356.82	-356.82	-356.82	**-357.33**	**-357.33**	**-357.33**
	bound	-Inf	-Inf	-359.23	-359.23	-359.23	**-357.33**	**-357.33**	**-357.33**
	runtime	0.12	0.05	0.01	0.01	0.01	0.05	0.09	0.09
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
8003131	value	-619.14	-619.14	**-650.66**	**-650.66**	**-650.66**	**-650.66**	**-650.66**	**-650.66**
	bound	-Inf	-Inf	**-650.66**	**-650.66**	**-650.66**	**-650.66**	**-650.66**	**-650.66**
	runtime	1.89	0.65	0.20	0.18	0.16	0.21	0.67	0.83
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
8003536	value	-285.00	-285.00	-317.29	-317.29	-317.29	**-317.98**	**-317.98**	**-317.98**
	bound	-Inf	-Inf	-317.99	-317.99	-317.99	**-317.98**	**-317.98**	**-317.98**
	runtime	0.10	0.04	0.01	0.01	0.01	0.04	0.17	0.24
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
8003952	value	-939.46	-939.46	-951.44	-951.44	-951.44	**-959.64**	**-959.64**	**-959.64**
	bound	-Inf	-Inf	-959.89	-959.89	-959.89	**-959.64**	**-959.64**	**-959.64**
	runtime	0.27	0.13	0.01	0.01	0.01	0.04	0.13	0.16
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
8004573	value	-264.96	-264.96	**-348.40**	**-348.40**	**-348.40**	**-348.40**	**-348.40**	**-348.40**
	bound	-Inf	-Inf	-348.58	-348.58	-348.58	**-348.40**	**-348.40**	**-348.40**
	runtime	0.04	0.02	0.02	0.01	0.01	0.06	0.36	0.29
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
8005616	value	-202.77	-202.77	-228.75	-228.75	-228.75	**-229.32**	**-229.32**	**-229.32**
	bound	-Inf	-Inf	-230.49	-230.49	-230.49	**-229.32**	**-229.32**	**-229.32**
	runtime	0.14	0.05	0.02	0.01	0.01	0.09	0.18	0.23
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
8006160	value	-334.96	-334.96	**-401.07**	**-401.07**	**-401.07**	**-401.07**	**-401.07**	**-401.07**
	bound	-Inf	-Inf	**-401.07**	**-401.07**	**-401.07**	**-401.07**	**-401.07**	**-401.07**
	runtime	0.06	0.03	0.01	0.01	0.01	0.01	0.07	0.07
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
8006302	value	-511.51	-511.51	-555.13	-555.13	-555.13	**-557.46**	**-557.46**	**-557.46**
	bound	-Inf	-Inf	-557.68	-557.68	-557.68	**-557.46**	**-557.46**	**-557.46**
	runtime	0.19	0.09	0.01	0.01	0.01	0.03	0.06	0.06
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
8008699	value	-981.46	-981.46	-984.46	-984.46	-984.46	**-984.30**	**-984.30**	**-984.30**
	bound	-Inf	-Inf	-984.46	-984.46	-984.46	**-984.30**	**-984.30**	**-984.30**
	runtime	1.87	1.18	0.03	0.02	0.02	0.19	0.25	0.41
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
8008545	value	-675.55	-675.55	-699.82	-699.82	-699.82	**-700.15**	**-700.15**	**-700.15**
	bound	-Inf	-Inf	-700.44	-700.44	-700.44	**-700.15**	**-700.15**	**-700.15**
	runtime	0.57	0.28	0.09	0.07	0.07	1.25	0.87	1.19
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
8008698	value	-440.74	-440.74	-459.31	-459.31	-459.31	**-461.55**	**-461.55**	**-461.55**
	bound	-Inf	-Inf	-461.55	-461.55	-461.55	**-461.55**	**-461.55**	**-461.55**
	runtime	0.35	0.15	0.02	0.01	0.01	0.58	0.35	0.42
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9000001	value	-740.50	-740.50	-784.97	-784.97	-784.97	**-792.68**	**-792.68**	**-792.68**
	bound	-Inf	-Inf	-793.68	-793.68	-793.68	**-792.68**	**-792.68**	**-792.68**
	runtime	0.91	0.46	0.07	0.10	0.11	0.46	0.52	0.51
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.03 GB
9000002	value	-593.19	-593.19	-597.19	**-597.19**	**-597.19**	**-597.19**	**-597.19**	**-597.19**
	bound	-Inf	-Inf	-599.40	-599.40	-599.40	**-597.19**	**-597.19**	**-597.19**
	runtime	0.26	0.12	0.03	0.02	0.02	0.05	0.14	0.19
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
9000003	value	-701.62	-701.62	**-766.60**	**-766.60**	**-766.60**	**-766.60**	**-766.60**	**-766.60**
	bound	-Inf	-Inf	**-766.60**	**-766.60**	**-766.60**	**-766.60**	**-766.60**	**-766.60**
	runtime	2.12	1.21	0.15	0.14	0.15	0.14	2.17	1.36
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB
9000029	value	-553.12	-553.12	-580.98	-580.98	-580.98	**-581.56**	**-581.56**	**-581.56**
	bound	-Inf	-Inf	-581.56	-581.56	-581.56	**-581.56**	**-581.56**	**-581.56**
	runtime	0.64	0.28	0.06	0.05	0.05	0.03	0.27	0.97
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
9000127	value	-764.69	-764.69	-836.84	-836.84	-836.84	**-837.29**	**-837.29**	**-837.29**
	bound	-Inf	-Inf	-838.10	-838.10	-838.10	**-837.29**	**-837.29**	**-837.29**
	runtime	0.99	0.48	0.18	0.13	0.13	0.49	2.24	1.98
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9000136	value	-736.09	-736.09	-772.98	-772.98	-772.98	**-776.21**	**-776.21**	**-776.21**
	bound	-Inf	-Inf	-777.44	-777.44	-777.44	**-776.21**	**-776.21**	**-776.21**
	runtime	0.91	0.46	0.07	0.06	0.06	0.46	0.51	0.51
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9000210	value	-520.21	-520.21	-568.03	-568.03	-568.03	**-568.26**	**-568.26**	**-568.26**
	bound	-Inf	-Inf	-568.70	-568.70	-568.70	**-568.26**	**-568.26**	**-568.26**
	runtime	1.20	0.57	0.06	0.04	0.04	1.26	1.43	1.75
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9000288	value	-447.32	-447.32	-485.52	-485.52	-485.52	**-485.65**	**-485.65**	**-485.65**
	bound	-Inf	-Inf	-496.07	-496.07	-496.07	**-485.65**	**-485.65**	**-485.65**
	runtime	0.39	0.16	0.02	0.01	0.01	0.03	0.05	0.05
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9000395	value	-370.16	-370.16	-413.77	-413.77	-413.77	**-414.96**	**-414.96**	**-414.96**
	bound	-Inf	-Inf	-415.94	-415.94	-415.94	**-414.96**	**-414.96**	**-414.96**
	runtime	0.49	0.21	0.03	0.02	0.02	0.23	0.39	0.43
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9000747	value	-461.47	-461.47	**-495.73**	**-495.73**	**-495.73**	**-495.73**	**-495.73**	**-495.73**
	bound	-Inf	-Inf	-496.41	-496.41	-496.41	**-495.73**	**-495.73**	**-495.73**
	runtime	0.61	0.30	0.03	0.02	0.02	0.03	0.13	0.16
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
9000868	value	-669.31	-669.31	-710.00	-710.00	-710.00	**-711.39**	**-711.39**	**-711.39**
	bound	-Inf	-Inf	-712.15	-712.15	-712.15	**-711.39**	**-711.39**	**-711.39**
	runtime	0.40	0.18	0.06	0.05	0.05	0.27	0.83	0.75
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9000875	value	-581.82	-581.82	-609.26	-609.26	-609.26	**-613.28**	**-613.28**	**-613.28**
	bound	-Inf	-Inf	-614.54	-614.54	-614.54	**-613.28**	**-613.28**	**-613.28**
	runtime	1.05	0.53	0.23	0.18	0.19	0.70	1.50	3.40
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9000928	value	-634.59	-634.59	-698.59	-698.59	-698.59	**-704.44**	**-704.44**	**-704.44**
	bound	-Inf	-Inf	-705.95	-705.95	-705.95	**-704.44**	**-704.44**	**-704.44**
	runtime	1.09	0.47	0.12	0.07	0.08	0.74	1.91	2.51
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.05 GB
9000933	value	-472.61	-472.61	-515.47	-515.47	-515.64	**-515.70**	**-515.70**	**-515.70**
	bound	-Inf	-Inf	-517.06	-517.06	-517.06	**-515.70**	**-515.70**	**-515.70**
	runtime	0.29	0.13	0.04	0.03	0.03	0.03	0.09	0.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB	0.04 GB
9000989	value	-583.70	-583.70	-659.08	-659.08	-659.08	**-660.46**	**-660.46**	**-660.46**
	bound	-Inf	-Inf	-662.10	-662.10	-662.10	**-660.46**	**-660.46**	**-660.46**
	runtime	0.43	0.19	0.02	0.02	0.02	0.06	0.17	0.18
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9001001	value	-668.82	-668.82	-713.70	-713.70	-713.70	**-717.57**	**-717.57**	**-717.57**
	bound	-Inf	-Inf	-720.40	-720.40	-720.40	**-717.57**	-717.65	**-717.57**
	runtime	0.47	0.21	0.11	0.09	0.10	0.28	0.69	0.86
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9001034	value	-643.05	-645.05	-670.78	-670.78	-671.95	**-671.52**	**-671.52**	**-671.52**
	bound	-Inf	-Inf	-671.95	-671.95	-671.95	**-671.52**	**-671.52**	**-671.52**
	runtime	0.62	0.27	0.29	0.18	0.19	0.62	0.99	0.90
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9001071	value	-806.46	-806.46	-860.19	-860.19	-860.19	**-861.29**	**-861.29**	**-861.29**
	bound	-Inf	-Inf	-861.52	-861.52	-861.52	**-861.29**	**-861.29**	**-861.29**
	runtime	0.74	0.33	0.21	0.16	0.16	0.43	0.80	0.68
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9001184	value	-466.54	-466.54	-503.91	-503.91	-503.91	**-505.01**	**-505.01**	**-505.01**
	bound	-Inf	-Inf	-505.44	-505.44	-505.44	**-505.01**	**-505.01**	**-505.01**
	runtime	0.39	0.16	0.02	0.01	0.02	0.03	0.05	0.05
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9001300	value	-828.19	-828.19	-837.40	-837.40	-837.40	**-837.41**	**-837.41**	**-837.41**
	bound	-Inf	-Inf	-838.44	-838.44	-838.44	**-837.41**	**-837.41**	**-837.41**
	runtime	2.04	1.23	0.24	0.22	0.22	0.78	0.73	1.00
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.05 GB	0.04 GB	0.04 GB
9001317	value	-458.32	-458.32	-517.24	-517.24	-517.24	**-519.21**	**-519.21**	**-519.21**
	bound	-Inf	-Inf	-519.60	-519.60	-519.60	**-519.21**	**-519.21**	**-519.21**
	runtime	0.61	0.30	0.03	0.02	0.02	0.22	0.43	0.44
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
9001619	value	-464.60	-464.60	-509.55	-509.55	-509.55	**-510.62**	**-510.62**	**-510.62**
	bound	-Inf	-Inf	-513.14	-513.14	-513.14	**-510.62**	**-510.62**	**-510.62**
	runtime	0.95	0.44	0.10	0.08	0.08	2.24	2.93	2.70
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9001713	value	-536.89	-536.89	-563.64	-563.64	-563.64	**-566.01**	**-566.01**	**-566.01**
	bound	-Inf	-Inf	-566.30	-566.30	-566.30	**-566.01**	**-566.01**	**-566.01**
	runtime	1.76	0.80	0.14	0.07	0.07	0.56	1.72	1.78
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9001991	value	-353.04	-355.04	-376.36	-376.36	-376.36	**-377.14**	**-377.14**	**-377.14**
	bound	-Inf	-Inf	-377.56	-377.56	-377.56	**-377.14**	**-377.14**	**-377.14**
	runtime	0.36	0.16	0.02	0.01	0.01	0.38	0.59	0.55
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
9002004	value	-536.16	-536.16	-561.45	-561.45	**-561.70**	**-561.70**	**-561.70**	**-561.70**
	bound	-Inf	-Inf	-561.74	-561.74	**-561.70**	**-561.70**	**-561.70**	**-561.70**
	runtime	0.29	0.13	0.01	0.01	0.01	0.03	0.07	0.11
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
9002021	value	-824.88	-824.88	-885.14	-885.14	-885.14	**-884.88**	**-884.88**	**-884.88**
	bound	-Inf	-Inf	-885.14	-885.14	-885.14	**-884.88**	**-884.88**	**-884.88**
	runtime	0.99	0.48	0.06	0.03	0.04	1.02	1.37	1.25
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9002090	value	-340.99	-340.99	-403.27	-403.27	-403.27	**-403.53**	**-403.53**	**-403.53**
	bound	-Inf	-Inf	-403.55	-403.55	-403.55	**-403.53**	**-403.53**	**-403.53**
	runtime	0.50	0.21	0.02	0.02	0.02	0.08	0.27	0.22
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9002114	value	-397.31	-397.31	-429.71	-429.71	-429.71	**-430.41**	**-430.41**	**-430.41**
	bound	-Inf	-Inf	-430.71	-430.71	-430.71	**-430.41**	**-430.41**	**-430.41**
	runtime	0.18	0.08	0.03	0.02	0.02	0.24	0.35	0.31
	memory (GB)	0.00 GB	0.00 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB
9002474	value	-689.62	-689.62	**-726.44**	**-726.44**	**-726.44**	**-726.44**	**-726.44**	**-726.44**
	bound	-Inf	-Inf	**-726.44**	**-726.44**	**-726.44**	**-726.44**	**-726.44**	**-726.44**
	runtime	1.13	0.52	0.04	0.03	0.03	0.03	0.45	0.31
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
9002577	value	-858.29	-858.29	-874.43	-874.43	-874.43	**-874.29**	**-874.29**	**-874.29**
	bound	-Inf	-Inf	-874.43	-874.43	-874.43	**-874.29**	**-874.29**	**-874.29**
	runtime	0.73	0.36	0.03	0.02	0.03	0.31	0.54	0.55
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9002827	value	-540.73	-540.73	-561.10	-561.10	-561.10	**-561.63**	**-561.63**	**-561.63**
	bound	-Inf	-Inf	-565.85	-565.85	-565.85	**-561.63**	**-561.63**	**-561.63**
	runtime	1.29	0.60	0.17	0.16	0.17	0.42	0.82	0.82
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9002861	value	-608.82	-608.82	**-661.79**	**-661.79**	**-661.79**	**-661.79**	**-661.79**	**-661.79**
	bound	-Inf	-Inf	-662.04	-662.04	-662.04	**-661.79**	**-661.79**	**-661.79**
	runtime	1.86	0.87	0.23	0.16	0.16	0.70	1.27	1.40
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.05 GB
9002972	value	-804.51	-804.51	-824.15	-824.15	-824.15	**-824.71**	**-824.71**	**-824.71**
	bound	-Inf	-Inf	-825.68	-825.68	-825.68	**-824.71**	**-824.71**	**-824.71**
	runtime	0.83	0.37	0.09	0.09	0.09	0.22	1.14	1.11
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9003116	value	-323.89	-323.89	-330.70	-330.70	-330.70	**-338.25**	**-338.25**	**-338.25**
	bound	-Inf	-Inf	-339.83	-339.83	-339.83	**-338.25**	**-338.25**	**-338.25**
	runtime	1.90	0.94	0.09	0.07	0.07	0.19	0.19	0.19
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9003135	value	-706.30	-706.30	-731.92	-731.92	-731.92	**-733.39**	**-733.39**	**-733.39**
	bound	-Inf	-Inf	-734.02	-734.02	-734.02	**-733.39**	**-733.39**	**-733.39**
	runtime	0.46	0.21	0.08	0.06	0.07	0.72	1.70	1.86
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9003234	value	-622.90	-622.90	-661.44	-661.44	-661.44	**-683.07**	**-683.07**	**-683.07**
	bound	-Inf	-Inf	-664.12	-664.12	-664.12	**-683.07**	**-683.07**	**-683.07**
	runtime	1.27	0.62	0.14	0.13	0.13	1.69	3.07	3.07
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.05 GB	0.04 GB
9003250	value	-405.26	-405.26	**-478.34**	**-478.34**	**-478.34**	**-478.34**	**-478.34**	**-478.34**
	bound	-Inf	-Inf	-478.41	-478.41	-478.41	**-478.34**	**-478.34**	**-478.34**
	runtime	0.11	0.05	0.01	0.00	0.01	0.13	0.19	0.18
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB
9003301	value	-1144.79	-1144.79	-1170.45	-1170.45	-1170.45	**-1178.36**	**-1178.36**	**-1178.36**
	bound	-Inf	-Inf	-1180.33	-1180.33	-1180.33	**-1178.36**	**-1178.36**	**-1178.36**
	runtime	0.91	0.46	0.07	0.06	0.06	0.05	0.14	0.19
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.05 GB	0.05 GB
9003333	value	-422.63	-422.63	**-464.55**	**-464.55**	**-464.55**	**-464.55**	**-464.55**	**-464.55**
	bound	-Inf	-Inf	**-464.55**	**-464.55**	**-464.55**	**-464.55**	**-464.55**	**-464.55**
	runtime	0.28	0.12	0.02	0.02	0.01	0.02	0.14	0.15
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB
9003339	value	-761.46	-761.46	-809.21	-809.21	-809.21	**-810.44**	**-810.44**	**-810.44**
	bound	-Inf	-Inf	-811.38	-811.38	-811.38	**-810.44**	**-810.44**	**-810.44**
	runtime	0.64	0.28	0.06	0.05	0.05	0.03	0.27	0.97
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9003378	value	-572.27	-572.27	**-591.56**	**-591.56**	**-591.56**	**-591.56**	**-591.56**	**-591.56**
	bound	-Inf	-Inf	**-591.56**	**-591.56**	**-591.56**	**-591.56**	**-591.56**	**-591.56**
	runtime	0.24	0.10	0.01	0.01	0.01	0.01	0.09	0.09
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
9003423	value	-625.90	-625.90	**-716.50**	**-716.50**	**-716.50**	**-716.50**	**-716.50**	**-716.50**
	bound	-Inf	-Inf	**-716.50**	**-716.50**	**-716.50**	**-716.50**	**-716.50**	**-716.50**
	runtime	0.24	0.10	0.01	0.01	0.01	0.03	0.09	0.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
9003585	value	-361.44	-361.44	-415.97	-415.97	-415.97	**-418.38**	**-418.38**	**-418.38**
	bound	-Inf	-Inf	-419.06	-419.06	-419.06	**-418.38**	**-418.38**	**-418.38**
	runtime	0.73	0.33	0.05	0.03	0.03	0.29	0.50	0.41
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9003635	value	-452.44	-452.44	-495.38	-495.38	-495.38	**-501.31**	**-501.31**	**-501.31**
	bound	-Inf	-Inf	-501.55	-501.55	-501.55	**-501.31**	**-501.31**	**-501.31**
	runtime	0.35	0.14	0.03	0.03	0.03	0.09	0.26	0.21
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9003836	value	-1206.85	-1206.85	-1260.18	-1260.18	-1260.18	**-1261.74**	**-1261.74**	**-1261.74**
	bound	-Inf	-Inf	-1264.59	-1264.59	-1264.59	**-1261.74**	**-1261.74**	**-1261.74**
	runtime	2.65	1.45	0.14	0.11	0.12	0.74	1.51	0.97
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.05 GB
9004060	value	-558.31	-558.31	-600.54	-600.54	-600.54	**-602.99**	**-602.99**	**-602.99**
	bound	-Inf	-Inf	-604.19	-604.

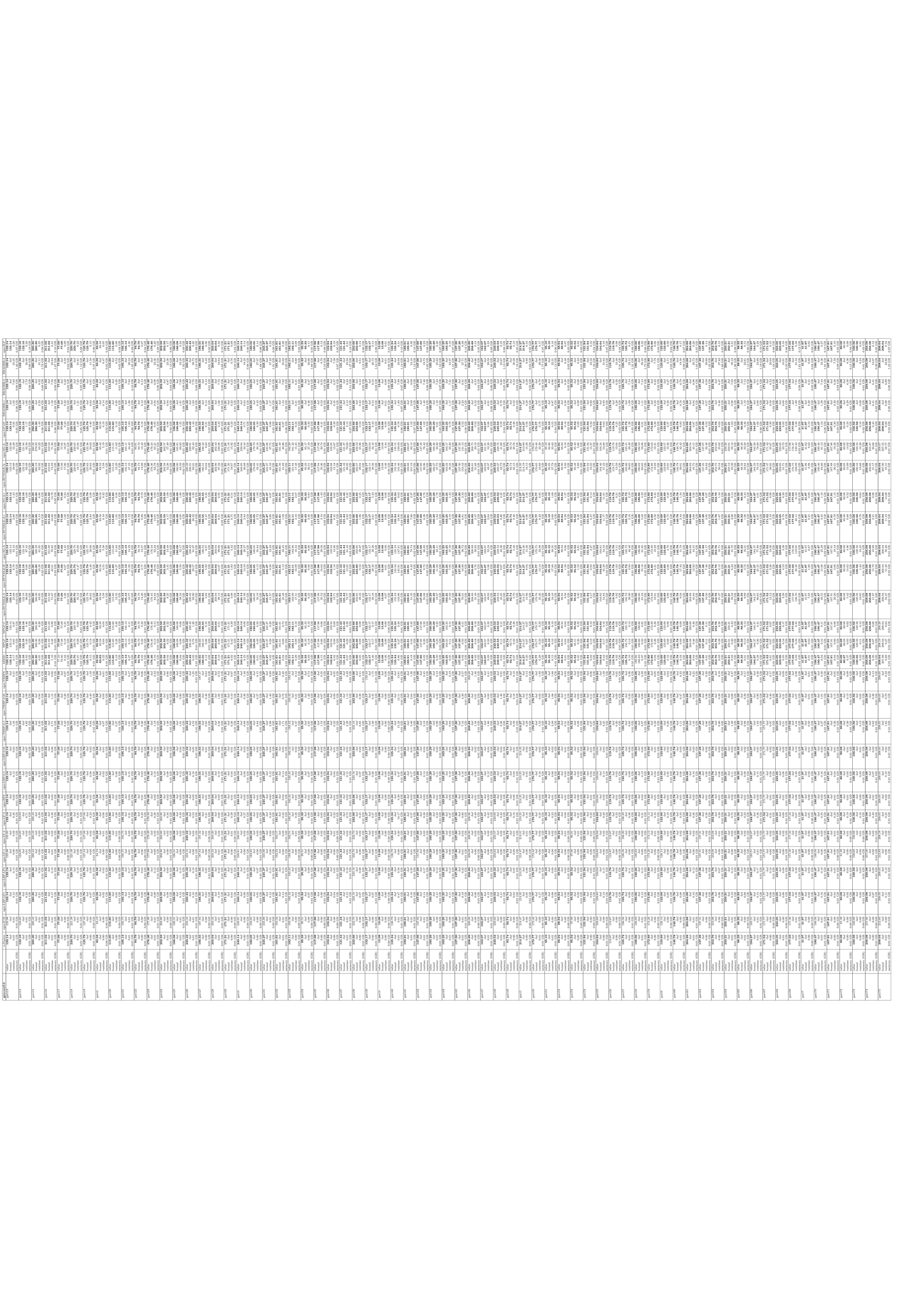
correlation+clustering		ogm-ICM	ogm-LF-1	MCR-CC	MCR-CCPD	MCR-CCPD-OWC	MCL-CCPD-CCPD	MCLCC1	MCLCCPD
9004294	value	-386.86	-386.86	**-407.78**	**-407.78**	**-407.78**	**-407.78**	**-407.78**	**-407.78**
	bound	-Inf	-Inf	**-407.78**	**-407.78**	**-407.78**	**-407.78**	**-407.78**	**-407.78**
	runtime	0.24	0.12	0.03	0.02	0.02	0.02	0.26	0.21
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB
9004353	value	-628.77	-628.77	-658.80	-658.80	-658.80	**-658.85**	**-658.85**	**-658.85**
	bound	-Inf	-Inf	-658.80	-658.80	-658.80	**-658.85**	**-658.85**	**-658.85**
	runtime	1.13	0.56	0.24	0.23	0.22	0.44	0.89	1.45
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9004368	value	-302.82	-302.82	-350.94	-350.94	-350.94	**-351.69**	**-351.69**	**-351.69**
	bound	-Inf	-Inf	-352.13	-352.13	-352.13	**-351.69**	**-351.69**	**-351.69**
	runtime	0.80	0.41	0.04	0.02	0.03	1.19	1.47	1.65
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9004383	value	-594.59	-594.59	-624.06	-624.06	-624.06	**-624.33**	**-624.33**	**-624.33**
	bound	-Inf	-Inf	-625.10	-625.10	-625.10	**-624.33**	**-624.33**	**-624.33**
	runtime	0.61	0.30	0.03	0.02	0.02	0.03	0.03	0.04
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB
9004427	value	-455.52	-455.52	-484.27	-484.27	-484.27	**-488.45**	**-488.45**	**-488.45**
	bound	-Inf	-Inf	-489.07	-489.07	-489.07	**-488.45**	**-488.45**	**-488.45**
	runtime	0.79	0.37	0.06	0.05	0.05	0.15	0.52	0.52
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9004520	value	-362.21	-362.21	-438.62	-438.62	-439.65	**-441.12**	**-441.12**	**-441.12**
	bound	-Inf	-Inf	-441.12	-441.12	-441.12	**-441.12**	**-441.12**	**-441.12**
	runtime	0.51	0.25	0.03	0.02	0.02	0.12	0.39	0.52
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9004581	value	-364.06	-364.06	**-399.15**	**-399.15**	**-399.15**	**-399.15**	**-399.15**	**-399.15**
	bound	-Inf	-Inf	**-399.15**	**-399.15**	**-399.15**	**-399.15**	**-399.15**	**-399.15**
	runtime	0.17	0.07	0.01	0.01	0.01	0.06	0.06	0.06
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
9004766	value	-282.36	-282.36	**-353.88**	**-353.88**	**-353.88**	**-353.88**	**-353.88**	**-353.88**
	bound	-Inf	-Inf	**-353.88**	**-353.88**	**-353.88**	**-353.88**	**-353.88**	**-353.88**
	runtime	0.29	0.14	0.02	0.01	0.01	0.03	0.03	0.04
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB
9004879	value	-1100.45	-1100.45	-1157.13	-1157.13	-1157.13	**-1159.74**	**-1159.74**	**-1159.74**
	bound	-Inf	-Inf	-1164.04	-1164.04	-1164.04	**-1159.74**	**-1159.74**	**-1159.74**
	runtime	3.41	1.53	0.24	0.28	0.29	1.98	6.56	7.58
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.06 GB	0.06 GB
9004965	value	-1216.62	-1216.62	-1253.73	-1253.73	-1253.73	**-1255.38**	**-1255.38**	**-1255.38**
	bound	-Inf	-Inf	-1255.86	-1255.86	-1255.86	**-1255.38**	**-1255.38**	**-1255.38**
	runtime	1.03	0.51	0.03	0.03	0.03	0.44	2.46	1.85
	memory (GB)	0.01 GB	0.01 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.06 GB	0.06 GB
9004971	value	-683.65	-683.65	-753.65	-753.65	-753.65	**-755.08**	**-755.08**	**-755.08**
	bound	-Inf	-Inf	-755.95	-755.95	-755.95	**-755.08**	**-755.08**	**-755.08**
	runtime	0.63	0.27	0.10	0.07	0.07	0.90	0.91	1.01
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
9005011	value	-502.58	-502.58	-543.91	-543.91	-543.91	**-546.15**	**-546.15**	**-546.15**
	bound	-Inf	-Inf	-546.78	-546.78	-546.78	**-546.15**	**-546.15**	**-546.15**
	runtime	0.61	0.30	0.04	0.03	0.03	0.07	0.08	0.09
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9005105	value	-580.68	-580.68	-610.25	-610.25	-610.25	**-610.41**	**-610.41**	**-610.41**
	bound	-Inf	-Inf	-610.53	-610.53	-610.53	**-610.41**	**-610.41**	**-610.41**
	runtime	1.89	0.90	0.05	0.04	0.05	0.17	0.41	0.39
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
9005245	value	-549.19	-549.19	-587.14	-587.14	-587.14	**-587.27**	**-587.27**	**-587.27**
	bound	-Inf	-Inf	-588.08	-588.08	-588.08	**-587.27**	**-587.27**	**-587.27**
	runtime	0.43	0.20	0.10	0.06	0.07	0.72	0.52	0.72
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.04 GB
9005273	value	-317.11	-317.11	-405.17	-405.17	-406.09	**-405.88**	**-405.88**	**-405.88**
	bound	-Inf	-Inf	-406.09	-406.09	-406.09	**-405.88**	**-405.88**	**-405.88**
	runtime	0.78	0.39	0.04	0.03	0.03	0.30	0.55	0.68
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.04 GB
9005294	value	-467.09	-467.09	-488.55	-488.55	-488.55	**-492.95**	**-492.95**	**-492.95**
	bound	-Inf	-Inf	-493.94	-493.94	-493.94	**-492.95**	**-492.95**	**-492.95**
	runtime	0.38	0.16	0.09	0.06	0.06	0.56	0.76	0.86
	memory (GB)	0.00 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
mean energy		-585.07	-585.07	-626.90	-626.90	-627.17	-628.16	-628.16	-628.16
mean bound		-Inf	-Inf	-628.90	-628.90	-628.90	-628.16	-628.16	-628.16
mean runtime		1.28	0.63	0.12	0.08	0.08	0.78	1.28	1.25
best value		0.00	0.00	22.94	22.94	22.94	100.00	100.00	100.00
best bound		0.00	0.00	13.71	13.71	13.99	99.72	98.88	99.72
verified opt		0.00	0.00	13.71	13.71	13.99	99.72	98.88	99.72

Table 9: dtf-chinesecheckers

dtf-chinesecheckers	LSA-TR (exec.)	LSA-TR (basis.)	ogm-ICM	ogm-LBP-LF2	ogm-LF-1	ogm-LF-2	ogm-LF-3	ogm-TRBS-LF1	ogm-TRWS-LF2	BPS-TAR	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDD	MFLP	ogm-ASSAL	ogm-DUNDEE-A	ogm-DUNDEE-B	ogm-SG-A	ogm-BUNDLE-A	ogm-BUNDLE-B	ogm-SG-A	ogm-BUNDLE-A	ogm-SG-A+	ogm-BUNDLE-A	ogm-SG-A+	ogm-LP-LF	QPBO	TRWS-TABI	TBWS-pct	ogm-ILP-pct	MCBC-pct	ogm-ILP	SA
TST_inst_0001_88_115	value	-56045.40	-56013.83	-56010.55	-56008.53	-56010.55	-56045.40	-56045.40	-56010.55	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56297.92	-56494.26	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	-56045.40	**-56545.58**	**-56545.58**	-56545.58	**-56545.58**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-56545.58**	**-56545.58**	-56545.58	**-56545.58**
	runtime	0.05	0.04	0.22	574.70	0.09	8.49	627.97	131.81	119.19	65.38	0.22	567.45	614.36	618.09	553.50	552.05	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	1050.69	0.22	0.22	0.22	0.22
	memory (GB)	0.06 GB	0.06 GB	0.07 GB	0.50 GB	0.09 GB	0.14 GB	1.16 GB	0.13 GB	0.17 GB	0.07 GB	0.40 GB	0.40 GB	0.40 GB	0.42 GB	0.42 GB	0.42 GB	0.43 GB	0.46 GB	0.18 GB	0.40 GB	0.42 GB	0.28 GB	0.40 GB	0.28 GB	0.42 GB	0.28 GB	0.43 GB	0.28 GB	0.42 GB	0.28 GB	1.76 GB	0.07 GB	0.07 GB	0.07 GB	0.33 GB	NaN GB	4.23 GB	NaN GB
TST_inst_0002_88_105	value	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24	-59299.24
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.06	0.07	0.17	516.75	0.18	7.62	722.04	84.14	67.03	59.76	0.17	189.17	522.92	568.89	573.08	499.60	4.70	305.30	519.03	439.93	439.74	382.08	460.79	419.19	429.51	387.91	2344.39	0.22	59.24	5.03	3566.32	3605.18	3542.51	NaN
	memory (GB)	0.06 GB	0.06 GB	0.06 GB	0.46 GB	0.09 GB	0.13 GB	1.15 GB	0.12 GB	0.16 GB	0.06 GB	0.37 GB	0.38 GB	0.38 GB	0.38 GB	0.38 GB	0.38 GB	0.38 GB	0.38 GB	0.14 GB	0.37 GB	0.38 GB	0.29 GB	0.35 GB	0.26 GB	0.38 GB	0.29 GB	0.35 GB	0.26 GB	0.38 GB	0.29 GB	1.66 GB	0.07 GB	0.07 GB	0.07 GB	0.70 GB	NaN GB	3.82 GB	NaN GB
TST_inst_0003_64_99	value	-31704.02	-31666.31	-31674.76	-31667.35	-31674.76	-31681.68	-31704.13	-31674.76	-31681.68	-31661.23	-31629.02	-31647.92	-31631.38	-31620.93	-31594.41	-31587.83	-30249.14	-30881.30	-31588.94	-31584.17	-31637.27	-31599.24	-31584.17	-31590.30	-31590.73	-31608.76	-31638.41	-31646.50	-31696.27	-31649.35	**-31705.26**	-31704.27	-31687.37	-31665.90
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-31705.26**	-Inf	**-31954.55**	-Inf	-Inf	-Inf
	runtime	0.04	0.08	0.15	536.28	0.15	4.99	652.84	86.91	56.24	36.75	0.27	327.23	344.86	354.50	379.58	382.89	333.56	5.60	240.70	764.36	295.04	297.11	285.67	255.58	243.20	285.31	235.40	2435.02	0.26	44.47	6.94	3572.80	3603.75	3540.29	NaN
	memory (GB)	0.04 GB	0.04 GB	0.04 GB	0.46 GB	0.09 GB	0.13 GB	0.59 GB	0.11 GB	0.14 GB	0.06 GB	0.11 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.11 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	0.25 GB	1.36 GB	0.07 GB	0.07 GB	0.07 GB	0.71 GB	NaN GB	2.66 GB	NaN GB
TST_inst_0004_88_97	value	-14497.71	-14493.40	-14493.40	-14464.69	-14493.40	-14497.27	**-14497.71**	-14437.80	-14453.64	-14480.08	-14427.13	-14455.04	-14496.77	-14461.38	-14440.30	-14435.90	-12800.21	-13617.29	-14493.40	-14443.95	-14417.07	-14421.15	-14439.59	-14432.27	-14439.59	-14421.83	-14330.30	-14481.97	-14433.56	-14427.14	**-14497.71**	**-14497.71**	-14490.43	-14473.90
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-14497.71**	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.04	0.07	0.19	547.47	0.17	6.47	677.86	89.14	69.81	66.28	0.17	202.42	532.34	525.85	496.38	494.67	502.34	5.60	266.31	1066.33	354.46	354.21	357.26	326.84	326.84	354.21	314.15	2435.02	0.26	64.63	5.58	3567.20	3603.75	3542.51	NaN
	memory (GB)	0.05 GB	0.05 GB	0.06 GB	0.41 GB	0.08 GB	0.10 GB	1.14 GB	0.11 GB	0.16 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.35 GB	0.35 GB	0.35 GB	0.35 GB	0.35 GB	0.15 GB	0.35 GB	0.35 GB	0.32 GB	0.35 GB	0.32 GB	0.35 GB	0.32 GB	0.35 GB	0.32 GB	0.35 GB	0.32 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.71 GB	NaN GB	3.53 GB	NaN GB
TST_inst_0005_120_97	value	-42975.31	-42937.26	-42736.97	-42736.97	-42971.10	-42971.10	-42974.12	-42736.97	-42971.10	-42971.10	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12	-42974.12
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.08	0.08	0.20	0.46	0.13	0.18	2.25	0.26	0.18	0.10	0.55	0.55	0.55	0.55	0.57	0.57	0.57	0.59	0.52	0.59	0.24	0.52	0.52	0.35	0.29	0.29	0.10	0.18	0.10	0.10	0.10	0.10	1.06	NaN	5.81	NaN
	memory (GB)	0.05 GB	0.05 GB	0.05 GB	0.45 GB	0.08 GB	0.12 GB	1.13 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	1.06 GB	NaN GB	3.23 GB	NaN GB
TST_inst_0031_64_78	value	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	-25495.64	**-25495.64**	-25495.64	-25495.64	-25495.64
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-25495.64**	-Inf	-Inf	-Inf
	runtime	0.04	0.04	0.11	0.46	0.13	0.18	2.25	0.26	0.18	0.10	0.55	0.55	0.55	0.55	0.57	0.57	0.57	0.59	0.52	0.59	0.24	0.52	0.52	0.35	0.29	0.29	0.10	0.18	0.10	0.10	0.10	0.10	1.06	NaN	5.81	NaN
	memory (GB)	0.05 GB	0.05 GB	0.05 GB	0.45 GB	0.08 GB	0.12 GB	1.13 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	1.06 GB	NaN GB	3.23 GB	NaN GB
TST_inst_0032_64_89	value	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88	-28530.88
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.04	0.06	0.14	514.23	0.14	5.39	334.80	58.09	53.60	33.96	294.72	314.22	303.75	345.81	325.49	287.92	28.35	356.66	572.89	256.75	252.07	251.96	255.24	229.75	256.02	247.29	2534.46	0.20	51.20	7.83	3576.77	3610.28	3582.61	NaN
	memory (GB)	0.04 GB	0.04 GB	0.04 GB	0.41 GB	0.08 GB	0.12 GB	0.57 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.69 GB	2.86 GB	3.12 GB	NaN GB
TST_inst_0033_88_84	value	-37740.83	-37704.15	-37690.36	-37711.62	-37690.36	-37698.24	-37729.40	-37633.71	-37682.60	-37712.51	-37701.86	-37705.85	-37703.57	-37683.91	-37720.76	-37685.44	-36586.62	-37133.72	-37710.52	-37639.43	-37655.33	-37642.32	-37639.43	-37639.43	-37639.43	-37602.03	-36602.71	-37651.99	-37584.59	-37537.40	-37711.67	-37743.30	-37731.32	**-37752.21**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.08	0.07	0.18	490.95	0.20	6.47	783.10	87.81	71.41	51.64	396.75	411.62	413.22	460.89	450.40	396.65	4.84	325.76	722.75	361.81	357.90	305.91	363.84	325.73	352.69	335.67	2708.93	0.21	62.79	5.38	3574.39	3603.33	3585.05	NaN
	memory (GB)	0.04 GB	0.04 GB	0.04 GB	0.41 GB	0.08 GB	0.12 GB	0.57 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.69 GB	2.86 GB	3.12 GB	NaN GB
TST_inst_0034_112_94	value	-58765.34	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	-58665.76	**-58707.37**	**-58707.37**	-58707.37	**-58707.37**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-58707.37**	**-58707.37**	-58707.37	**-58707.37**
	runtime	0.06	0.06	0.17	0.46	0.13	0.18	2.25	0.26	0.18	0.10	0.55	0.55	0.55	0.55	0.57	0.57	0.57	0.59	0.52	0.59	0.24	0.52	0.52	0.35	0.29	0.29	0.10	0.18	0.10	0.10	0.10	0.10	1.06	NaN	5.81	NaN
	memory (GB)	0.05 GB	0.05 GB	0.05 GB	0.45 GB	0.08 GB	0.12 GB	1.13 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.69 GB	2.86 GB	3.12 GB	NaN GB
TST_inst_0035_80_86	value	-37750.14	-37739.49	-37750.14	-37750.14	-37750.14	-37750.14	**-37750.37**	**-37750.37**	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	-37750.14	**-37750.37**	**-37750.37**	-37750.37	-37750.14
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-37750.37**	**-37750.37**	-37750.37	-Inf
	runtime	0.03	0.03	0.15	0.46	0.13	0.18	2.25	0.26	0.18	0.10	0.55	0.55	0.55	0.55	0.57	0.57	0.57	0.59	0.52	0.59	0.24	0.52	0.52	0.35	0.29	0.29	0.10	0.18	0.10	0.10	0.10	0.10	1.06	NaN	5.81	NaN
	memory (GB)	0.04 GB	0.04 GB	0.05 GB	0.45 GB	0.08 GB	0.12 GB	1.13 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.69 GB	2.86 GB	3.12 GB	NaN GB
TST_inst_0036_72_114	value	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52	-38353.52
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.04	0.04	0.19	0.46	0.13	0.18	2.25	0.26	0.18	0.10	0.55	0.55	0.55	0.55	0.57	0.57	0.57	0.59	0.52	0.59	0.24	0.52	0.52	0.35	0.29	0.29	0.10	0.18	0.10	0.10	0.10	0.10	1.06	NaN	5.81	NaN
	memory (GB)	0.04 GB	0.04 GB	0.05 GB	0.45 GB	0.08 GB	0.12 GB	1.13 GB	0.10 GB	0.14 GB	0.06 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.14 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	0.34 GB	1.34 GB	0.07 GB	0.07 GB	0.07 GB	0.69 GB	2.86 GB	3.12 GB	NaN GB
TST_inst_0037_80_101	value	-44483.62	-44483.62	-44448.78	-44483.83	-44448.78	-44459.01	-44484.32	-44453.72	-44459.01	-44483.28	-44476.25	-44459.01	-44482.97	-44475.04	-44475.04	-44482.97	-44475.04	-44475.04	-44339.93	-44396.24	-44396.24	-44396.24	-44412.21	-44419.90	-44446.40	-44446.40	-44453.23	-44453.23	-44484.57

Table 16 (part 5)

| open-source | open-TIGER | open-TIGER | open-TIGER



| | | |

geo-surf-3		cs-Fusion	cgns-ICM	cgns-LBP-LF1	cgns-LBP-LF2	cgns-LF-1	cgns-LF-2	cgns-LF-3	cgns-RPS	cgns-LBP-0.5	cgns-LBP-0.95	cgns-TRBP-0.5	cgns-TRBP-0.95	cgns-TRBPS	ADDD	MPLP	MPLP-C	cgns-BUNDLE-A	cgns-BUNDLE-H	cgns-SG-A	cgns-BUNDLE-A+	cgns-SG-A+	cgns-BUNDLE-A-	cgns-SG-A-	cgns-LP-LF	BRADIR-1	BRADIR-2	BRADIR-3	cgns-ILP
geo02	value	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**73.07**	73.07	73.07	73.07	73.07	73.07	**73.07**	73.07	**73.07**	73.07	**73.07**	**73.07**	**73.07**	**73.07**	**73.07**
	runtime	0.00	0.00	0.02	0.03	0.00	0.01	0.24	0.00	0.02	0.01	0.12	0.05	0.03	0.01	0.10	4.10	0.64	0.14	0.95	0.04	0.64	6.61	9.64	0.09	0.09	6.09	6.31	0.06
	memory (GB)	0.01 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB
geo03	value	**131.11**	131.22	**131.11**	**131.11**	131.22	131.22	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**	**131.11**
	runtime	0.01	0.01	0.18	0.18	0.01	0.03	0.72	0.02	0.16	0.06	4.57	1.01	0.98	0.34	3.10	13.10	79.50	46.92	73.59	6.04	9.52	74.38	75.64	1.42	1.76	5.78	16.07	0.38
	memory (GB)	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.02 GiB	0.06 GiB	0.07 GiB	0.08 GiB	1.69 GiB	0.10 GiB
geo04	value	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**	**142.79**
	runtime	0.01	0.00	0.05	0.07	0.00	0.02	0.22	0.01	0.05	0.02	0.15	0.03	0.03	0.01	0.09	0.53	2.94	1.24	4.38	0.41	0.29	25.01	27.87	0.50	0.90	0.90	26.33	0.16
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.05 GiB	0.05 GiB	1.66 GiB	0.06 GiB
geo05	value	**117.56**	117.56	119.13	117.89	117.89	117.56	117.56	117.56	117.56	117.56	117.56	117.56	117.56	**117.56**	**117.56**	**117.56**	117.56	**117.56**	**117.56**	117.53	117.46	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**117.56**	**117.56**	**117.56**	117.56	**117.56**	**117.56**	117.53	117.46	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**	**117.56**
	runtime	0.01	0.00	0.05	0.06	0.00	0.01	0.24	0.01	0.05	0.02	0.15	0.03	0.03	0.01	0.09	0.53	2.94	1.24	4.38	0.41	0.29	25.01	27.87	0.50	0.90	0.90	26.33	0.16
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.05 GiB	0.05 GiB	1.66 GiB	0.06 GiB
geo06	value	**155.35**	156.70	**155.35**	**155.35**	156.70	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**	**155.35**
	runtime	0.01	0.01	0.18	0.21	0.01	0.04	0.40	0.03	0.18	0.05	2.98	1.21	0.43	0.05	0.07	0.07	45.48	8.31	34.27	2.78	1.40	80.66	80.49	2.56	2.45	6.47	29.05	0.32
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.07 GiB	0.14 GiB	0.14 GiB	1.75 GiB	0.11 GiB
geo07	value	**235.76**	235.94	**235.76**	**235.76**	235.94	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**	**235.76**
	runtime	0.01	0.02	0.36	0.66	0.02	0.07	0.72	0.04	0.34	0.09	0.72	0.16	0.08	0.02	0.18	0.66	29.05	4.84	19.60	1.61	0.75	38.03	37.72	0.75	0.68	1.61	6.41	0.06
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.41 GiB	0.06 GiB
geo08	value	**57.40**	57.78	**57.40**	**57.40**	57.78	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	57.40	57.40	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**	**57.40**
	runtime	0.00	0.00	0.03	0.03	0.00	0.01	0.21	0.01	0.03	0.01	0.21	0.07	0.03	0.01	0.07	0.00	2.18	0.07	2.42	0.15	0.11	11.94	11.33	0.20	0.20	0.20	6.52	0.07
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.04 GiB
geo09	value	**76.17**	77.53	**76.17**	**76.17**	77.53	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	76.17	76.16	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**	**76.17**
	runtime	0.01	0.01	0.09	0.09	0.00	0.02	0.30	0.02	0.07	0.03	1.14	0.60	0.49	0.12	0.15	0.56	21.88	9.57	22.37	2.78	1.80	31.74	31.96	0.55	0.66	6.67	17.99	0.13
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.04 GiB	0.03 GiB
mean energy		132.14	132.77	132.17	132.17	132.77	132.36	132.21	132.17	132.17	132.16	132.16	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14
mean bound		132.14	132.77	132.17	132.17	132.77	132.36	132.21	132.17	132.17	132.16	132.16	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.14	132.09	131.98	132.14	132.14	132.14	132.14	132.14
mean runtime		0.00	0.00	0.07	0.09	0.00	0.02	0.30	0.02	0.07	0.03	1.14	0.60	0.49	0.12	0.15	0.56	21.88	9.57	22.37	2.78	1.80	31.74	31.96	0.55	0.66	6.67	17.99	0.13
best value		96.50	41.33	90.87	91.33	44.33	69.00	87.87	90.87	90.87	90.87	94.37	94.37	94.37	96.50	96.50	96.50	100.00	100.00	100.00	96.50	96.50	96.50	96.50	99.33	99.67	100.00	100.00	100.00
best bound		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.00	62.33	64.00	65.00	92.00	66.33	97.33	97.33	3.00	2.00	97.33	0.00	0.00	97.33
verified opt		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	96.00	63.00	64.67	66.67	92.33	51.67	100.00	100.00	3.00	2.00	100.00	0.00	0.00	100.00

Table 11: geo-surf-7

geo-surf-7		cs-Fusion	cgns-ICM	cgns-LBP-LF1	cgns-LBP-LF2	cgns-LF-1	cgns-LF-2	cgns-LF-3	cgns-RPS	cgns-LBP-0.5	cgns-LBP-0.95	cgns-TRBP-0.5	cgns-TRBP-0.95	cgns-TRBPS	ADDD	MPLP	MPLP-C	cgns-BUNDLE-A	cgns-BUNDLE-H	cgns-SG-A	cgns-BUNDLE-A+	cgns-SG-A+	cgns-BUNDLE-A-	cgns-SG-A-	cgns-LP-LF	BRADIR-1	BRADIR-2	BRADIR-3	cgns-ILP
geo1	value	**499.52**	1160.88	535.22	535.42	1107.75	970.36	856.12	535.42	535.42	535.42	537.68	537.68	537.68	**499.52**	**499.52**	**499.52**	499.36	**499.52**	498.58	**499.52**	**499.52**	500.15	502.51	**499.52**	**499.52**	**499.52**	**499.52**	**499.52**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**499.52**	**499.52**	**499.52**	499.36	**499.52**	498.58	**499.52**	**499.52**	494.82	483.92	**499.52**	499.52	499.52	499.52	**499.52**
	runtime	0.01	0.00	0.47	0.51	0.01	0.03	0.19	0.00	0.46	0.01	0.20	0.19	0.06	0.01	0.03	0.03	57.12	0.39	54.28	19.41	0.93	57.72	0.01	1.24	7.24	30.93	2.56	0.03
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.01 GiB	0.02 GiB	0.03 GiB	0.02 GiB	0.03 GiB	0.17 GiB	0.05 GiB	0.05 GiB	0.85 GiB	0.29 GiB
geo10	value	802.92	1160.88	535.22	535.42	1107.75	970.36	856.12	535.42	535.42	535.42	537.68	537.68	537.68	**499.52**	**499.52**	**499.52**	499.36	**499.52**	498.58	**499.52**	**499.52**	500.15	502.51	**499.52**	**499.52**	**499.52**	**499.52**	**499.52**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**499.52**	**499.52**	**499.52**	499.36	**499.52**	498.58	**499.52**	**499.52**	494.82	483.92	**499.52**	499.52	499.52	499.52	**499.52**
	runtime	0.04	0.02	1.83	1.92	0.01	0.55	30.85	22.96	1.78	0.73	26.16	26.11	25.16	1.61	3.30	3.27	115.55	102.47	113.21	83.61	97.79	115.30	113.15	31.75	3600.00	3600.01	3600.00	4.76
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.38 GiB	0.11 GiB	0.11 GiB	1.17 GiB	0.07 GiB
geo100	value	**220.96**	424.71	227.29	227.29	424.71	375.32	297.22	227.29	227.29	227.29	227.29	227.29	227.29	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	227.29	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**	216.96	218.91	**220.96**	**220.96**	**220.96**	**220.96**	**220.96**
	runtime	0.00	0.00	0.19	0.21	0.00	0.04	2.55	0.05	0.20	0.06	2.89	3.50	1.64	0.12	0.15	0.15	15.70	12.09	15.27	3.16	5.58	15.64	15.07	0.57	1.56	7.58	42.33	0.25
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.04 GiB	0.04 GiB	0.03 GiB	0.04 GiB
geo101	value	**150.85**	362.28	165.39	165.39	362.28	165.02	157.52	165.39	165.39	165.39	157.15	157.15	157.15	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**	**150.85**
	runtime	0.00	0.00	0.07	0.07	0.00	0.04	0.93	0.01	0.07	0.02	0.54	0.12	0.05	0.01	0.05	0.07	14.99	4.67	9.44	0.85	0.89	14.79	6.67	0.08	0.89	0.91	24.74	0.10
	memory (GB)	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.04 GiB	0.04 GiB	0.57 GiB	0.05 GiB
geo102	value	**582.56**	1332.43	628.35	628.35	1332.43	900.04	752.33	628.35	628.35	628.35	631.07	631.14	631.07	**582.56**	**582.56**	**582.56**	579.00	**582.56**	582.07	**582.56**	**582.56**	582.07	596.42	**582.56**	580.89	**582.56**	**582.56**	**582.56**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**582.56**	**582.56**	**582.56**	579.00	**582.56**	573.67	581.49	581.11	570.73	541.37	**582.56**	268.82	321.64	39.69	**582.56**
	runtime	0.01	0.01	0.65	0.67	0.01	0.17	1.37	0.03	0.71	0.03	9.79	9.64	9.79	0.03	3.32	3.30	60.65	0.19	59.27	60.11	58.97	60.14	58.58	3.53	268.82	321.64	39.69	1.25
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.16 GiB	0.12 GiB	0.11 GiB	1.16 GiB	0.28 GiB
geo103	value	**305.34**	306.52	**305.34**	**305.34**	306.52	**305.34**	**305.34**	**305.34**	565.78	492.46	**305.34**	**305.34**	**305.34**	**305.34**	**305.34**	**305.34**	311.10	**305.34**	305.34	**305.34**	**305.34**	305.34	305.34	**305.34**	**305.34**	**305.34**	**305.34**	**305.34**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**305.34**	**305.34**	**305.34**	305.34	**305.34**	305.06	**305.34**	**305.34**	303.39	298.55	**305.34**	303.39	303.39	303.39	**305.34**
	runtime	0.01	0.01	0.13	0.13	0.01	0.03	1.93	0.01	0.14	0.06	1.96	1.97	0.14	0.06	0.09	0.09	3.61	0.01	3.61	0.03	0.03	3.61	3.61	0.03	3.33	8.89	5.11	0.03
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	1.80 GiB	0.12 GiB
geo104	value	**499.64**	676.95	501.61	501.61	676.95	501.61	501.61	501.61	501.61	501.61	501.61	499.64	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	501.61	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	**499.64**	497.79	491.31	**499.64**	**499.64**	**499.64**	**499.64**
	runtime	0.01	0.02	0.59	0.63	0.01	0.11	2.41	0.19	0.57	0.21	8.03	1.87	3.19	0.16	0.42	0.42	39.76	8.80	41.49	8.15	8.86	39.43	40.53	1.23	1.67	7.75	40.90	0.65
	memory (GB)	0.01 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.07 GiB	0.03 GiB
geo105	value	231.75	326.06	**231.49**	**231.49**	326.06	323.65	323.65	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	231.75	232.37	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**	**231.49**
	runtime	0.00	0.00	0.45	0.45	0.00	0.02	1.16	0.14	0.47	0.12	2.12	0.56	0.52	0.13	0.41	0.42	13.65	0.14	13.17	11.61	11.94	12.47	11.93	0.17	0.15	7.59	36.94	0.12
	memory (GB)	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.06 GiB	0.07 GiB	0.07 GiB	0.81 GiB	0.10 GiB
geo106	value	**96.87**	120.65	99.65	**96.87**	120.65	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.87**	**96.85**	**96.87**	**96.87**	95.72	94.44	**96.87**	**96.87**	**96.87**	**96.87**
	runtime	0.00	0.00	0.03	0.03	0.00	0.02	0.27	0.01	0.03	0.01	0.08	0.03	0.03	0.01	0.01	0.01	3.49	0.06	3.31	0.22	0.18	3.44	3.37	0.11	0.92	6.92	7.13	0.06
	memory (GB)	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.04 GiB	0.04 GiB
geo107	value	**551.16**	615.41	576.52	576.52	971.13	907.92	705.61	576.52	576.52	576.52	576.52	576.52	576.52	551.16	551.16	551.16	551.16	551.16	551.09	551.16	551.16	552.18	557.92	551.16	551.16	551.16	551.16	551.16
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**551.16**	**551.16**	**551.16**	**551.16**	**551.16**	**551.16**	**551.16**	**551.16**	541.90	517.58	**551.16**	541.90	541.90	541.90	**551.16**
	runtime	0.01	0.01	0.18	0.18	0.01	0.03	0.72	0.02	0.16	0.06	4.57	1.01	0.98	0.34	3.10	13.10	79.50	46.92	73.59	6.04	9.52	74.38	75.64	1.42	1.76	5.78	16.07	0.38
	memory (GB)	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.02 GiB	0.06 GiB	0.07 GiB	0.08 GiB	1.69 GiB	0.10 GiB
geo108	value	**362.39**	369.44	381.90	381.90	369.44	369.44	369.44	381.90	381.90	381.90	374.37	363.35	363.35	**362.39**	**362.39**	**362.39**	**362.39**	**362.39**	**362.39**	**362.39**	**362.39**	364.90	373.96	**362.39**	**362.39**	**362.39**	**362.39**	**362.39**
	bound	-Inf	-Inf	-Inf	-Inf

Table 12: image-seg-3rdorder

image-seg-3rdorder		cgm-ICM	cgm-LF-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OMC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
101065.bmp	value	7610.70	7603.92	6930.97	6930.97	6930.97	**6718.03**	**6718.03**	**6718.03**
	bound	-Inf	-Inf	6514.43	6514.43	6514.43	**6718.03**	**6718.03**	**6718.03**
	runtime	7.17	2.32	3.50	2.77	3.01	14.08	21.13	25.48
	memory (GB)	0.01 GB	0.01 GB	0.14 GB	0.14 GB	0.14 GB	0.35 GB	0.36 GB	0.36 GB
101087.bmp	value	3864.10	3864.10	3476.75	3476.75	3476.75	**3476.75**	**3476.75**	**3476.75**
	bound	-Inf	-Inf	3385.10	3385.10	3385.10	**3476.75**	**3476.75**	**3476.75**
	runtime	2.02	0.68	7.19	6.83	6.21	7.70	7.03	7.08
	memory (GB)	0.01 GB	0.01 GB	0.08 GB	0.08 GB	0.08 GB	0.17 GB	0.19 GB	0.19 GB
102061.bmp	value	4256.45	4256.45	4070.68	4070.68	4070.68	**3804.87**	**3804.87**	**3804.87**
	bound	-Inf	-Inf	3721.83	3721.83	3721.83	**3804.87**	**3804.87**	**3804.87**
	runtime	2.31	0.88	1.17	0.98	1.06	12.00	18.16	19.04
	memory (GB)	0.01 GB	0.01 GB	0.08 GB	0.08 GB	0.08 GB	0.20 GB	0.20 GB	0.20 GB
103070.bmp	value	5553.07	5553.07	5369.09	5369.09	5369.09	**5224.05**	**5224.05**	**5224.05**
	bound	-Inf	-Inf	5007.54	5007.54	5007.54	**5224.05**	**5224.05**	**5224.05**
	runtime	3.74	1.23	11.03	7.82	9.05	44.25	74.56	78.95
	memory (GB)	0.01 GB	0.01 GB	0.11 GB	0.11 GB	0.11 GB	0.27 GB	0.28 GB	0.28 GB
105025.bmp	value	7621.13	7618.42	7426.46	7425.20	7425.20	**7216.97**	**7216.97**	**7216.97**
	bound	-Inf	-Inf	7055.05	7055.05	7055.25	**7216.97**	**7216.97**	**7216.97**
	runtime	7.34	2.44	52.28	40.67	39.01	80.03	29.54	32.85
	memory (GB)	0.01 GB	0.01 GB	0.16 GB	0.15 GB	0.15 GB	0.35 GB	0.39 GB	0.39 GB
106024.bmp	value	1993.14	1993.14	2053.21	2053.21	2053.21	**1945.82**	**1945.82**	**1945.82**
	bound	-Inf	-Inf	1903.68	1903.68	1903.68	**1945.82**	**1945.82**	**1945.82**
	runtime	0.61	0.20	0.43	0.35	0.36	3.16	5.11	5.14
	memory (GB)	0.01 GB	0.01 GB	0.05 GB	0.05 GB	0.05 GB	0.10 GB	0.11 GB	0.11 GB
106005.bmp	value	9354.28	9351.23	9087.20	9084.81	9084.81	**8784.54**	**8784.54**	**8784.54**
	bound	-Inf	-Inf	8496.68	8496.68	8496.68	**8784.54**	**8784.54**	**8784.54**
	runtime	12.69	4.75	5.06	3.65	4.42	21.82	18.29	18.44
	memory (GB)	0.01 GB	0.01 GB	0.17 GB	0.17 GB	0.17 GB	0.41 GB	0.43 GB	0.43 GB
108070.bmp	value	10999.22	10997.34	10683.85	10683.85	10683.85	**10376.90**	**10376.90**	**10376.90**
	bound	-Inf	-Inf	10683.85	10683.85	10683.85	**10376.90**	**10376.90**	**10376.90**
	runtime	16.24	5.41	36.68	21.28	23.87	110.39	78.23	74.10
	memory (GB)	0.01 GB	0.01 GB	0.21 GB	0.21 GB	0.21 GB	0.54 GB	0.55 GB	0.55 GB
108082.bmp	value	6697.80	6696.91	6400.58	6399.77	6399.77	**6164.81**	**6164.81**	**6164.81**
	bound	-Inf	-Inf	5941.16	5941.16	5941.45	**6164.81**	**6164.81**	**6164.81**
	runtime	5.68	2.01	11.76	9.85	11.36	23.91	31.62	30.54
	memory (GB)	0.01 GB	0.01 GB	0.13 GB	0.13 GB	0.13 GB	0.30 GB	0.32 GB	0.32 GB
109053.bmp	value	5991.21	5991.21	5942.63	5942.63	5942.63	**5607.71**	**5607.71**	**5607.71**
	bound	-Inf	-Inf	5439.80	5439.80	5439.80	**5607.71**	**5607.71**	**5607.71**
	runtime	3.61	1.27	5.71	4.61	5.01	18.66	17.61	17.61
	memory (GB)	0.01 GB	0.01 GB	0.11 GB	0.12 GB	0.12 GB	0.30 GB	0.30 GB	0.30 GB
110082.bmp	value	6757.99	6756.21	6381.36	6381.36	6381.36	**6161.48**	**6161.48**	**6161.48**
	bound	-Inf	-Inf	5961.30	5961.30	5961.62	**6161.48**	**6161.48**	**6161.48**
	runtime	6.79	2.51	0.94	0.93	1.08	15.21	14.87	18.53
	memory (GB)	0.01 GB	0.01 GB	0.12 GB	0.12 GB	0.12 GB	0.30 GB	0.31 GB	0.29 GB
12084.bmp	value	10317.71	10326.25	9846.82	9846.82	9846.53	**9614.53**	**9614.53**	**9614.53**
	bound	-Inf	-Inf	9372.90	9372.90	9373.23	**9614.53**	**9614.53**	**9614.53**
	runtime	10.67	7.01	4.69	3.61	4.02	27.12	25.48	27.03
	memory (GB)	0.01 GB	0.01 GB	0.16 GB	0.18 GB	0.18 GB	0.40 GB	0.40 GB	0.40 GB
122074.bmp	value	4775.48	4775.48	4728.78	4728.78	4728.78	**4589.79**	**4589.79**	**4589.79**
	bound	-Inf	-Inf	4482.60	4482.60	4482.61	**4589.79**	**4589.79**	**4589.79**
	runtime	2.78	0.84	41.79	25.14	27.51	54.05	74.67	90.17
	memory (GB)	0.01 GB	0.01 GB	0.11 GB	0.10 GB	0.10 GB	0.24 GB	0.26 GB	0.26 GB
126007.bmp	value	3841.21	3841.14	3710.68	3710.68	3710.68	**3613.57**	**3613.57**	**3613.57**
	bound	-Inf	-Inf	3501.20	3501.20	3502.18	**3613.57**	**3613.57**	**3613.57**
	runtime	1.63	0.55	2.07	1.53	1.68	9.17	8.31	8.31
	memory (GB)	0.01 GB	0.01 GB	0.07 GB	0.07 GB	0.07 GB	0.17 GB	0.17 GB	0.17 GB
130026.bmp	value	6802.29	6780.83	6703.11	6757.35	6757.35	**6496.35**	**6496.35**	**6496.35**
	bound	-Inf	-Inf	6321.76	6321.76	6321.76	**6496.35**	**6496.35**	**6496.35**
	runtime	6.83	2.26	100.52	64.24	72.52	143.39	88.36	80.78
	memory (GB)	0.01 GB	0.01 GB	0.15 GB	0.15 GB	0.15 GB	0.35 GB	0.36 GB	0.35 GB
134035.bmp	value	9134.98	9117.69	8956.43	8960.25	8960.25	**8704.63**	**8704.63**	**8704.63**
	bound	-Inf	-Inf	8520.64	8520.64	8520.89	**8704.63**	**8704.63**	**8704.63**
	runtime	14.28	5.10	49.47	41.26	45.38	75.95	29.28	39.68
	memory (GB)	0.01 GB	0.01 GB	0.21 GB	0.21 GB	0.21 GB	0.41 GB	0.43 GB	0.43 GB
14037.bmp	value	1851.05	1848.67	1799.90	1799.90	1800.38	**1744.11**	**1744.11**	**1744.11**
	bound	-Inf	-Inf	1705.79	1705.79	1706.02	**1744.11**	**1744.11**	**1744.11**
	runtime	0.48	0.17	3.42	1.94	2.23	3.30	9.09	8.60
	memory (GB)	0.01 GB	0.01 GB	0.05 GB	0.04 GB	0.04 GB	0.09 GB	0.10 GB	0.09 GB
143060.bmp	value	2601.33	2601.33	2399.47	2399.47	2399.47	**2245.49**	**2245.49**	**2245.49**
	bound	-Inf	-Inf	2170.85	2170.85	2170.85	**2245.49**	**2245.49**	**2245.49**
	runtime	0.63	0.22	1.25	0.87	0.95	2.15	1.06	1.28
	memory (GB)	0.01 GB	0.01 GB	0.05 GB	0.05 GB	0.05 GB	0.11 GB	0.12 GB	0.12 GB
145086.bmp	value	4149.22	4012.57	4057.87	4057.87	4057.86	**4149.22**	**4149.22**	**4149.22**
	bound	-Inf	-Inf	4057.87	4057.87	4057.86	**4149.22**	**4149.22**	**4149.22**
	runtime	3.74	1.35	1.10	0.76	0.88	9.19	12.15	5.15
	memory (GB)	0.01 GB	0.01 GB	0.09 GB	0.09 GB	0.09 GB	0.22 GB	0.24 GB	0.24 GB
147091.bmp	value	5321.15	5321.15	5189.59	5190.72	5190.72	**4965.09**	**4965.09**	**4965.09**
	bound	-Inf	-Inf	4805.19	4805.19	4805.19	**4965.09**	**4965.09**	**4965.09**
	runtime	3.48	1.21	7.35	5.35	5.23	14.72	13.95	14.16
	memory (GB)	0.01 GB	0.01 GB	0.13 GB	0.13 GB	0.10 GB	0.24 GB	0.25 GB	0.25 GB
148026.bmp	value	11352.14	11352.14	10962.36	10962.36	10960.74	**10578.90**	**10578.90**	**10578.90**
	bound	-Inf	-Inf	10962.36	10962.36	10962.36	**10578.90**	**10578.90**	**10578.90**
	runtime	20.07	7.35	4.95	3.61	4.15	28.38	26.19	29.19
	memory (GB)	0.01 GB	0.01 GB	0.21 GB	0.20 GB	0.21 GB	0.52 GB	0.52 GB	0.52 GB
149089.bmp	value	8730.03	8731.11	8526.22	8526.22	8526.22	**8195.57**	**8195.57**	**8195.57**
	bound	-Inf	-Inf	7910.51	7910.51	7910.51	**8195.57**	**8195.57**	**8195.57**
	runtime	12.26	4.23	34.64	22.23	23.85	43.62	21.52	23.42
	memory (GB)	0.01 GB	0.01 GB	0.16 GB	0.16 GB	0.16 GB	0.39 GB	0.43 GB	0.43 GB
150065.bmp	value	6796.27	6794.43	6621.54	6621.54	6621.54	**6421.54**	**6421.54**	**6421.54**
	bound	-Inf	-Inf	6265.73	6265.73	6265.73	**6421.54**	**6421.54**	**6421.54**
	runtime	6.31	2.24	56.33	37.14	40.40	66.04	44.77	38.33
	memory (GB)	0.01 GB	0.01 GB	0.15 GB	0.14 GB	0.14 GB	0.33 GB	0.36 GB	0.36 GB
157055.bmp	value	6375.46	6384.05	6041.59	6041.59	6041.59	**5890.28**	**5890.28**	**5890.28**
	bound	-Inf	-Inf	5761.03	5761.03	5761.20	**5890.28**	**5890.28**	**5890.28**
	runtime	6.64	2.39	2.62	2.54	2.65	10.52	11.15	11.63
	memory (GB)	0.01 GB	0.01 GB	0.12 GB	0.12 GB	0.12 GB	0.29 GB	0.29 GB	0.30 GB
159008.bmp	value	6901.52	6900.83	6400.54	6400.54	6409.09	**6176.19**	**6176.19**	**6176.19**
	bound	-Inf	-Inf	5954.40	5954.40	5954.39	**6176.19**	**6176.19**	**6176.19**
	runtime	6.57	2.39	53.46	42.89	50.28	76.98	78.64	78.10
	memory (GB)	0.01 GB	0.01 GB	0.12 GB	0.12 GB	0.12 GB	0.30 GB	0.32 GB	0.32 GB
160068.bmp	value	4453.72	4453.72	4227.34	4229.47	4229.47	**4130.10**	**4130.10**	**4130.10**
	bound	-Inf	-Inf	4015.72	4015.72	4015.86	**4130.10**	**4130.10**	**4130.10**
	runtime	2.95	1.04	3.06	2.14	2.32	7.31	5.30	5.55
	memory (GB)	0.01 GB	0.01 GB	0.09 GB	0.08 GB	0.09 GB	0.20 GB	0.20 GB	0.20 GB
16077.bmp	value	6219.05	6220.63	5742.43	5742.43	5742.43	**5542.57**	**5542.57**	**5542.57**
	bound	-Inf	-Inf	5363.95	5363.95	5363.95	**5542.57**	**5542.57**	**5542.57**
	runtime	4.62	1.67	5.84	4.74	5.01	12.77	11.52	11.52
	memory (GB)	0.01 GB	0.01 GB	0.11 GB	0.11 GB	0.11 GB	0.27 GB	0.27 GB	0.28 GB
163065.bmp	value	5835.01	5834.91	5846.64	5846.64	5866.08	**5547.78**	**5547.78**	**5547.78**
	bound	-Inf	-Inf	5334.24	5334.24	5334.40	**5547.78**	**5547.78**	**5547.78**
	runtime	4.11	1.44	11.40	8.89	9.09	26.19	18.29	21.78
	memory (GB)	0.01 GB	0.01 GB	0.11 GB	0.11 GB	0.11 GB	0.28 GB	0.28 GB	0.28 GB
167062.bmp	value	1561.26	1561.26	1498.21	1497.46	1497.46	**1468.58**	**1468.58**	**1468.58**
	bound	-Inf	-Inf	1414.58	1414.58	1414.58	**1468.58**	**1468.58**	**1468.58**
	runtime	0.55	0.18	1.56	0.19	0.21	1.15	1.22	1.68
	memory (GB)	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.09 GB	0.09 GB	0.10 GB
167083.bmp	value	10925.98	10927.48	10492.60	10492.60	10492.60	**10154.37**	**10154.37**	**10154.37**
	bound	-Inf	-Inf	9932.48	9932.48	9932.60	**10154.37**	**10154.37**	**10154.37**
	runtime	16.96	6.08	151.42	103.51	103.50	160.52	39.81	30.49
	memory (GB)	0.01 GB	0.01 GB	0.23 GB	0.22 GB	0.22 GB	0.59 GB	0.52 GB	0.52 GB
170057.bmp	value	4216.62	4216.62	4100.49	4100.49	4100.49	**3979.26**	**3979.26**	**3979.26**
	bound	-Inf	-Inf	3888.03	3888.03	3888.20	**3979.26**	**3979.26**	**3979.26**
	runtime	2.29	0.77	29.04	24.79	19.05	36.66	14.29	12.05
	memory (GB)	0.01 GB	0.01 GB	0.10 GB	0.09 GB	0.09 GB	0.19 GB	0.23 GB	0.22 GB

mouse_energy										
	0.00 CDT	0.01 CDT	0.12 CDT	0.13 CDT	0.03 CDT	0.04 CDT	0.11 CDT	0.15 CDT	0.16 CDT	0.17 CDT
treasure	0.06	0.03	0.03	0.12	0.03	0.03	0.03	0.12	0.35	0.37
forest	-Inf	-Inf	2.55	2.73	-Inf	-Inf	-Inf	2.73	3.09	3.15
water	-Inf	-Inf	2.55	2.73	-Inf	-Inf	-Inf	2.73	3.09	3.15
energy

image-seg-3rdorder	ogm-ICM	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
mean runtime	6.30	2.23	32.56	20.85	21.63	46.68	70.54	50.78
best value	0.00	0.00	0.00	0.00	0.00	100.00	99.00	99.00
best bound	0.00	0.00	0.00	0.00	0.00	99.00	97.00	100.00
verified opt	0.00	0.00	0.00	0.00	0.00	100.00	98.00	100.00

Table 13: image-seg

image-seg		ogm-ICM	ogm-KL	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
101065.bmp	value	5846.60	5503.35	5846.60	**5207.50**	**5207.50**	**5207.50**	**5207.50**	**5207.50**	**5207.50**
	bound	-Inf	-Inf	-Inf	**5207.50**	**5207.50**	**5207.50**	**5207.50**	**5207.50**	**5207.50**
	runtime	1.44	0.83	1.53	0.94	0.51	0.34	0.32	0.51	0.72
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB
101087.bmp	value	2990.00	2861.00	2990.00	**2789.00**	**2789.00**	**2789.00**	**2789.00**	**2789.00**	**2789.00**
	bound	-Inf	-Inf	-Inf	2789.68	2789.68	2789.68	**2789.00**	**2789.00**	**2789.00**
	runtime	1.27	0.13	0.44	0.36	0.11	0.12	0.20	0.40	0.42
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
102061.bmp	value	3129.02	3077.77	3129.02	2941.74	2947.74	2947.74	**2943.77**	**2943.77**	**2943.77**
	bound	-Inf	-Inf	-Inf	2943.48	2943.48	2943.48	**2943.77**	**2943.77**	**2943.77**
	runtime	1.50	0.53	0.51	0.64	0.38	0.39	0.50	1.51	1.17
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
103070.bmp	value	4554.03	4442.75	4554.03	**4199.38**	**4199.38**	**4199.38**	**4199.38**	**4199.38**	**4199.38**
	bound	-Inf	-Inf	-Inf	**4199.38**	**4199.38**	**4199.38**	**4199.38**	**4199.38**	**4199.38**
	runtime	2.38	0.89	0.82	6.24	4.03	3.95	1.15	1.15	1.29
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.03 GiB
105025.bmp	value	6345.17	6306.79	6345.17	6073.28	6075.28	6073.28	**6055.33**	**6055.33**	**6055.33**
	bound	-Inf	-Inf	-Inf	6054.14	6054.14	6054.14	**6055.33**	**6055.33**	**6055.33**
	runtime	4.46	1.27	1.55	13.57	6.43	4.44	5.89	3.05	2.97
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.06 GiB	0.03 GiB	0.03 GiB	0.07 GiB	0.05 GiB	0.04 GiB
106024.bmp	value	1872.00	1614.00	1872.00	1599.25	1599.25	1599.25	**1599.25**	**1599.25**	**1599.25**
	bound	-Inf	-Inf	-Inf	1598.60	1598.60	1598.60	**1599.25**	**1599.25**	**1599.25**
	runtime	0.37	0.03	0.13	0.57	0.28	0.29	0.46	0.60	0.40
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.03 GiB	0.02 GiB
108005.bmp	value	6977.02	6912.42	6977.02	6581.96	6581.96	6581.96	**6578.03**	**6578.03**	**6578.03**
	bound	-Inf	-Inf	-Inf	6577.75	6577.75	6577.75	**6578.03**	**6578.03**	**6578.03**
	runtime	1.76	4.78	3.70	6.94	4.41	4.66	4.05	2.12	3.67
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.05 GiB	0.06 GiB	0.04 GiB
108070.bmp	value	8714.08	8636.17	8714.08	8426.08	8426.08	8426.08	**8422.24**	**8422.24**	**8422.24**
	bound	-Inf	-Inf	-Inf	8422.24	8422.24	8422.24	**8422.24**	**8422.24**	**8422.24**
	runtime	10.45	3.76	3.52	59.95	21.70	12.15	12.66	3.73	5.84
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.07 GiB	0.04 GiB	0.04 GiB	0.07 GiB	0.06 GiB	0.05 GiB
108082.bmp	value	5133.62	5037.91	5133.02	**4800.15**	**4800.15**	**4800.15**	**4800.15**	**4800.15**	**4800.15**
	bound	-Inf	-Inf	-Inf	**4800.15**	**4800.15**	**4800.15**	**4800.15**	**4800.15**	**4800.15**
	runtime	3.76	1.68	1.30	6.97	3.48	3.27	3.23	2.88	1.98
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.05 GiB	0.03 GiB
109053.bmp	value	4894.16	4606.11	4894.16	4425.56	4425.56	4425.56	**4421.13**	**4421.13**	**4421.13**
	bound	-Inf	-Inf	-Inf	4421.03	4421.03	4421.03	**4421.13**	**4421.13**	**4421.13**
	runtime	2.52	0.61	0.61	1.58	0.77	0.77	0.77	0.77	0.77
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB	0.03 GiB
115082.bmp	value	4803.34	4705.51	4803.34	**4530.71**	**4530.71**	**4530.71**	**4530.71**	**4530.71**	**4530.71**
	bound	-Inf	-Inf	-Inf	**4530.71**	**4530.71**	**4530.71**	**4530.71**	**4530.71**	**4530.71**
	runtime	4.23	1.34	1.42	0.42	0.09	0.09	0.10	0.13	0.39
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB
12084.bmp	value	7610.99	7456.34	7610.89	7288.30	7288.30	7288.30	**7284.45**	**7284.45**	**7284.45**
	bound	-Inf	-Inf	-Inf	7284.11	7284.11	7284.11	**7284.45**	**7284.45**	**7284.45**
	runtime	10.87	13.94	3.65	7.70	2.68	2.68	2.68	1.15	1.42
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB	0.04 GiB
123074.bmp	value	4133.22	4059.26	4130.47	**3842.74**	**3842.74**	**3842.74**	**3842.74**	**3842.74**	**3842.74**
	bound	-Inf	-Inf	-Inf	**3842.74**	**3842.74**	**3842.74**	**3842.74**	**3842.74**	**3842.74**
	runtime	1.82	0.23	0.59	9.07	5.33	5.28	5.37	4.29	3.14
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.03 GiB	0.03 GiB	0.03 GiB	0.05 GiB	0.03 GiB
126007.bmp	value	2824.58	2701.58	2824.58	2686.11	2686.11	2686.11	**2684.83**	**2684.83**	**2684.83**
	bound	-Inf	-Inf	-Inf	2684.75	2684.75	2684.75	**2684.83**	**2684.83**	**2684.83**
	runtime	2.54	0.47	0.47	0.92	0.41	0.41	0.41	0.55	0.55
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.03 GiB	0.02 GiB
130026.bmp	value	5628.36	5594.57	5628.36	5366.20	5366.20	5366.20	**5350.83**	**5350.83**	**5350.83**
	bound	-Inf	-Inf	-Inf	5349.80	5349.80	5349.80	**5350.83**	**5350.83**	**5350.83**
	runtime	3.86	1.63	1.30	32.39	14.08	14.34	17.89	5.27	4.83
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.07 GiB	0.03 GiB	0.03 GiB	0.08 GiB	0.07 GiB	0.05 GiB
134035.bmp	value	6970.83	6678.92	6969.83	6590.30	6590.30	6590.30	**6578.98**	**6578.98**	**6578.98**
	bound	-Inf	-Inf	-Inf	6577.99	6577.99	6577.99	**6578.98**	**6578.98**	**6578.98**
	runtime	4.01	9.23	3.61	6.22	3.72	3.72	3.72	2.74	3.72
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.03 GiB	0.03 GiB	0.07 GiB	0.05 GiB	0.04 GiB
14037.bmp	value	1499.33	1444.97	1499.33	**1383.14**	**1383.14**	**1383.14**	**1383.14**	**1383.14**	**1383.14**
	bound	-Inf	-Inf	-Inf	**1383.14**	**1383.14**	**1383.14**	**1383.14**	**1383.14**	**1383.14**
	runtime	0.32	0.02	0.11	0.13	0.08	0.08	0.08	0.11	0.06
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
143090.bmp	value	1845.79	1784.06	1845.79	**1714.38**	**1714.38**	**1714.38**	**1714.38**	**1714.38**	**1714.38**
	bound	-Inf	-Inf	-Inf	**1714.38**	**1714.38**	**1714.38**	**1714.38**	**1714.38**	**1714.38**
	runtime	0.84	0.06	0.20	0.23	0.11	0.11	0.11	0.16	0.14
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB
145086.bmp	value	3613.88	3395.28	3611.88	**3322.21**	**3322.21**	**3322.21**	**3322.21**	**3322.21**	**3322.21**
	bound	-Inf	-Inf	-Inf	**3322.21**	**3322.21**	**3322.21**	**3322.21**	**3322.21**	**3322.21**
	runtime	1.96	0.23	0.67	0.21	0.05	0.05	0.05	0.15	0.14
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB
147091.bmp	value	4129.09	4085.05	4134.20	3976.61	3976.61	3976.61	**3973.71**	**3973.71**	**3973.71**
	bound	-Inf	-Inf	-Inf	3973.48	3973.48	3973.48	**3973.71**	**3973.71**	**3973.71**
	runtime	1.57	0.68	0.67	1.99	1.10	1.06	1.23	0.76	1.07
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
148026.bmp	value	8705.09	8463.81	8705.46	8308.02	8309.02	8304.02	**8205.98**	**8205.98**	**8205.98**
	bound	-Inf	-Inf	-Inf	8205.78	8205.78	8205.78	**8205.98**	**8205.98**	**8205.98**
	runtime	11.91	1.80	3.96	1.56	0.57	0.59	6.80	1.04	1.27
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.05 GiB	0.04 GiB
148089.bmp	value	8759.63	8691.00	8759.63	8442.57	8442.57	8442.57	**6439.58**	**6439.58**	**6439.58**
	bound	-Inf	-Inf	-Inf	6439.34	6439.34	6439.34	**6439.58**	**6439.58**	**6439.58**
	runtime	7.18	5.01	7.51	5.00	2.37	2.18	4.37	2.11	1.55
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.05 GiB	0.04 GiB
156065.bmp	value	5472.00	5412.00	5472.00	5254.11	5254.11	5254.11	**5254.15**	**5254.15**	**5254.15**
	bound	-Inf	-Inf	-Inf	5254.11	5254.11	5254.11	**5254.15**	**5254.15**	**5254.15**
	runtime	4.98	2.46	1.34	7.80	4.00	4.10	4.36	4.73	2.95
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.05 GiB	0.03 GiB
157055.bmp	value	4968.82	4798.78	4968.82	4686.20	4686.20	4686.20	**4685.17**	**4685.17**	**4685.17**
	bound	-Inf	-Inf	-Inf	4684.78	4684.78	4684.78	**4685.17**	**4685.17**	**4685.17**
	runtime	4.10	0.68	1.28	0.41	0.14	0.15	0.35	0.51	0.59
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB
159008.bmp	value	4981.27	4783.12	4981.27	4545.66	4545.66	4545.66	**4540.87**	**4540.87**	**4540.87**
	bound	-Inf	-Inf	-Inf	4540.87	4540.87	4540.87	**4540.87**	**4540.87**	**4540.87**
	runtime	5.91	2.29	1.39	1.17	0.52	0.53	0.56	1.41	1.05
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.04 GiB	0.03 GiB
160068.bmp	value	3254.26	3216.91	3254.26	3001.15	3001.15	3001.15	**3089.32**	**3089.32**	**3089.32**
	bound	-Inf	-Inf	-Inf	3089.08	3089.08	3089.08	**3089.32**	**3089.32**	**3089.32**
	runtime	1.70	0.33	0.57	1.14	0.72	0.74	0.98	0.59	0.40
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
19077.bmp	value	4505.41	4364.00	4505.41	4227.88	4227.88	4227.88	**4227.88**	**4227.88**	**4227.88**
	bound	-Inf	-Inf	-Inf	4227.65	4227.65	4227.65	**4227.88**	**4227.88**	**4227.88**
	runtime	3.16	0.76	1.08	1.46	0.61	0.63	0.65	0.55	0.51
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.04 GiB	0.03 GiB
193065.bmp	value	4627.94	4558.56	4627.94	4392.93	4392.93	4392.26	**4381.13**	**4381.13**	**4381.13**
	bound	-Inf	-Inf	-Inf	4380.39	4380.39	4380.41	**4381.13**	**4381.13**	**4381.13**
	runtime	2.86	0.79	1.00	5.80	2.49	3.11	4.09	1.84	1.53
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB	0.03 GiB
197062.bmp	value	1297.98	1274.78	1297.98	1276.13	1276.13	1276.13	**1273.72**	**1273.72**	**1273.72**
	bound	-Inf	-Inf	-Inf	1273.72	1273.72	1273.72	**1273.72**	**1273.72**	**1273.72**
	runtime	0.30	0.05	0.10	0.16	0.05	0.05	0.11	0.08	0.10
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB
197083.bmp	value	8652.97	8572.76	8650.26	8333.88	8333.88	8333.88	**8331.63**	**8331.63**	**8331.63**
	bound	-Inf	-Inf	-Inf	8331.41	8331.41	8331.41	**8331.63**	**8331.63**	**8331.63**
	runtime	10.78	3.71	3.58	21.68	11.48	11.46	11.72	2.71	2.83
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.03 GiB	0.03 GiB	0.06 GiB	0.05 GiB	0.04 GiB
170057.bmp	value	3405.82	3345.02	3405.82	3298.55	3298.55	3298.55	**3266.17**	**3266.17**	**3266.17**
	bound	-Inf	-Inf	-Inf	3266.17	3266.17	3266.17	**3266.17**	**3266.17**	**3266.17**
	runtime	1.52	0.58	0.52	6.52	4.83	4.75	4.04	3.06	4.19
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.05 GiB	0.04 GiB
175032.bmp	value	11925.49	11888.55	11925.49	11577.50	11577.50	11577.50	**11542.63**	**11542.63**	**11542.63**
	bound	-Inf	-Inf	-Inf	11540.02	11540.02	11540.02	**11542.63**	**11542.63**	**11542.63**
	runtime	17.48	4.20	5.89	126.50	76.83	77.29	99.30	41.64	45.41
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.15 GiB	0.06 GiB	0.06 GiB	0.17 GiB	0.14 GiB	0.09 GiB
175043.bmp	value	8389.84	8260.10	8389.84	**7816.92**	**7816.92**	**7816.92**	**7816.92**	**7816.92**	**7816.92**
	bound	-Inf	-Inf	-Inf	**7816.92**	**7816.92**	**7816.92**	**7816.92**	**7816.92**	**7816.92**
	runtime	10.23	3.64	3.61	6.04	2.92	2.92	3.02	1.96	2.68
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.04 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB
182053.bmp	value	3833.66	3751.47	3833.66	3581.74	3581.74	3581.31	**3579.24**	**3579.24**	**3579.24**
	bound	-Inf	-Inf	-Inf	3578.90	3578.90	3579.02	**3579.24**	**3579.24**	**3579.24**
	runtime	2.18	0.47	0.75	2.36	1.45	1.46	1.68	2.74	2.41
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.04 GiB	0.03 GiB
189080.bmp	value	1177.12	1103.64	1177.12	1080.02	1080.02	1080.02	**1077.47**	**1077.47**	**1077.47**
	bound	-Inf	-Inf	-Inf	1077.42	1077.42	1077.42	**1077.47**	**1077.47**	**1077.47**
	runtime	0.30	0.03	0.07	0.09	0.04	0.04	0.06	0.07	0.06
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB
19021.bmp	value	4744.05	4608.54	4744.05	4520.06	4520.06	4517.83	**4515.08**	**4515.08**	**4515.08**
	bound	-Inf	-Inf	-Inf	4514.97	4514.97	4514.98	**4515.08**	**4515.08**	**4515.08**
	runtime	3.30	2.27	1.12	3.54	1.71	1.71	1.82	0.84	1.37
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.04 GiB	0.04 GiB
196073.bmp	value	578.45	534.98	578.45	**545.47**	**545.47**	**545.47**	**545.47**	**545.47**	**545.47**
	bound	-Inf	-Inf	-Inf	**545.47**	**545.47**	**545.47**	**545.47**	**545.47**	**545.47**
	runtime	0.04	0.01	0.00	0.02	0.01	0.01	0.01	0.02	0.02
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
197017.bmp	value	3039.02	2917.21	3039.02	**2798.77**	**2798.77**	**2798.77**	**2798.77**	**2798.77**	**2798.77**
	bound	-Inf	-Inf	-Inf	**2798.77**	**2798.77**	**2798.77**	**2798.77**	**2798.77**	**2798.77**
	runtime	1.47	0.34	0.51	0.40	0.07	0.07	0.07	0.15	0.10
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB
208001.bmp	value	6650.26	6609.41	6650.26	6276.25	6276.25	6276.25	**6272.68**	**6272.68**	**6272.68**
	bound	-Inf	-Inf	-Inf	6272.23	6272.23	6272.23	**6272.68**	**6272.68**	**6272.68**
	runtime	4.79	1.05	1.91	4.19	1.73	1.28	1.72	39.03	3.05
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.03 GiB	0.03 GiB	0.06 GiB	0.05 GiB	0.04 GiB
210088.bmp	value	2083.07	2016.80	2085.07	**1895.44**	**1895.44**	**1895.44**	**1895.44**	**1895.44**	**1895.44**
	bound	-Inf	-Inf	-Inf	**1895.44**	**1895.44**	**1895.44**	**1895.44**	**1895.44**	**1895.44**
	runtime	0.73	0.09	0.26	0.18	0.08	0.08	0.08	0.06	0.13
	memory (GiB)	0.00 GiB	0.00 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
21077.bmp	value	3121.08	3060.34	3121.07	**2946.71**	**2946.71**	**2946.71**	**2946.71**	**2946.71**	**2946.71**
	bound	-Inf	-Inf	-Inf	**2946.71**	**2946.71**	**2946.71**	**2946.71**	**2946.71**	**2946.71**
	runtime	1.71	0.09	0.58	0.19	0.06	0.06	0.06	0.06	0.08
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.02 GiB
216081.bmp	value	4158.73	4158.73	4158.73	4158.73	4158.73	4158.73	**4158.73**	**4158.73**	**4158.73**
	bound	-Inf	-Inf	-Inf	4158.67	4158.67	4158.67	**4158.73**	**4158.73**	**4158.73**
	runtime	3.43	0.56	1.15	0.27	0.10	0.12	0.17	0.46	0.35
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
219090.bmp	value	2598.23	2576.46	2598.23	**2501.27**	**2501.27**	**2501.27**	**2501.27**	**2501.27**	**2501.27**
	bound	-Inf	-Inf	-Inf	**2501.27**	**2501.27**	**2501.27**	**2501.27**	**2501.27**	**2501.27**
	runtime	1.20	0.28	0.42	0.20	0.06	0.06	0.06	0.08	0.07
	memory (GiB)	0.00 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
220075.bmp	value	3272.66	3152.56	3272.66	**3115.95**	**3115.95**	**3115.95**	**3115.95**	**3115.95**	**3115.95**
	bound	-Inf	-Inf	-Inf	**3115.95**	**3115.95**	**3115.95**	**3115.95**	**3115.95**	**3115.95**
	runtime	1.99	0.13	0.68	0.12	0.03	0.03	0.03	0.06	0.15
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
223061.bmp	value	6827.06	6789.08	6827.06	6585.28	6585.28	6585.28	**6576.83**	**6576.83**	**6576.83**
	bound	-Inf	-Inf	-Inf	6575.97	6575.97	6575.97	**6576.83**	**6576.83**	**6576.83**
	runtime	5.23	1.15	1.78	60.19	27.14	27.34	29.53	10.46	7.63
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.09 GiB	0.05 GiB	0.05 GiB	0.10 GiB	0.09 GiB	0.05 GiB
227092.bmp	value	2064.00	2014.00	2064.00	**1998.46**	**1998.46**	**1998.46**	**1998.46**	**1998.46**	**1998.46**
	bound	-Inf	-Inf	-Inf	1997.85	1997.85	1997.85	**1998.46**	**1998.46**	**1998.46**
	runtime	0.48	0.03	0.17	1.24	1.05	1.05	1.51	0.55	0.66
	memory (GiB)	0.00 GiB	0.00 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.03 GiB	0.03 GiB
229036.bmp	value	6406.52	6267.68	6404.33	**6125.73**	**6125.73**	**6125.73**	**6125.73**	**6125.73**	**6125.73**
	bound	-Inf	-Inf	-Inf	**6125.73**	**6125.73**	**6125.73**	**6125.73**	**6125.73**	**6125.73**
	runtime	4.39	1.69	2.14	3.59	1.01	1.74	1.88	0.68	1.28
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB
236037.bmp	value	9566.43	9479.07	9566.43	9072.01	9072.01	9072.01	**9060.84**	**9060.84**	**9060.84**
	bound	-Inf	-Inf	-Inf	9060.84	9060.84	9060.84	**9060.84**	**9060.84**	**9060.84**
	runtime	12.16	5.24	4.13	62.02	16.56	16.01	19.82	9.55	6.69
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.08 GiB	0.03 GiB	0.03 GiB	0.07 GiB	0.07 GiB	0.05 GiB
24077.bmp	value	5000.10	4807.75	5000.10	4766.37	4766.37	4765.36	**4761.98**	**4761.98**	**4761.98**
	bound	-Inf	-Inf	-Inf	4761.87	4761.87	4761.87	**4761.98**	**4761.98**	**4761.98**
	runtime	4.59	0.79	1.58	0.64	0.29	0.32	0.45	0.82	0.91
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB	0.03 GiB	0.04 GiB	0.03 GiB
241004.bmp	value	1215.29	1113.52	1215.29	**1057.14**	**1057.14**	**1057.14**	**1057.14**	**1057.14**	**1057.14**
	bound	-Inf	-Inf	-Inf	**1057.14**	**1057.14**	**1057.14**	**1057.14**	**1057.14**	**1057.14**
	runtime	0.19	0.01	0.07	0.05	0.03	0.03	0.07	0.02	0.03
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB
241048.bmp	value	5132.49	4896.07	5132.49	4740.05	4740.05	4740.05	**4730.95**	**4730.95**	**4730.95**
	bound	-Inf	-Inf	-Inf	4729.82	4729.82	4729.82	**4730.95**	**4730.95**	**4730.95**
	runtime	3.67	0.70	1.17	2.08	1.08	1.15	1.51	1.49	1.03
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.03 GiB	0.02 GiB	0.02 GiB	0.04 GiB	0.04 GiB	0.03 GiB
253027.bmp	value	6971.10	6899.47	6970.90	6609.76	6609.76	6609.76	**6606.62**	**6606.62**	**6606.62**
	bound	-Inf	-Inf	-Inf	6606.62	6606.62	6606.62	**6606.62**	**6606.62**	**6606.62**
	runtime	8.27	2.18	2.76	8.35	3.28	3.28	3.65	2.05	2.52
	memory (GiB)	0.01 GiB	0.01 GiB	0.01 GiB	0.05 GiB	0.02 GiB	0.02 GiB	0.05 GiB	0.07 GiB	0.04 GiB
253055.bmp	value	1905.50	1853.88	1920.50	1503.91	1503.91	1503.91	**1502.16**	**1502.16**	**1502.16**
	bound	-Inf	-Inf	-Inf	1502.04	1502.04	1502.04	**1502.16**	**1502.16**	**1502.16**
	runtime	0.41	0.04	0.15	0.17	0.07	0.07	0.11	0.06	0.17
	memory (GiB)	0.00 GiB	0.00 GiB	0.00 GiB	0.02 GiB	0.01 GiB	0.01 GiB	0.02 GiB	0.02 GiB	0.02 GiB
260058.bmp	value	1195.03	1110.91	1195.03	**1084.26**	**1084.26**	**1084.26**	**1084.26**	**1084.26**	**1084.26**
	bound	-Inf	-Inf	-Inf	**1084.26**	**1084.26**	**1084.26**	**1

image-org		ogm-ICM	ogm-KL	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCFD	MCI-CCI	MCI-CCIFD
33039.bmp	value	8721.65	8582.09	8721.65	8102.26	8102.26	8102.26	**8069.07**	**8069.07**	**8069.07**
	bound	-Inf	-Inf	-Inf	8068.45	8068.45	8068.45	**8069.07**	**8069.07**	**8069.07**
	runtime	11.31	3.32	4.06	15.55	9.57	9.51	10.82	6.80	5.72
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.06 GB	0.03 GB	0.03 GB	0.06 GB	0.06 GB	0.04 GB
351093.bmp	value	6447.23	6337.46	6447.23	6111.09	6111.09	6111.09	**6105.28**	**6105.28**	**6105.28**
	bound	-Inf	-Inf	-Inf	6104.28	6104.28	6104.28	**6105.28**	**6105.28**	**6105.28**
	runtime	5.61	1.97	1.91	7.95	5.41	5.36	5.78	5.26	5.49
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
361010.bmp	value	3532.89	3440.32	3532.89	**3361.02**	**3361.02**	**3361.02**	**3361.02**	**3361.02**	**3361.02**
	bound	-Inf	-Inf	-Inf	**3361.02**	**3361.02**	**3361.02**	**3361.02**	**3361.02**	**3361.02**
	runtime	2.14	0.57	0.74	0.43	0.10	0.10	0.10	0.20	0.27
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
37073.bmp	value	2096.96	2050.17	2099.22	**1975.00**	**1975.00**	**1975.00**	**1975.00**	**1975.00**	**1975.00**
	bound	-Inf	-Inf	-Inf	**1975.00**	**1975.00**	**1975.00**	**1975.00**	**1975.00**	**1975.00**
	runtime	0.64	0.16	0.22	0.02	0.01	0.01	0.01	0.02	0.02
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
376043.bmp	value	6130.66	6099.05	6131.41	**5863.83**	**5863.83**	**5863.83**	**5863.83**	**5863.83**	**5863.83**
	bound	-Inf	-Inf	-Inf	**5863.83**	**5863.83**	**5863.83**	**5863.83**	**5863.83**	**5863.83**
	runtime	4.99	1.12	1.64	12.07	7.34	7.83	7.17	2.29	1.64
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.04 GB
38082.bmp	value	8377.71	8480.50	8377.71	8081.97	8081.97	8081.97	**8060.34**	**8060.34**	**8060.34**
	bound	-Inf	-Inf	-Inf	8060.34	8060.34	8060.34	**8060.34**	**8060.34**	**8060.34**
	runtime	8.84	2.79	3.01	28.83	16.64	17.40	9.68	4.84	4.52
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.09 GB	0.04 GB	0.04 GB	0.07 GB	0.07 GB	0.05 GB
38092.bmp	value	4381.18	4310.52	4381.18	**4071.86**	**4071.86**	**4071.86**	**4071.86**	**4071.86**	**4071.86**
	bound	-Inf	-Inf	-Inf	**4071.86**	**4071.86**	**4071.86**	**4071.86**	**4071.86**	**4071.86**
	runtime	2.54	0.87	0.92	1.91	1.25	1.33	1.22	0.30	0.43
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
385039.bmp	value	3960.00	3870.70	3960.70	3747.90	3747.90	3747.90	**3745.53**	**3745.53**	**3745.53**
	bound	-Inf	-Inf	-Inf	3744.80	3744.80	3744.80	**3745.53**	**3745.53**	**3745.53**
	runtime	2.54	0.61	0.80	1.64	1.02	1.02	0.92	0.25	0.27
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
41033.bmp	value	2193.07	2086.82	2193.07	**1994.24**	**1994.24**	**1994.24**	**1994.24**	**1994.24**	**1994.24**
	bound	-Inf	-Inf	-Inf	**1994.24**	**1994.24**	**1994.24**	**1994.24**	**1994.24**	**1994.24**
	runtime	0.71	0.06	0.24	0.36	0.13	0.13	0.13	0.09	0.26
	memory (GB)	0.00 GB	0.00 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
41069.bmp	value	5315.93	5269.75	5315.93	5114.98	5114.98	5114.98	**5110.96**	**5110.96**	**5110.96**
	bound	-Inf	-Inf	-Inf	5110.96	5110.96	5110.96	**5110.96**	**5110.96**	**5110.96**
	runtime	1.40	0.28	0.37	17.06	1.02	1.74	1.66	14.31	0.29
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.03 GB	0.04 GB	0.06 GB	0.04 GB
42012.bmp	value	3696.78	3485.93	3696.78	3258.85	3258.85	3258.85	**3248.70**	**3248.70**	**3248.70**
	bound	-Inf	-Inf	-Inf	3248.12	3248.12	3248.12	**3248.70**	**3248.70**	**3248.70**
	runtime	1.64	0.17	0.55	1.87	1.35	1.37	1.60	2.78	2.04
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
42049.bmp	value	1149.62	1090.54	1149.62	**1069.22**	**1069.22**	**1069.22**	**1069.22**	**1069.22**	**1069.22**
	bound	-Inf	-Inf	-Inf	**1069.22**	**1069.22**	**1069.22**	**1069.22**	**1069.22**	**1069.22**
	runtime	0.62	0.07	0.09	0.03	0.01	0.01	0.01	0.02	0.02
	memory (GB)	0.00 GB	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB
43074.bmp	value	2399.48	2390.74	2399.48	2343.74	2343.74	2343.74	**2332.83**	**2332.83**	**2332.83**
	bound	-Inf	-Inf	-Inf	2332.48	2332.49	2332.49	**2332.83**	**2332.83**	**2332.83**
	runtime	0.67	0.10	0.22	2.11	1.58	1.60	1.66	0.40	0.38
	memory (GB)	0.00 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.02 GB
45096.bmp	value	1017.58	1008.08	1017.58	**977.78**	**977.78**	**977.78**	**977.78**	**977.78**	**977.78**
	bound	-Inf	-Inf	-Inf	**977.78**	**977.78**	**977.78**	**977.78**	**977.78**	**977.78**
	runtime	0.20	0.02	0.07	0.06	0.02	0.02	0.02	0.03	0.11
	memory (GB)	0.00 GB	0.00 GB	0.00 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB
51082.bmp	value	3940.05	3923.38	3940.05	3798.52	3798.52	3798.52	**3796.36**	**3796.36**	**3796.36**
	bound	-Inf	-Inf	-Inf	3795.85	3795.85	3795.85	**3796.36**	**3796.36**	**3796.36**
	runtime	1.84	0.58	0.61	8.18	3.51	3.75	4.19	0.97	1.54
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.05 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB
52073.bmp	value	8285.39	8300.94	8286.72	7841.24	7841.24	7841.24	**7835.96**	**7835.96**	**7835.96**
	bound	-Inf	-Inf	-Inf	7835.37	7835.37	7835.37	**7835.96**	**7835.96**	**7835.96**
	runtime	5.68	2.42	2.92	32.86	9.56	9.68	5.79	3.01	3.61
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.08 GB	0.03 GB	0.03 GB	0.04 GB	0.06 GB	0.05 GB
58060.bmp	value	10190.8	10124.2	10190.8	**9881.46**	**9881.46**	**9881.46**	**9881.46**	**9881.46**	**9881.46**
	bound	-Inf	-Inf	-Inf	9881.46	9881.46	9881.46	**9881.46**	**9881.46**	**9881.46**
	runtime	14.83	9.33	4.92	18.29	9.89	9.97	10.96	5.12	3.41
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.07 GB	0.04 GB	0.04 GB	0.08 GB	0.07 GB	0.05 GB
62096.bmp	value	3544.29	3473.37	3544.29	**3419.40**	**3419.40**	**3419.40**	**3419.40**	**3419.40**	**3419.40**
	bound	-Inf	-Inf	-Inf	**3419.40**	**3419.40**	**3419.40**	**3419.40**	**3419.40**	**3419.40**
	runtime	1.72	0.81	0.57	2.30	1.48	1.47	1.49	0.42	0.57
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
65033.bmp	value	7926.42	7580.96	7926.42	7365.34	7365.34	7365.34	**7364.57**	**7364.57**	**7364.57**
	bound	-Inf	-Inf	-Inf	7364.57	7364.57	7364.57	**7364.57**	**7364.57**	**7364.57**
	runtime	9.03	2.86	3.03	3.14	1.76	1.77	1.75	1.64	1.31
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.02 GB	0.05 GB	0.05 GB	0.04 GB
66053.bmp	value	4544.94	4524.61	4542.31	4429.11	4429.11	4429.11	**4427.25**	**4427.25**	**4427.25**
	bound	-Inf	-Inf	-Inf	4427.16	4427.16	4427.16	**4427.25**	**4427.25**	**4427.25**
	runtime	3.05	0.54	1.05	3.53	2.72	2.66	2.87	0.69	1.37
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
69015.bmp	value	4284.44	4164.27	4284.44	4024.09	4024.09	4024.09	**4024.45**	**4024.45**	**4024.45**
	bound	-Inf	-Inf	-Inf	4024.09	4024.09	4024.09	**4024.45**	**4024.45**	**4024.45**
	runtime	2.79	0.63	0.93	1.71	0.88	0.90	1.01	1.89	1.31
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.04 GB
69020.bmp	value	5571.99	5453.10	5571.99	5192.68	5192.68	5192.68	**5179.29**	**5179.29**	**5179.29**
	bound	-Inf	-Inf	-Inf	5178.67	5178.67	5178.67	**5179.29**	**5179.29**	**5179.29**
	runtime	3.85	0.93	1.20	8.69	4.49	4.56	4.26	1.60	1.88
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.04 GB
69040.bmp	value	8318.29	8254.71	8318.29	7987.78	7987.78	7987.78	**7974.58**	**7974.58**	**7974.58**
	bound	-Inf	-Inf	-Inf	7974.58	7974.58	7974.58	**7974.58**	**7974.58**	**7974.58**
	runtime	8.38	4.06	2.83	70.55	24.62	23.56	74.97	8.59	8.57
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.10 GB	0.05 GB	0.05 GB	0.10 GB	0.08 GB	0.06 GB
76053.bmp	value	4972.60	4805.56	4972.00	4534.40	4534.40	4533.53	**4514.99**	**4514.99**	**4514.99**
	bound	-Inf	-Inf	-Inf	4512.85	4512.85	4512.85	**4514.99**	**4514.99**	**4514.99**
	runtime	3.22	0.57	1.10	2.81	1.68	1.79	2.11	2.61	1.38
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.05 GB	0.03 GB
78004.bmp	value	3538.94	3412.31	3528.94	3256.19	3256.19	3256.19	**3254.61**	**3254.61**	**3254.61**
	bound	-Inf	-Inf	-Inf	3254.30	3254.30	3254.30	**3254.61**	**3254.61**	**3254.61**
	runtime	2.54	0.57	0.80	1.64	1.02	1.02	0.92	0.25	0.27
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB
80233.bmp	value	4137.15	4123.48	4137.15	4034.70	4034.70	4034.70	4034.70	**4023.38**	**4023.38**
	bound	-Inf	-Inf	-Inf	4023.11	4023.11	4023.11	4023.11	**4023.38**	**4023.38**
	runtime	2.05	0.65	0.69	18.55	8.67	8.51	8.85	5.47	4.82
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.06 GB	0.03 GB	0.03 GB	0.06 GB	0.05 GB	0.04 GB
85048.bmp	value	6388.89	6182.61	6389.64	5857.10	5857.10	5857.10	**5851.38**	**5851.38**	**5851.38**
	bound	-Inf	-Inf	-Inf	5851.38	5851.38	5851.38	**5851.38**	**5851.38**	**5851.38**
	runtime	6.13	2.24	2.16	1.18	0.55	0.58	0.43	0.42	0.58
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.04 GB	0.04 GB	0.03 GB
86000.bmp	value	4897.04	4718.61	4897.04	4637.53	4637.53	4637.53	**4633.86**	**4633.86**	**4633.86**
	bound	-Inf	-Inf	-Inf	4633.52	4633.52	4633.52	**4633.86**	**4633.86**	**4633.86**
	runtime	4.50	0.60	1.52	0.64	0.32	0.33	0.92	0.77	0.91
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
86016.bmp	value	6821.84	6601.14	6821.84	6620.72	6620.72	6620.72	**6618.85**	**6618.85**	**6618.85**
	bound	-Inf	-Inf	-Inf	6618.83	6618.83	6618.83	**6618.85**	**6618.85**	**6618.85**
	runtime	3.51	1.26	1.14	17.50	7.61	7.78	7.81	2.12	2.42
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.05 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.04 GB
86068.bmp	value	5407.69	5324.07	5407.69	**5198.87**	**5198.87**	**5198.87**	**5198.87**	**5198.87**	**5198.87**
	bound	-Inf	-Inf	-Inf	**5198.87**	**5198.87**	**5198.87**	**5198.87**	**5198.87**	**5198.87**
	runtime	3.51	1.26	1.14	17.50	7.61	7.78	7.81	2.12	2.42
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.05 GB	0.03 GB	0.03 GB	0.03 GB	0.05 GB	0.04 GB
87046.bmp	value	4590.21	4557.04	4590.21	**4315.53**	**4315.53**	**4315.53**	**4315.53**	**4315.53**	**4315.53**
	bound	-Inf	-Inf	-Inf	**4315.53**	**4315.53**	**4315.53**	**4315.53**	**4315.53**	**4315.53**
	runtime	1.93	1.09	0.96	4.23	2.09	2.42	1.32	0.34	0.42
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.04 GB	0.03 GB
89072.bmp	value	4339.28	4171.06	4332.96	3935.25	3935.25	3935.25	**3933.75**	**3933.75**	**3933.75**
	bound	-Inf	-Inf	-Inf	3933.32	3933.32	3933.32	**3933.75**	**3933.75**	**3933.75**
	runtime	2.90	1.57	0.99	0.63	0.17	0.18	0.48	0.31	0.47
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
97033.bmp	value	4722.23	4553.50	4722.23	**4320.69**	**4320.69**	**4320.69**	**4320.69**	**4320.69**	**4320.69**
	bound	-Inf	-Inf	-Inf	**4320.69**	**4320.69**	**4320.69**	**4320.69**	**4320.69**	**4320.69**
	runtime	3.16	0.37	0.99	1.25	0.56	0.55	0.76	0.18	0.32
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB
mean energy		4702.27	4608.49	4705.01	4446.69	4446.67	4446.51	4442.64	4442.64	4442.64
mean bound		-Inf	-Inf	-Inf	4442.34	4442.34	4442.34	4442.64	4442.64	4442.64
mean runtime		3.98	1.46	1.35	8.54	4.34	4.34	4.89	2.43	2.24
best value		0.00	0.00	0.00	35.00	35.00	35.00	100.00	100.00	100.00
best bound		0.00	0.00	0.00	35.00	35.00	35.00	100.00	100.00	100.00
verified opt		0.00	0.00	0.00	35.00	35.00	35.00	100.00	100.00	100.00

Table 14: inclusion

inclusion		o-Fusion	ogm-ICM	ogm-LBP-LP1	ogm-LBP-LP2	ogm-LP-1	ogm-LP-2	ogm-LP-3	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDO	MPLP	MPLP-C	ogm-BUNDLE-A	ogm-BUNDLE-A+	ogm-SG-A+1	ogm-BUNDLE-A-	ogm-SG-A-	ogm-LP-LP	MCI-TC-MTC-TCI	BRAOBI-1	BRAOBI-2	BRAOBI-3	ogm-ILP
mosdlib-12-0.0-0.2	value	1595.53	1595.53	1595.53	1496.25	1479.02	1496.25	1472.44	**1415.94**	**1415.94**	1447.25	1447.25	1447.25	1447.25	1418.58	3416.08	3415.89	5427.91	5427.91	1410.07	3854139.92	1416.38	1416.38	1415.12	**1415.94**	1416.38	1416.38	**1415.94**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**1415.94**	**1415.94**	1415.37	1415.37	1415.37	1415.71	1415.71	1415.71	1415.71	1406.09	1415.71	1319.67	1404.16	1319.67	1415.68	1377.72	1415.71	**1415.94**	3600.00	3600.00	**1415.94**
	runtime	0.03	0.01	23.78	29.81	0.01	0.12	0.95	23.63	24.50	23.99	25.09	24.85	22.76	1.19	7.28	3600.14	70.62	75.78	80.16	75.84	76.09	20.15	131.28	3600.01	3600.00	3600.01	7.30
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.06 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB	0.01 GB	0.29 GB	0.51 GB	0.81 GB	0.80 GB	1.79 GB	0.74 GB
mosdlib-12-0.0-0.2	value	1603.85	1594.41	1423.68	1422.94	1594.41	1496.68	1479.54	2423.21	1423.42	1423.38	1423.42	6422.53	3423.03	4423.49	6422.84	3423.03	5436.16	1423.22	22488.86	1423.20	22488.86	7422.02	**1422.89**	inf	1425.39	1425.39	**1422.89**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**1422.89**	1422.79	1422.79	1422.79	1422.79	1422.79	1411.66	1326.11	1411.60	1326.11	1422.79	1374.21	1422.79	1374.21	1422.79	**1422.89**	3600.00	**1422.89**	3600.00	**1422.89**
	runtime	0.03	0.01	23.74	29.75	0.01	0.10	1.19	24.04	24.28	25.49	24.22	2.65	9.02	0.92	3.69	3600.76	13.85	9.36	81.13	80.52	77.00	17.26	76.18	19.01	0.73	3600.01	3600.00	3600.00	4.51
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.06 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.02 GB	0.01 GB	0.29 GB	0.39 GB	0.38 GB	1.44 GB	0.54 GB	0.73 GB	0.73 GB
mosdlib-34-0.0-0.2	value	1596.11	1516.26	**1381.14**	**1381.14**	1516.26	1435.69	1419.62	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	4389.78	**1381.14**	3391.17	25491.03	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	**1381.14**	1295.44	**1381.14**	1372.31	1295.44	1381.10	1380.26	**1381.14**	**1381.14**	**1381.14**	**1381.14**
	runtime	0.03	0.01	1.25	1.29	0.01	0.09	1.32	9.36	4.02	2.42	3.83	0.09	0.75	0.46	1.25	0.46	0.09	0.15	75.75	28.19	61.04	70.20	77.85	7.22	1.86	22.35	3600.01	3600.00	3600.00	2.02
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.02 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.29 GB	0.49 GB	0.39 GB	0.38 GB	1.44 GB	0.54 GB	0.73 GB
mosdlib-34-0.0-0.2	value	1595.12	1595.12	1595.12	1427.32	**1427.17**	1501.88	1514.84	1498.41	1427.61	1427.17	1427.17	1427.17	1426.48	5426.89	5427.96	3427.17	5426.48	4389.78	2432.97	22502.90	1427.27	4443.81	1426.34	1426.28	1426.28	1426.28	**1427.17**
	bound	-Inf	-Inf	-Inf	-Inf	**1427.17**	-Inf	-Inf	-Inf	1426.58	1426.58	1426.58	1426.58	1426.58	1426.58	1426.58	1426.58	1416.80	1330.85	1426.57	1416.80	1330.85	1426.57	1406.73	1426.58	**1427.17**	**1427.17**	**1427.17**	**1427.17**
	runtime	0.03	0.01	22.65	29.75	0.01	0.08	1.21	24.34	24.17	24.45	24.31	2.64	9.30	1.59	3.70	3704.64	80.33	76.04	76.95	79.42	75.98	18.15	79.42	75.98	3600.01	3600.00	3600.01	2.04
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.06 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.29 GB	0.38 GB	0.32 GB	0.33 GB	1.40 GB	0.72 GB	0.73 GB
mosdlib-50-0.0-0.2	value	1588.33	1571.96	1402.45	1402.45	1571.96	1478.84	1455.31	2402.49	2402.41	4402.16	2402.30	3402.15	2402.62	2402.17	2402.60	5401.70	3406.27	1402.96	34489.58	3413.11	34489.58	3402.44	**1402.34**	inf	1402.78	1402.78	**1402.34**	**1383.69**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1402.02	1401.77	1402.01	1399.05	1399.05	1399.05	1399.05	1399.05	1315.05	1401.94	1315.05	1401.80	1393.06	1402.01	**1402.34**	1384.15	1384.15	1384.15	**1383.69**
	runtime	0.03	0.01	23.96	24.00	0.01	0.09	1.31	23.89	24.13	24.09	25.27	24.86	22.77	2.66	13.17	4800.28	81.07	76.68	77.62	80.98	83.04	71.11	77.02	20.64	129.20	3600.01	3600.00	3600.00	4.22
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.06 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.29 GB	0.50 GB	0.40 GB	0.39 GB	1.43 GB	0.72 GB	0.73 GB
mosdlib-7-0.0-0.2	value	1583.33	1579.17	1403.25	**1403.25**	1579.17	1481.42	1461.33	1403.25	1403.25	4403.25	1403.25	1404.61	1403.25	1403.58	1403.25	1403.25	3406.27	1402.96	20405.17	1402.96	20405.17	1402.96	**1403.25**	inf	1404.29	1404.29	**1403.25**	**1383.33**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1399.87	1310.40	1399.87	1310.40	1399.87	1358.17	**1403.25**	**1403.25**	1383.33	1383.33	1383.33	**1383.33**
	runtime	0.03	0.01	23.80	23.84	0.01	0.09	1.11	23.92	24.88	24.17	25.27	24.91	22.69	2.62	13.21	4332.48	71.53	77.27	81.20	77.46	81.55	17.94	80.36	81.58	34.63	3600.00	3600.01	6.60	3.46
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.07 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.29 GB	0.50 GB	0.29 GB	0.28 GB	1.42 GB	0.74 GB	0.73 GB
mosdlib-8-0.0-0.2	value	1571.67	1551.35	**1368.33**	**1368.33**	1551.35	1442.26	1427.29	**1368.33**	**1368.33**	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	3368.33	inf	3369.53	3369.53	**1368.33**	**1368.33**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1368.15	1368.29	1368.33	1368.23	1368.29	1289.37	1359.76	1289.37	1368.24	1368.24	1368.29	**1368.33**	-Inf	-Inf	**1368.33**
	runtime	0.05	0.01	23.76	24.12	0.01	0.10	1.17	23.30	24.06	24.37	24.78	25.37	22.64	2.67	13.07	3.21	3600.00	77.68	80.61	81.05	81.27	75.53	75.56	75.47	17.47	15.23	3600.01	3600.01	5.00	3.54
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.02 GB	0.06 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.02 GB	0.01 GB	0.29 GB	0.48 GB	0.47 GB	0.38 GB	1.41 GB	0.73 GB	0.73 GB
mosdlib-9-0.0-0.2	value	1587.13	1556.20	1384.66	1384.66	1556.20	1426.39	1461.23	1384.66	1384.66	3384.66

Table 18: knott-3d-300

knott-3d-300		ogm-ICM	ogm-KL	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
gm_knott_3d_072	value	-29971.07	-30354.28	-29972.56	-32872.18	-32897.47	**-32900.85**	**-32900.85**	**-32900.85**	**-32900.85**
	bound	-Inf	-Inf	-Inf	-33032.20	-33000.58	**-32999.85**	**-32999.85**	**-32999.85**	**-32999.85**
	runtime	15.20	73.29	156.01	2691.50	22282.37	3546.06	1916.31	3616.15	3619.70
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.14 GB	0.33 GB	0.32 GB	0.47 GB	0.46 GB	0.46 GB
gm_knott_3d_073	value	-23706.70	-24308.30	-24135.32	-24346.21	-25849.06	**-25863.38**	**-25863.38**	**-25863.38**	**-25863.38**
	bound	-Inf	-Inf	-Inf	-25962.73	-25864.27	**-25863.38**	**-25863.38**	**-25863.38**	**-25863.38**
	runtime	84.70	13.99	30.47	3600.29	985.07	1084.76	858.31	1.46	1.46
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.15 GB	0.15 GB	0.37 GB	0.37 GB	0.37 GB
gm_knott_3d_074	value	-26867.24	-27493.46	-26867.24	-27007.83	-27570.87	**-27721.90**	**-27721.90**	**-27721.90**	**-27721.90**
	bound	-Inf	-Inf	-Inf	-27087.83	-25735.87	-Inf	-25735.87	**-27721.90**	**-27721.90**
	runtime	102.03	12.90	34.62	3600.28	3600.18	3600.24	3600.22	24.80	59.57
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.22 GB	0.14 GB
gm_knott_3d_075	value	-27756.49	-28813.15	-27756.49	**-30478.37**	**-30478.37**	**-30478.37**	**-30478.37**	**-30478.37**	**-30478.37**
	bound	-Inf	-Inf	-Inf	**-30478.37**	**-30478.37**	**-30478.37**	**-30478.37**	-30496.28	**-30478.37**
	runtime	96.22	13.44	32.95	2328.37	372.69	387.31	308.91	7.81	39.59
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.12 GB
gm_knott_3d_076	value	-25180.51	-25248.64	-25174.82	-27024.74	-27024.74	**-27056.99**	**-27056.99**	**-27056.99**	**-27056.99**
	bound	-Inf	-Inf	-Inf	-27056.99	-27056.99	**-27056.99**	**-27056.99**	**-27056.99**	**-27056.99**
	runtime	73.15	9.67	24.39	3439.26	692.79	690.81	378.29	20.09	48.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.57 GB	0.09 GB	0.09 GB	0.22 GB	0.20 GB	0.12 GB
gm_knott_3d_077	value	-27542.11	-27819.39	-27528.90	-29389.31	-29481.84	**-29482.24**	**-29482.24**	**-29482.24**	**-29482.24**
	bound	-Inf	-Inf	-Inf	-29513.05	-29482.26	**-29482.24**	**-29482.24**	**-29482.24**	**-29482.24**
	runtime	82.52	17.43	27.93	3600.27	402.34	532.80	406.06	20.10	50.07
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.10 GB	0.10 GB	0.21 GB	0.22 GB	0.13 GB
gm_knott_3d_078	value	-19040.24	-19114.57	-19067.47	-19794.84	-20228.71	-20052.31	-20241.86	**-20211.55**	**-20211.55**
	bound	-Inf	-Inf	-Inf	-19794.84	-20228.71	-20052.31	-20241.86	**-20211.55**	**-20211.55**
	runtime	101.34	13.40	35.53	3600.17	3600.09	3600.05	3600.02	1487.19	478.19
	memory (GB)	0.01 GB	0.01 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.28 GB	0.16 GB
gm_knott_3d_079	value	-24147.34	-24683.01	-24147.34	-26520.59	-26578.44	**-26607.98**	**-26607.98**	**-26607.98**	**-26607.98**
	bound	-Inf	-Inf	-Inf	-26767.43	-26609.83	**-26607.98**	**-26607.98**	**-26607.98**	**-26607.98**
	runtime	60.45	10.14	20.73	3600.32	896.52	848.65	647.84	128.55	72.10
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.13 GB	0.13 GB	0.34 GB	0.25 GB	0.13 GB
mean energy		-25196.51	-25556.93	-25243.76	-26786.99	-27276.38	-27287.49	-27280.79	-27302.78	-27302.78
mean bound		-Inf	-Inf	-Inf	-27141.36	-27107.22	-27100.02	-27300.19	-27305.02	-27302.78
mean runtime		84.37	13.16	29.68	3423.65	1338.99	1367.03	1261.99	220.30	104.55
best value		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
best bound		0.00	0.00	0.00	12.50	12.50	75.00	75.00	87.50	100.00
verified opt		0.00	0.00	0.00	12.50	12.50	75.00	75.00	87.50	100.00

Table 19: knott-3d-450

knott-3d-450		ogm-ICM	ogm-KL	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
gm_knott_3d_096	value	-82367.79	-83436.56	-82369.50	-85024.54	-85024.54	-85024.54	-85024.54	**-89959.41**	**-89959.41**
	bound	-Inf	-Inf	-Inf	-94633.08	-94633.08	-94633.08	-94633.08	**-89959.41**	**-89959.41**
	runtime	814.51	144.49	281.09	9081.37	5297.25	5543.17	5486.84	399.33	500.23
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.34 GB	0.17 GB	0.17 GB	0.17 GB	0.79 GB	0.44 GB
gm_knott_3d_097	value	-80807.14	-69302.32	-80807.14	-4626.25	-4626.25	-4626.25	-4626.25	**-73477.55**	**-73477.55**
	bound	-Inf	-Inf	-Inf	-78026.62	-78026.62	-78026.62	-78026.62	**-73477.55**	**-73477.55**
	runtime	939.68	179.05	301.59	9744.11	7460.86	7455.94	851.56	472.64	477.53
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.38 GB	0.18 GB	0.18 GB	0.34 GB	0.69 GB	0.43 GB
gm_knott_3d_098	value	-81022.57	-81676.37	-81022.57	-5807.00	-5807.00	-5807.00	-5807.00	**-86593.97**	**-86593.97**
	bound	-Inf	-Inf	-Inf	-91051.86	-91051.86	-91051.86	-91051.86	**-86593.97**	**-86593.97**
	runtime	857.08	170.94	292.12	8452.81	5158.60	5265.15	3217.56	497.05	916.37
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.34 GB	0.17 GB	0.17 GB	0.17 GB	0.80 GB	0.46 GB
gm_knott_3d_099	value	-90622.15	-81173.62	-90622.15	-5800.63	-5800.63	-5800.63	-5800.63	**-85808.57**	**-85808.57**
	bound	-Inf	-Inf	-Inf	-91020.66	-91020.66	-91020.66	-91020.66	**-86421.04**	**-86381.14**
	runtime	982.61	144.37	327.21	9744.04	6977.74	6977.74	3600.05	1487.19	478.19
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.34 GB	0.17 GB	0.18 GB	0.17 GB	0.77 GB	0.46 GB
gm_knott_3d_100	value	-70029.23	-69788.31	-70009.39	-4148.97	-4148.97	-4148.97	-4148.97	**-76699.37**	**-76699.37**
	bound	-Inf	-Inf	-Inf	-81763.95	-81763.95	-81763.95	-81763.95	**-76699.37**	**-76699.37**
	runtime	811.50	176.43	274.46	8070.32	5251.77	5273.36	3219.51	407.26	484.70
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.32 GB	0.17 GB	0.17 GB	0.17 GB	0.79 GB	0.44 GB
gm_knott_3d_101	value	-68581.62	-69409.01	-68581.62	-4104.14	-4104.14	-4104.14	-4104.14	**-74529.51**	**-74529.51**
	bound	-Inf	-Inf	-Inf	-79463.73	-79463.73	-79463.73	-79463.73	**-74529.51**	**-74529.51**
	runtime	944.63	197.53	317.99	10864.02	7383.78	7390.57	7304.28	451.31	617.73
	memory (GB)	0.04 GB	0.03 GB	0.04 GB	0.34 GB	0.17 GB	0.17 GB	0.17 GB	0.79 GB	0.44 GB
gm_knott_3d_102	value	-66622.60	-61743.77	-66761.21	-3926.11	-3926.11	-3926.11	-3926.11	**-66482.68**	**-66482.68**
	bound	-Inf	-Inf	-Inf	-71160.72	-71160.72	-71160.72	-71160.72	**-66482.68**	**-66482.68**
	runtime	850.08	192.10	297.73	10359.02	7392.13	7439.46	7285.17	311.22	889.78
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.36 GB	0.20 GB	0.19 GB	0.19 GB	0.76 GB	0.47 GB
gm_knott_3d_103	value	-68403.19	-68962.60	-68403.19	-4221.22	-4221.22	-4221.22	-4221.22	-71431.63	**-71790.55**
	bound	-Inf	-Inf	-Inf	-79062.14	-79062.14	-79062.14	-79062.14	-73394.87	**-73934.38**
	runtime	869.31	196.54	294.45	10182.73	7162.02	7164.09	7048.99	3968.80	3604.91
	memory (GB)	0.03 GB	0.03 GB	0.04 GB	0.36 GB	0.18 GB	0.18 GB	0.18 GB	0.83 GB	0.54 GB
mean energy		-77183.58	-73777.24	-77183.58	-40272.85	-40272.85	-40272.85	-79118.55	-79207.05	-79207.05
mean bound		-Inf	-Inf	-Inf	-84272.85	-84272.85	-84272.85	-79118.55	-79207.05	-79207.05
mean runtime		883.63	186.89	298.07	9814.45	6493.34	6455.21	6404.14	1196.62	1379.90
best value		0.00	0.00	0.00	0.00	0.00	0.00	0.00	87.50	87.50
best bound		0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.00	100.00
verified opt		0.00	0.00	0.00	0.00	0.00	0.00	0.00	75.00	75.00

Table 20: matching

matching		o-Fusion	ogm-SCM	ogm-LBP-LF1	ogm-LBP-LF2	ogm-LF-1	ogm-LF-2	ogm-LP-3	ogm-TRWS-LF1	ogm-TRWS-LF2	BPS-TAB	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDD	MPLP	MPLP-C	ogm-ADSAL	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SG-A	ogm-BUNDLE-A+	ogm-SG-A+	ogm-BUNDLE-A-ccp	ogm-SG-A-ccp	ogm-LP-LF	TRWS-TAB	BIAGSB-1	BIAGSB-2	BIAGSB-3	ADDD-ZH	ogm-ASTAR	ogm-ConstrLP	ogm-ILP
matching9	value	113000000000000.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	11.02	-Inf	11.02	-Inf	-Inf	-Inf	11.56	10.96	**19.36**	9.95	11.27	11.37	9.17	10.39	9.17	11.37	9.17	11.56	11.02	**19.36**	-Inf	-Inf	-Inf	-Inf	**19.36**	**19.36**	**19.36**	**19.36**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	11.02	-Inf	11.02	-Inf	-Inf	-Inf	11.56	10.96	**19.36**	9.95	11.27	11.37	9.17	10.39	9.17	11.37	9.17	11.56	11.02	**19.36**	**19.36**	**19.36**	**19.36**	**19.36**	**19.36**	**19.36**	**19.36**
	runtime	0.00	0.00	0.00	0.00	0.08	0.00	12.12	12.09	0.02	0.18	0.18	0.02	0.02	0.96	0.96	0.96	1.05	0.96	0.12	1.42	1.05	0.96	0.12	1.42	1.05	0.96	0.02	0.37	7.50	30.28	1.84	0.43	123.51	1039.99	0.02	0.37
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.05 GB	0.03 GB	0.03 GB	0.03 GB	0.32 GB	0.01 GB	0.03 GB	0.19 GB	0.11 GB
matching1	value	171000000000000.00	-Inf	131.43	142.90	**23.58**	131.43	41.02	41.02	37.05	**23.58**	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	**23.58**	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	170000000000.50	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**
	bound	-Inf	-Inf	-Inf	-Inf	**23.58**	-Inf	-Inf	41.02	-Inf	**23.58**	-Inf	-Inf	-Inf	-Inf	-Inf	**23.58**	-Inf	-Inf	-Inf	**23.58**	-Inf	-Inf	-Inf	**23.58**	-Inf	-Inf	-Inf	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**	**23.58**
	runtime	0.00	0.00	0.00	0.00	0.24	0.00	0.48	12.09	0.02	0.18	0.18	0.02	0.02	0.96	0.96	0.96	1.05	0.96	0.12	1.42	1.05	0.96	0.12	1.42	1.05	0.96	0.02	0.37	7.50	30.28	1.84	0.43	123.51	1039.99	0.02	0.37
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.05 GB	0.03 GB	0.03 GB	0.03 GB	0.32 GB	0.01 GB	0.03 GB	0.19 GB	0.11 GB
matching2	value	190000000000000.00	164.21	132.20	59.42	164.21	56.86	56.86	51.59	36.42	59.82	110000000096.00	110000000096.00	110000000096.00	110000000096.00	110000000096.00	110000000096.00	110000000096.00	20000000020.59	30000000025.18	**26.08**	26.57	20000000043.93	10000000040.64	20000000094.35	20000000043.93	10000000029.63	20000000033.93	30000000197.56	20000000026.59	51.59	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	21.18	-Inf	21.18	-Inf	-Inf	-Inf	22.97	-Inf	21.07	**26.08**	22.91	19.87	22.09	8.06	19.87	8.06	19.87	22.99	21.18	-Inf	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**	**26.08**
	runtime	0.00	0.00	0.00	0.31	0.00	0.46	18.89	0.06	0.46	0.15	0.00	0.00	0.00	0.00	0.00	0.00	0.49	4.88	372.97	2.26	2.28	2.10	2.26	1.99	2.26	1.04	35.18	0.06	3.20	9.21	63.91	1538.68	1.73	85.81	73.94	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB	0.01 GB	0.06 GB	0.06 GB	0.81 GB	0.02 GB	0.65 GB	0.17 GB	0.17 GB
matching3	value	112000000000000.00	39.10	42.68	36.44	39.10	28.07	28.07	32.06	25.60	26.98	80000000066.03	80000000066.03	80000000066.03	80000000066.03	80000000066.03	80000000066.03	80000000066.03	13000000053.70	90000000051.81	**15.86**	30.67	10000000042.92	10000000055.67	81.55	10000000042.92	90.94	10000000042.92	79.61	17000000087.82	41.92	**15.86**	**15.86**	**15.86**	150000000049.50	**15.86**	**15.86**	**15.86**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	15.22	-Inf	15.22	-Inf	-Inf	-Inf	-Inf	-Inf	16.35	15.16	21.22	15.62	14.47	15.91	9.27	14.25	9.27	14.49	9.27	16.35	15.22	-Inf	-Inf	-Inf	-Inf	21.22	21.22	21.22
	runtime	33317.60	33317.60	33317.60	31323.23	33317.60	65189.74	65189.74	64997.24	33317.60	33317.60	0.00	0.00	0.00	0.00	0.00	0.00	0.26	15.11	1380.53	1.56	1.57	1.57	1.38	1.56	1.57	1.57	1.32	21.38	0.03	2.05	8.00	41.69	1192.02	0.80	313.52	302.00
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB
mean energy		146500000000000.00	35.73	85.37	38.07	35.73	40.79	28.81	31.46	33.37	97500000044.52	97500000044.52	97500000044.52	97500000044.52	97500000044.52	97500000044.52	97500000044.52	10500000039.23	32500000039.98	12.27	32.47	10000000041.82	7500000053.90	75000000059.44	10000000041.72	75000000059.46	10000000048.72	75000000096.76	16.38	21.22	21.22	21.22	21.22	37500000029.63	21.22	21.22	21.22
mean bound		-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	15.22	15.22	-Inf	15.22	-Inf	-Inf	-Inf	-Inf	-Inf	16.35	15.16	21.22	15.62	14.47	15.91	9.27	14.25	9.27	14.49	9.27	16.35	15.22	-Inf	-Inf	-Inf	-Inf	21.22	21.22	21.22
mean runtime		0.00	0.00	0.00	0.19	0.00	0.29	12.35	0.03	0.32	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.26	15.11	1380.53	1.56	1.57	1.57	1.38	1.56	1.57	1.57	1.32	21.38	0.03	2.05	8.00	41.69	1192.02	0.80	313.52	302.00
best value		0.00	0.00	0.00	25.00	0.00	0.00	0.00	0.00	25.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	100.00	100.00
best bound		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	100.00	100.00
verified opt		0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00	100.00	100.00	100.00

Table 21: modularity-clustering

modularity-clustering		ogm-ICM	ogm-KL	ogm-LP-1	MCR-CC	MCR-CCFDB	MCR-CCFDB-OWC	MCI-CCFDB-CCIFD	MCI-CCI	MCI-CCIFD
adjnoun	value	0.0000	**-0.2980**	0.0000	-0.1273	-0.1273	-0.1112	0.0000	0.0000	0.0000
	bound	-Inf	-Inf	-Inf	-0.4113	-0.4113	**-0.3368**	-0.3727	-0.4374	-0.4096
	runtime	0.16	0.01	0.05	246.57	8.44	3660.51	3601.32	3601.25	3601.25
	memory (GB)	0.00 GB	0.01 GB	0.00 GB	0.02 GB	0.02 GB	0.02 GB	0.03 GB	0.03 GB	0.03 GB
dolphins	value	0.0000	-0.5268	0.0000	-0.5197	-0.5197	**-0.5267**	**-0.5285**	**-0.5285**	**-0.5285**
	bound	-Inf	-Inf	-Inf	-0.5315	-0.5315	**-0.5285**	**-0.5285**	**-0.5285**	**-0.5285**
	runtime	0.03	0.00	0.01	2.06	0.17	1.04	0.27	14.92	8.23
	memory (GB)	0.00 GB	0.00 GB	0.00 GB	0.03 GB	0.02 GB	0.02 GB	0.03 GB	0.04 GB	0.03 GB
football	value	0.0000	**-0.6046**	0.0000	-0.5878	-0.5878	**-0.6046**	**-0.6046**	**-0.6046**	**-0.6046**
	bound	-Inf	-Inf	-Inf	-0.6056	-0.6056	**-0.6046**	**-0.6046**	**-0.6046**	**-0.6046**
	runtime	0.18	0.01	0.06	158.57	0.95	6.99	2.08	12.38	12.89
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.14 GB	0.03 GB	0.03 GB	0.06 GB	0.13 GB	0.07 GB
karate	value	0.0000	**-0.4198**	0.0000	-0.3815	-0.3815	**-0.4198**	**-0.4198**	**-0.4198**	**-0.4198**
	bound	-Inf	-Inf	-Inf	**-0.4198**	**-0.4198**	**-0.4198**	**-0.4198**	**-0.4198**	**-0.4198**
	runtime	0.00	0.00	0.00	0.06	0.01	0.01	0.01	0.13	0.05
	memory (GB)	0.00 GB	0.00 GB	0.00 GB	0.02 GB	0.02 GB	0.02 GB	0.01 GB	0.02 GB	0.02 GB
lesmis	value	0.0000	-0.5443	0.0000	-0.5549	-0.5549	**-0.5600**	**-0.5600**	**-0.5600**	**-0.5600**
	bound	-Inf	-Inf	-Inf	-0.5609	-0.5609	**-0.5600**	**-0.5600**	**-0.5600**	**-0.5600**
	runtime	0.05	0.00	0.02	1.05	0.15	1.01	0.21	2.62	0.55
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.02 GB	0.02 GB	0.03 GB	0.06 GB	0.03 GB
polbooks	value	0.0000	-0.5226	0.0000	-0.5166	-0.5166	**-0.5272**	**-0.5272**	-0.4743	-0.5266
	bound	-Inf	-Inf	-Inf	-0.5309	-0.5309	**-0.5272**	**-0.5272**	**-0.5272**	**-0.5272**
	runtime	0.13	0.00	0.04	193.02	3.16	7.64	2.06	3601.13	3601.22
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.10 GB	0.03 GB	0.03 GB	0.05 GB	2.66 GB	2.11 GB
mean energy		0.0000	-0.4660	0.0000	-0.4619	-0.4643	-0.4582	-0.4613	-0.4312	-0.4399
mean bound		-Inf	-Inf	-Inf	-0.5094	-0.5094	-0.4962	-0.5021	-0.5158	-0.5176
mean runtime		0.09	0.01	0.03	190.37	2.15	692.75	601.38	1207.07	1204.03
best value		0.00	50.00	0.00	33.33	16.67	83.33	83.33	66.67	66.67
best bound		0.00	0.00	0.00	16.67	16.67	100.00	83.33	66.67	66.67
verified opt		0.00	0.00	0.00	16.67	16.67	83.33	83.33	66.67	66.67

Table 22: mrf-inpainting

mrf-inpainting		FastPD	mrf-o-Exp-TAB	mrf-o-Exp-TwL	mrf-o-Swap-TAB	mrf-o-Swap-TwL	mrf-PassFD-LF2	mrf-PassFD-LF2	ogm-ICM	ogm-LF-1	ogm-LF-2	ogm-SRMP	ogm-BPS	ogm-BPS-TwL	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ogm-TRWS-TAB	ogm-ADSAL	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SG-A	ogm-BUNDLE-A+	ogm-SG-A+	ogm-BUNDLE-A-ccp	ogm-SG-A-ccp	ogm-ConstrLP
house-gm	value	4932710.00	3860233.00	3760630.00	3760630.00	-Inf	-Inf	-Inf	3760630.00	-Inf	-Inf	3760630.00	-Inf	-Inf	3760630.00	-Inf	-Inf	3760630.00	-Inf	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00	3760630.00
	bound	0.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**37560520.00**	-Inf	-Inf	**37560520.00**	-Inf	-Inf	**37560520.00**	-Inf	-Inf	**37560520.00**	-Inf	**37560520.00**	37563556.37	34486586.06	34066514.56	34936077.92	34504335.35	33314360.50	34362611.31	34310743.44	33756514.13
	runtime	15.20	73.29	156.01	2691.50	22282.37	3546.06	1916.31	3616.15	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3616.15	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70	3619.70
	memory (GB)	0.19 GB	0.30 GB	0.33 GB	4.14 GB	4.15 GB	0.47 GB	0.29 GB	0.31 GB	0.92 GB	0.92 GB	0.47 GB	0.63 GB	0.82 GB	0.47 GB	0.47 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB	2.97 GB
penguin-gm	value	1655.00	1550291.00	1550291.00	1550291.00	-Inf	-Inf	-Inf	1550291.00	-Inf	-Inf	1550291.00	-Inf	-Inf	1550291.00	-Inf	-Inf	1550291.00	-Inf	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00	1550291.00
	bound	0.00	-Inf	-Inf	-2815355187.50	-2815355187.50	-Inf	-Inf	1534893181.12	-Inf	-Inf	1534893181.12	-Inf	-Inf	1534893181.12	-Inf	-Inf	**1534893181.12**	-Inf	**1534893181.12**	1535072243.61	1449079.15	1437602.24	1450021.83	1449079.15	1364648.50	1449079.15	1364648.50	1534329445.65
	runtime	2.25	10.84	26.12	78.58	2246.09	101.06	35.12	3600.14	360.63	1223.75	294.48	347.00	3619.79	3630.26	3616.09	3618.00	3622.23	3629.46	304.35	3616.18	3606.27	361.12	3605.31	3607.24	3616.30	3613.45	3692.07	12823.52
	memory (GB)	1.37 GB	0.10 GB	0.14 GB	1.38 GB	1.38 GB	0.12 GB	0.10 GB	0.10 GB	0.10 GB	0.21 GB	0.28 GB	0.99 GB	0.99 GB	0.99 GB	0.99 GB	0.99 GB	0.99 GB	0.99 GB	0.99 GB	0.28 GB	1.16 GB	0.99 GB	0.91 GB	0.27 GB	0.92 GB	0.27 GB	0.90 GB	0.27 GB	0.83 GB
mean energy		32939340.00	27266166.50	27055552.00	27559127.00	27559149.50	32467486.00	32467900.00	31188870.50	29461015.00	29463929.00

protein-folding		ogm-ICM	ogm-LBP-LF1	ogm-LBP-LF2	ogm-LF-1	ogm-LF-2	ogm-LF-3	ogm-TRWS-LF1	ogm-TRWS-LF2	BPS-TAB	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDO	MPLP	MPLP-C	ogm-ADSAL	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SG-A	ogm-BUNDLE-A+	ogm-SG-A+	ogm-BUNDLE-A-	ogm-SG-A-	LBP-LF	TRWS-TAB	BRAOBB-1	BRAOBB-2	BRAOBB-3	ADSD-BB	ogm-CombiLP	ogm-ILP
2FOT	value	-11107.30	-11869.77	-11926.85	-11107.30	-11704.17	-11706.45	-11842.47	-11883.06	-11888.96	-11888.96	-11326.75	-11318.86	-11343.62	-11696.35	-10733.78	-9115.92	-11282.40	-11107.05	-11894.35	-11010.16	-11446.37	-11013.61	-11362.41	-11085.01	-10993.42	-10926.45	-11842.46	NaN	NaN	NaN	NaN	NaN	**-11941.22**	
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-12079.69	-12079.69	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-21048.40	-12083.10	-11951.33	-Inf	-Inf	-10183.85	-12027.35	-11344.16	-12063.62	-11974.35	-11978.29	-11616.91	-12070.68	NaN	NaN	NaN	NaN	NaN	**-11941.22**	
	runtime	0.15	212.83	292.32	0.08	118.52	34159.91	41.05	66.57	57.88	263.74	217.92	229.99	210.48	211.20	209.48	381.81	1190.79	3728.02	779.00	814.22	798.62	827.95	798.99	771.71	797.84	824.95	59.70	NaN	NaN	NaN	NaN	NaN	803.94	
	memory (GB)	0.13 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.53 GB	0.37 GB	7.06 GB	0.14 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	0.15 GB	NaN GB	NaN GB	NaN GB	NaN GB	NaN GB	18.08 GB	
2HQW	value	-12588.70	**-13501.93**	**-13501.93**	-12588.76	-12869.27	-12977.29	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	**-13501.93**	-9652.97	-12587.95	-13501.06	-13330.31	-12220.91	-12221.36	-11929.73	-12125.97	-12255.33	-11524.02	-10976.62	-13369.60	NaN	NaN	NaN	NaN	NaN	**-13501.93**	
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-13501.93**	**-13501.93**	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	NaN	NaN	NaN	NaN	NaN	**-13501.93**	
	runtime	0.09	13.25	38.13	0.04	70.35	43618.72	33.09	56.76	55.29	144	15.20	5.94	193.07	194.16	191.32	267.08	1095.95	4093.77	673.13	698.78	142.25	388.08	143.10	168.38	961.50	780.74	52.96	NaN	NaN	NaN	NaN	NaN	531.17	
	memory (GB)	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.47 GB	0.34 GB	7.68 GB	0.30 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	0.14 GB	NaN GB	NaN GB	NaN GB	NaN GB	NaN GB	13.62 GB	
2O60	value	-13611.10	-13330.53	-13359.76	-12611.10	-12958.27	-13075.27	-13354.50	**-13472.42**	-13354.30	-13349.36	-13349.56	-13309.88	-13468.76	-12101.50	-12680.07	-8646.64	-13202.88	-13072.33	-12543.64	-12505.49	-12972.67	-12348.07	-12909.17	-12348.07	-12909.17	-11988.45	-10910.00	-13343.31	NaN	NaN	NaN	NaN	NaN	**-13472.43**	
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-13584.50	-13584.50	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-23293.45	-13560.03	-13506.60	-13577.39	-14589.83	-14435.04	-14993.87	-14385.62	-14864.17	-15451.02	-16530.74	-13584.50	NaN	NaN	NaN	NaN	NaN	**-13472.43**	
	runtime	0.09	1904.11	2624.03	0.05	111.87	42989.74	47.40	117.12	44.33	304.60	275.90	272.05	254.85	254.36	252.67	1633.83	3607.39	8166.27	1144.23	1171.62	1084.92	1215.09	1078.85	1215.09	1078.85	1093.98	1104.94	65.22	NaN	NaN	NaN	NaN	NaN	551.66	
	memory (GB)	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.19 GB	0.19 GB	0.19 GB	0.19 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.64 GB	0.47 GB	7.44 GB	0.39 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	0.18 GB	NaN GB	NaN GB	NaN GB	NaN GB	NaN GB	18.50 GB	
3HXL	value	-13611.10	**-14101.32**	**-14101.32**	-13609.56	-13609.56	-13609.56	-13901.56	-13929.34	**-14101.32**	**-14101.32**	**-14101.32**	**-14101.32**	**-14101.32**	**-14101.32**	**-14101.32**	-9652.97	-13141.66	-13969.56	-14101.32	-14128.35	-14193.34	-13876.98	-13892.56	-13892.56	-13876.98	-13892.56	-13876.98	-13892.56	NaN	NaN	NaN	NaN	NaN	**-14101.32**	
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-14436.64	-14436.64	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-21960.56	-14128.51	-14128.51	-Inf	-Inf	-14128.51	-14193.34	-14748.99	-15406.73	-14892.54	-15406.73	-14892.54	-15406.73	NaN	NaN	NaN	NaN	NaN	**-14101.32**	
	runtime	0.09	1904.11	2624.03	0.05	111.87	42989.74	47.40	117.12	44.33	304.60	275.90	272.05	254.85	254.36	252.67	1633.83	3607.39	8166.27	1144.23	1171.62	1084.92	1215.09	1078.85	1215.09	1078.85	1093.98	1104.94	65.22	NaN	NaN	NaN	NaN	NaN	551.66	
	memory (GB)	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.64 GB	0.47 GB	7.44 GB	0.39 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	0.17 GB	NaN GB	NaN GB	NaN GB	NaN GB	NaN GB	18.50 GB	
pdb1l25	value	4643.63	3956.06	3931.58	4643.63	3991.68	3927.29	3951.31	3936.47	3945.86	4143.45	4079.86	4047.05	3975.23	4038.24	4101.65	4063.55	3976.88	3922.80	3932.02	3999.74	3998.70	4015.53	3978.85	4136.48	4051.76	4370.47	3951.31	4263.78	4067.37	4075.98	**3922.80**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	3907.40	3907.40	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	3907.40	3905.78	3922.80	3907.40	3905.78	3903.64	3899.66	3904.12	3901.40	3876.42	3751.55	3907.40	NaN	NaN	NaN	NaN	NaN	**3922.80**	
	runtime	0.17	111.70	116.54	0.10	8.08	1614.64	15.24	18.36	21.32	106.74	108.07	111.27	141.48	140.53	134.44	26.53	68.35	137.70	436.63	140.97	140.33	172.33	188.42	174.41	182.58	171.10	14.31	3600.00	3600.00	3600.00	3600.00	102.33		
	memory (GB)	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.31 GB	0.24 GB	0.21 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	0.09 GB	1.74 GB	
pdb1d2c	value	1961.07	**1738.42**	**1738.42**	1961.07	1755.16	1753.07	1769.76	1754.17	1750.47	1811.19	**1738.42**	1816.15	1773.42	1816.15	1811.19	1836.43	1775.27	**1738.42**	1739.47	1771.26	1761.44	1773.48	1761.10	1768.75	1785.16	1790.37	1786.77	1836.34	1758.99	1760.07	**1738.42**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1738.42	1738.42	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1738.42	1738.42	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	NaN	NaN	NaN	NaN	NaN	**1738.42**	
	runtime	0.03	56.85	58.82	0.05	0.05	2.39	4.64	7.73	9.85	11.03	55.76	71.79	72.40	72.17	75.84	52.17	257.59	101.69	210.51	78.38	78.23	92.03	79.84	94.05	74.14	75.69	19.84	3600.00	3600.00	3600.00	3600.00	17.96		
	memory (GB)	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.16 GB	0.10 GB	0.54 GB	0.11 GB	0.07 GB	0.09 GB	0.05 GB	0.06 GB	0.10 GB	0.06 GB	0.05 GB	0.05 GB	0.45 GB	0.66 GB	0.32 GB			
pdb1fmj	bound	1127.99	983.80	962.06	1013.23	962.82	979.62	985.43	979.57	986.47	1012.87	1001.26	1031.26	1009.64	1030.70	1001.31	**978.05**	1000.82	982.04	980.49	1004.64	1000.74	1001.31	**978.05**	1000.82	982.04	980.49	1004.64	985.43	1011.88	981.34	**978.05**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	960.65	960.65	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	965.33	966.21	**978.05**	966.54	964.92	965.18	964.40	965.60	944.91	958.45	930.91	965.65	-Inf	-Inf	-Inf	**978.05**		
	runtime	0.03	28.93	29.63	0.03	1.79	247.22	2.11	2.88	4.93	5.14	28.99	28.58	36.30	36.30	36.30	26.41	68.54	36.30	109.26	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30	36.30		
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB
pdb1l24	value	**439.39**	**439.39**	**439.39**	1027.05	442.14	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**439.39**	**439.39**	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	435.58	**439.39**	**439.39**	**439.39**	436.90	439.30	438.99	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	**439.39**	436.72	428.32	**439.39**	3600.00	3600.00		
	runtime	0.07	42.75	44.20	0.04	2.05	320.70	1.20	2.48	8.07	41.33	41.57	42.27	54.15	54.69	50.97	4.52	7.49	19.10	44.64	56.83	59.87	55.03	57.15	54.33	56.32	55.27	1.10	3600.00	3600.00	3600.00	3600.00	6.37		
	memory (GB)	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.11 GB	0.07 GB	0.49 GB	0.08 GB	0.04 GB	0.05 GB	0.04 GB	0.04 GB	0.05 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.93 GB	0.93 GB	0.93 GB	0.93 GB	0.24 GB		
pdb1jmx	value	1444.94	1160.64	1158.69	1444.94	1175.21	**1154.61**	1160.24	1158.69	1159.88	1260.40	1178.70	1259.45	1191.38	1196.34	1169.96	1206.37	1190.08	**1154.61**	1158.69	1192.34	1159.23	1166.35	1159.46	1184.60	1180.19	1236.24	1162.34	1278.12	1159.18	1164.28	**1154.61**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1144.30	1144.30	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1144.29	1144.29	-Inf	-Inf	1143.54	1136.64	1144.29	1143.53	1144.29	1144.12	1137.02	1128.98	1109.92	1144.29	-Inf	-Inf	-Inf		
	runtime	0.07	34.71	36.19	0.03	1.28	1512.45	1.20	2.28	5.96	9.03	10.47	33.28	43.72	44.04	41.54	5.51	13.62	35.33	29.68	50.44	54.87	53.54	53.73	53.08	51.91	54.08	0.93	3600.00	3600.00	3600.00	3600.00	6.43		
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB
pdb1kgx	value	2226.07	1740.93	**1724.06**	2401.18	1768.31	1749.27	1729.53	1724.19	1744.61	1834.83	1751.45	1783.35	1754.24	1790.46	1773.85	1776.25	1746.21	**1724.06**	1732.45	1744.24	1725.15	1737.47	1744.24	1710.26	1716.34	1651.45	1700.98	1609.52	1717.86	1st	1st	1st	**1724.06**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1717.86	1717.86	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1711.90	1717.84	1724.06	1717.86	1715.24	1714.24	1710.26	1716.34	1651.45	1700.98	1609.52	1717.86	-Inf	-Inf	-Inf		
	runtime	0.08	41.84	43.78	0.04	2.73	349.52	2.52	3.57	6.89	41.38	52.66	52.64	41.37	41.33	52.56	4.89	41.17	41.33	52.66	54.34	52.84	55.21	55.03	54.34	52.69	2.39	56.54	52.69	2.39	3600.00	3600.00	3600.00	3600.00	11.66		
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.04 GB	0.04 GB	0.04 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.11 GB	0.07 GB	0.40 GB	0.07 GB	0.05 GB	0.03 GB	0.03 GB	0.04 GB	0.03 GB	0.03 GB	0.04 GB	0.03 GB	0.52 GB	0.54 GB	1.13 GB	0.29 GB		
pdb1kwk	value	707.41	707.41	707.41	707.41	707.41	**703.36**	707.41	707.41	707.41	707.41	707.41	707.41	707.41	707.41	707.41	707.41	707.41	**703.36**	707.41	707.41	710.16	706.11	716.37	**703.36**	707.41	707.41	707.41	707.41	707.41	707.41	707.41	**703.36**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	696.49	696.49	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	691.79	696.47	**703.36**	696.49	692.57	693.60	691.00	693.98	680.13	672.89	653.43	696.49	-Inf	-Inf	-Inf		
	runtime	0.07	212.83	292.32	0.08	118.52	34159.91	41.05	66.57	57.88	263.74	217.92	229.99	210.48	211.20	209.48	381.81	1190.79	3728.02	779.00	814.22	798.62	827.95	798.99	771.71	797.84	824.95	59.70	NaN	NaN	NaN	NaN	NaN	803.94	
	memory (GB)	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.53 GB	0.37 GB	7.06 GB	0.14 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	0.07 GB	18.08 GB	
pdb1m3j	value	2901.46	2432.79	**2427.43**	2861.46	2460.22	**2427.43**	2444.39	2430.52	2455.56	2497.24	2477.18	2464.41	2445.98	2471.42	2466.72	2601.02	2453.66	**2427.43**	2428.96	2464.53	2458.38	2489.95	2445.19	2513.14	2547.80	2656.69	2478.16	2646.07	2493.61	2458.39	**2427.43**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	2421.75	2421.75	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	2400.69	2421.75	**2427.43**	2421.75	2420.18	2419.26	2412.30	2421.26	2394.35	2388.37	2265.19	2421.75	-Inf	-Inf	-Inf	-Inf		
	runtime	0.08	41.28	42.73	0.05	2.89	308.65	2.21	3.45	7.82	39.17	39.85	40.77	48.06	51.91	48.77	12.09	15.28	33.11	94.07	61.62	60.60	54.59	58.31	55.20	57.99	54.81	2.11	3600.00	3600.00	3600.00	3600.00	6.41		
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.04 GB	0.04 GB	0.04 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.11 GB	0.07 GB	0.49 GB	0.08 GB	0.04 GB	0.05 GB	0.04 GB	0.04 GB	0.05 GB	0.04 GB	0.04 GB	0.04 GB	0.04 GB	0.83 GB	0.83 GB	0.83 GB	0.83 GB	0.24 GB		
pdb1jgks	value	1811.08	**1512.84**	**1512.84**	1811.08	1528.85	1522.07	1513.73	1513.75	1513.73	1546.29	1526.10	1520.10	1527.10	1527.10	1527.10	1654.57	1527.10	**1512.84**	1513.73	1522.78	1518.19	1525.47	**1512.84**	1518.28	1536.11	1622.04	1513.73	1672.19	1521.23	1526.36	**1512.84**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1512.05	1512.05	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	1507.47	1512.05	**1512.84**	1512.05	1512.05	1507.47	1506.91	1512.05	1507.77	1483.20	1445.86	1512.05	-Inf	-Inf	-Inf	-Inf	-Inf		
	runtime	0.07	30.13	36.65	0.04	2.75	338.78	0.43	1.48	5.50	34.31	35.20	34.31	38.03	38.08	13.12	61.90	9.18	6.50	17.19	53.45	47.03	47.42	42.62	54.86	32.25	42.62	42.07	3.36	3600.00	3600.01	3600.00	3600.00	4.88		
	memory (GB)	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.11 GB	0.07 GB	0.49 GB	0.08 GB	0.03 GB	0.04 GB	0.03 GB	0.03 GB	0.04 GB	0.03 GB	0.03 GB	0.03 GB	0.03 GB	0.47 GB	0.47 GB	0.47 GB	0.47 GB	0.24 GB		
mean energy		-5446.52	-5892.07	-5941.27	-5371.52	-5576.01	-5595.73	-5856.47	-5860.05	-5955.73	-5872.01	-5941.10	-5949.10	-5949.10	-5949.10	-5949.10	-5891.62	-5955.73	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10	-5949.10
mean bound		-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-6041.38	-6041.38	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-10384.42	-6033.98	-5984.52	-6128.90	-6477.78	-6413.00	-6706.79	-6395.59	-6494.35	-6996.84	-7076.56	-6041.38	NaN	NaN	NaN	NaN	NaN	-5955.73	
mean runtime		0.08	102.71	130.17	0.05	54.34	22422.07	22.93	54.96	29.44	119.08	99.20	102.41	94.48	94.42	200.19	510.90	1030.52	1814.80	447.58	459.77	449.47	474.96	451.93	453.93	22.19	NaN	NaN	NaN	NaN	NaN	NaN	NaN	NaN	700.10	
best value		0.00	38.10	57.14	0.00	0.00	19.05	14.29	28.57	52.38	28.57	61.90	47.62	42.86	38.10	33.33	0.00	4.76	4.76	4.76	4.76	4.76	4.76	4.76	4.76	9.52	4.76	0.00	0.00	9.52	0.00	4.76	4.76	90.24	
best bound		0.00	0.00	0.00	0.00	0.00	0.00	4.76	4.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.76	0.00	0.00	4.76	4.76	0.00	4.76	4.76	4.76	0.00	0.00	0.00	0.00	0.00	0.00	100.00	
verified opt		0.00	0.00	0.00	0.00	0.00	0.00	4.76	4.76	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	4.76	0.00	0.00	4.76	4.76	0.00	4.76	4.76	4.76	0.00	0.00	0.00	0.00	0.00	0.00	100.00	

Table 27: protein-prediction

protein-prediction		ogm-ICM	ogm-LBP-LF2	ogm-LF-1	ogm-LF-2	ogm-LF-3	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDO	MPLP	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SG-A	ogm-BUNDLE-A+	ogm-SG-A+	ogm-BUNDLE-A-	ogm-SG-A-	LBP-LF	ogm-ILP
disNOstcoverage1	value	32490.95	30307.07	32490.95	31919.98	31870.84	32019.15	32029.93	30402.78	30019.35	30458.92	31671.63	30443.06	30374.49	30538.61	30528.87	31170.19	31485.43	31562.07	**30703.23**	**30146.55**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30136.31	30136.31	30096.00	30133.66	30136.31	30096.00	30133.66	29911.36	29905.20	30136.36	30084.54	30114.33	**30147.59**		
	runtime	0.05	32.72	0.03	0.70	18.77	27.97	32.32	33.23	34.35	37.01	28.15	16.50	88.05	1122.65	1121.92	1131.73	1127.07	1126.51	1249.77	1211.85	264.96	3601.40		
	memory (GB)	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.25 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.25 GB	0.17 GB	0.25 GB	0.22 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB
disNOstcoverage2	value	32561.26	29324.08	32561.26	31873.74	31807.03	30966.11	29324.08	30295.69	29322.74	30092.85	31253.29	29313.70	29320.04	29358.84	29341.64	30346.42	30759.87	31792.35	29359.60	29960.03	29323.43	**29305.44**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**29305.44**		
	runtime	0.05	2.66	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		
	memory (GB)	0.01 GB	0.06 GB	0.02 GB	0.02 GB	0.06 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.27 GB	0.25 GB	0.16 GB	0.26 GB	0.18 GB	0.26 GB	0.17 GB	0.24 GB	0.54 GB		
disNOstcoverage3	value	32490.95	30307.07	32490.95	31919.98	31870.84	32019.15	32029.93	30402.78	30019.35	30458.92	31671.63	30443.06	30374.49	30538.61	30528.87	31170.19	31485.43	31562.07	30703.23	30146.55	30114.33	**30147.60**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	30136.31	30136.31	30096.00	30133.66	30136.31	30096.00	30133.66	29911.36	29905.20	30136.36	30084.54	30114.33	**30147.60**		
	runtime	0.05	2.66	0.03	0.03	0.02	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01		
	memory (GB)	0.01 GB	0.06 GB	0.02 GB	0.02 GB	0.06 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.05 GB	0.25 GB	0.26 GB	0.17 GB	0.22 GB	0.16 GB	0.25 GB	0.16 GB	0.24 GB	0.65 GB		
disNOstcoverage4	value	30314.29	28975.20	32506.25	32023.75	31914.29	30357.60	29071.20	29071.20	29071.20	29071.20	31176.57	29071.20	29071.20	29071.20	30286.36	31592.82	29690.34	30286.36	29690.34	31592.82	29852.94	**28952.08**		
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**28952.08**		
	runtime	0.04	0.16	0.03	0

scene-decomposition		n-Fusion	ogm-ICM	ogm-LBP-LP1	ogm-LBP-LP2	ogm-LP-1	ogm-LP-2	ogm-LP-3	ogm-TRWS-LP1	ogm-TRWS-LP2	BPS-TAB	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDD	MPLP	MPLP-C	ogm-ADSAL	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SC-A	ogm-BUNDLE-A+	ogm-SC-A+	ogm-BUNDLE-A-	ogm-SC-A-	ogm-LP-LP	TRWS-TAB	BRAOBB-1	BRAOBB-2	BRAOBB-3	ADDD-BB	ogm-CostedLP	ogm-HLP
1000993	value	-890.18	-889.10	-890.24	-889.10	-890.22	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24	-890.24
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.05	0.06	0.00	0.02	4.46	0.00	0.01	0.10	0.01	0.05	0.02	0.14	0.06	0.03	0.11	0.03	0.03	0.04	0.74	0.18	0.91	0.04	0.03	0.99	0.90	0.25	0.00	0.73	6.73	29.64	0.11	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001184	value	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96	-987.96
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.03	0.01	0.13	0.05	0.02	0.08	0.03	0.02	0.03	0.46	0.08	0.48	0.03	0.02	0.48	0.48	0.12	0.00	0.80	6.80	27.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.90 GB	0.01 GB	0.01 GB	0.07 GB
1001195	value	-732.72	-730.05	-732.85	-732.85	-730.05	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85	-732.72	-732.72	-732.85	-732.85	-732.85	-732.85	-732.85	-732.85
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.06	0.00	0.02	0.35	0.00	0.00	0.09	0.01	0.03	0.01	0.11	0.04	0.02	0.07	0.03	0.02	0.03	1.07	0.05	1.04	0.03	0.02	2.80	0.73	0.20	0.00	0.80	6.80	29.37	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.90 GB	0.01 GB	0.01 GB	0.07 GB
1001252	value	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02	-686.02
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.12	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001585	value	-708.46	-706.31	-708.46	-708.46	-706.31	-706.97	-706.97	-708.46	-708.46	-708.02	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.45	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46	-708.46
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.05	0.05	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.04	0.02	0.24	0.10	0.05	0.02	0.03	0.03	0.04	1.30	0.13	1.19	0.06	0.07	1.32	1.18	0.25	0.00	2.38	8.41	23.50	0.02	0.02	0.18
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001688	value	-711.48	-708.30	-710.39	-710.39	-708.30	-710.24	-710.39	-711.48	-711.48	-710.39	-710.39	-710.39	-710.39	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48	-711.48
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.05	0.05	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.04	0.02	0.24	0.10	0.05	0.02	0.03	0.03	0.04	1.30	0.13	1.19	0.06	0.07	1.32	1.18	0.25	0.00	2.38	8.41	23.50	0.02	0.02	0.18
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001770	value	-681.18	-679.27	-681.18	-679.27	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18	-681.18
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.03	0.00	0.02	0.35	0.00	0.00	0.09	0.01	0.03	0.01	0.11	0.04	0.02	0.07	0.03	0.02	0.03	0.51	0.05	0.51	0.03	0.02	1.12	0.64	0.16	0.00	0.65	6.65	29.29	0.01	0.01	0.14
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001791	value	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.24	-1091.28	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20	-1091.20
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001875	value	-787.56	-782.82	-787.56	-787.56	-782.82	-787.29	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.29	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56	-787.56
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.43	0.00	0.00	0.09	0.01	0.03	0.01	0.20	0.06	0.05	0.02	0.02	0.02	0.02	1.16	0.07	1.06	0.05	0.05	1.16	1.06	0.21	0.00	0.63	6.62	24.44	0.01	0.01	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1001944	value	-746.76	-746.41	-746.76	-746.41	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76	-746.76
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.03	0.01	0.12	0.04	0.02	0.08	0.03	0.02	0.03	0.46	0.08	0.48	0.03	0.02	0.48	0.48	0.12	0.00	0.80	6.80	27.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100001	value	-852.50	-851.41	-852.50	-852.50	-851.41	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50	-852.50
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.03	0.01	0.12	0.04	0.02	0.08	0.03	0.02	0.03	0.46	0.08	0.48	0.03	0.02	0.48	0.48	0.12	0.00	0.80	6.80	27.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100002	value	-703.77	-700.97	-703.77	-703.77	-700.97	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77	-703.77
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100003	value	-719.45	-718.17	-719.45	-719.45	-718.17	-719.29	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-718.41	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45	-719.45
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100004	value	-771.74	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91	-771.91
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.46	0.08	0.48	0.03	0.02	0.48	0.48	0.12	0.00	0.80	6.80	27.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100005	value	-694.20	-693.47	-694.20	-694.20	-693.47	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20	-694.59	-694.99	-694.20	-694.20	-694.20	-694.20	-694.20	-694.20
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.03	0.01	0.12	0.04	0.02	0.08	0.03	0.02	0.03	0.46	0.08	0.48	0.03	0.02	0.48	0.48	0.12	0.00	0.80	6.80	27.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100006	value	-15.12	-14.79	-15.12	-14.79	-14.79	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12	-15.12
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100007	value	-1010.03	-1010.01	-1010.03	-1010.03	-1010.01	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03	-1010.03
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.04	0.00	0.02	0.41	0.00	0.00	0.10	0.01	0.03	0.01	0.11	0.04	0.02	0.08	0.03	0.02	0.03	0.64	0.06	0.64	0.03	0.02	0.64	0.64	0.16	0.00	0.72	6.72	29.48	0.05	0.02	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
1100008	value	-1011.71	-1011.71	-1011.71	-1011.71	-1011.

scene-decomposition		n-Fusion	ogm-ICM	ogm-LBP-LP1	ogm-LBP-LP2	ogm-LP-1	ogm-LP-2	ogm-LP-3	ogm-TBW-LP1	ogm-TBW-LP2	BPS-TAB	ogm-BPS	ogm-LBP-0.5	ogm-LBP-0.95	ogm-TRBP-0.5	ogm-TRBP-0.95	ogm-TRBPS	ADDD	MPLP	MPLP-C	ogm-ADSAL	ogm-BUNDLE-A	ogm-BUNDLE-H	ogm-SG-A	ogm-SG-A+	ogm-BUNDLE-A+	ogm-SG-A+	ogm-BUNDLE-A+	ogm-SG-A+	ogm-LP-LP	TBW5-TAB	BRAOBB-1	BRAOBB-2	BRAOBB-3	ADDD-BB	ogm-CombiLP	ogm-ILP
6000039	value	**-866.86**	-863.73	-863.40	-866.40	-863.73	-863.73	-860.04	**-866.86**	**-866.86**	**-866.86**	-865.40	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-866.86**	**-866.86**	**-866.86**	-Inf	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**	**-866.86**
	runtime	0.00	0.00	0.04	0.05	0.07	0.01	0.02	4.54	0.00	0.10	0.02	0.05	0.02	0.38	0.15	0.07	0.01	0.05	0.05	0.05	1.29	0.09	1.16	0.07	0.06	1.29	1.20	0.24	0.00	0.87	6.86	29.63	0.01	0.02	0.17	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB
6000040	value	**-762.91**	**-762.91**	-762.91	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-762.91**	**-762.91**	**-762.91**	-Inf	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**	**-762.91**
	runtime	0.00	0.00	0.04	0.05	0.07	0.02	0.03	4.54	0.00	0.10	0.02	0.04	0.02	0.38	0.15	0.07	0.01	0.05	0.05	0.05	1.23	0.09	1.14	0.07	0.06	1.23	1.18	0.24	0.00	36.32	42.18	26.82	0.01	0.02	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.05 GB	0.85 GB	0.01 GB	0.01 GB	0.07 GB
6000041	value	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-923.10**	**-923.10**	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**	**-923.10**
	runtime	0.00	0.00	0.03	0.03	0.07	0.01	0.02	0.42	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.47	0.09	1.10	0.07	0.06	1.47	1.13	0.24	0.00	30.33	0.02	0.16	0.01	0.02	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.03 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	1.05 GB	0.01 GB	0.01 GB	0.07 GB	
6000042	value	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-848.62**	**-848.62**	**-848.62**	-Inf	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**	**-848.62**
	runtime	0.00	0.00	0.04	0.05	0.07	0.01	0.02	0.42	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.28	0.09	1.12	0.07	0.06	1.28	1.16	0.24	0.00	0.87	6.87	24.40	0.03	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000043	value	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-929.64**	**-929.64**	**-929.64**	-Inf	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**	**-929.64**
	runtime	0.00	0.00	0.03	0.04	0.06	0.01	0.02	0.41	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.27	0.09	1.16	0.07	0.06	1.27	1.16	0.24	0.00	0.87	6.87	24.40	0.03	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000044	value	**-762.38**	-761.68	**-762.38**	**-762.38**	-761.68	-761.68	-761.68	**-762.38**	**-762.38**	**-762.38**	-761.68	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-762.38**	**-762.38**	**-762.38**	-Inf	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**	**-762.38**
	runtime	0.00	0.00	0.03	0.04	0.06	0.01	0.02	0.41	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.22	0.09	1.14	0.07	0.06	1.22	1.14	0.24	0.00	0.87	6.87	24.40	0.03	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000045	value	-870.36	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-870.52**	**-870.52**	**-870.52**	-Inf	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**	**-870.52**
	runtime	0.00	0.00	0.05	0.06	0.07	0.01	0.02	0.44	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.22	0.09	1.14	0.07	0.06	1.22	1.14	0.24	0.00	0.87	6.87	24.40	0.03	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000046	value	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-827.31**	**-827.31**	**-827.31**	-Inf	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**	**-827.31**
	runtime	0.00	0.00	0.04	0.06	0.07	0.02	0.03	0.43	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.28	0.09	1.14	0.07	0.06	1.28	1.14	0.24	0.00	0.87	6.87	24.40	0.03	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000047	value	**-872.40**	-872.16	-872.20	-872.20	-872.16	-872.20	-872.20	**-872.40**	**-872.40**	**-872.40**	-872.20	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-872.40**	**-872.40**	**-872.40**	-Inf	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**	**-872.40**
	runtime	0.00	0.00	0.04	0.06	0.07	0.02	0.43	0.00	0.09	0.10	0.02	0.05	0.02	0.14	0.06	0.03	0.02	0.07	0.07	0.08	1.02	0.09	1.18	0.07	0.06	1.02	0.97	0.24	0.00	1.25	7.26	27.10	0.02	0.04	0.17	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000048	value	**-856.87**	-856.05	-856.70	-856.70	-856.05	-856.16	-856.70	**-856.87**	**-856.87**	-856.70	-856.70	-856.70	-856.70	-856.70	-856.70	-856.70	**-856.87**	-855.84	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-856.87**	**-856.87**	**-856.87**	-Inf	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**	**-856.87**
	runtime	0.00	0.00	0.04	0.06	0.07	0.01	0.02	0.44	0.00	0.11	0.02	0.05	0.02	0.14	0.06	0.03	0.02	0.07	0.07	0.08	1.23	0.09	1.15	0.07	0.06	1.23	1.15	0.24	0.00	1.22	7.25	26.86	0.01	0.04	0.17	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000049	value	**-853.92**	-851.13	-853.92	-853.92	-851.43	-853.92	**-853.92**	**-853.92**	**-853.92**	-853.92	-853.92	-853.92	-853.92	-853.92	-853.92	-853.92	**-853.92**	-852.95	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-853.92**	**-853.92**	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**	**-853.92**
	runtime	0.00	0.00	0.04	0.06	0.07	0.01	0.02	0.44	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.34	0.09	1.27	0.07	0.06	1.34	1.14	0.24	0.00	0.87	27.64	0.02	0.01	0.16		
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000050	value	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-885.48**	**-885.48**	**-885.48**	-Inf	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**	**-885.48**
	runtime	0.00	0.00	0.04	0.05	0.07	0.01	0.02	0.43	0.00	0.09	0.01	0.04	0.01	0.06	0.03	0.01	0.01	0.05	0.04	0.03	1.29	0.09	1.14	0.07	0.06	1.29	1.14	0.24	0.00	0.87	27.64	0.02	0.01	0.16		
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000051	value	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-920.98**	**-920.98**	**-920.98**	-Inf	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**	**-920.98**
	runtime	0.00	0.01	0.03	0.05	0.06	0.01	0.02	0.42	0.00	0.09	0.01	0.04	0.01	0.17	0.08	0.04	0.01	0.02	0.02	0.02	0.24	0.05	0.47	0.02	0.02	1.01	0.93	0.24	0.00	0.79	6.78	27.16	0.02	0.01	0.16	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000052	value	**-917.26**	-914.69	**-917.26**	**-917.26**	-914.69	-917.10	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-917.26**	**-917.26**	**-917.26**	-Inf	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**	**-917.26**
	runtime	0.00	0.00	0.07	0.08	0.08	0.02	0.03	4.57	0.00	0.10	0.02	0.07	0.03	0.26	0.10	0.05	0.02	0.06	0.06	0.06	1.27	0.09	1.15	0.07	0.06	1.27	1.15	0.24	0.00	0.87	6.87	24.92	0.01	0.07	0.17	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB
6000053	value	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	**-920.71**	**-920.71**	**-920.71**	-Inf	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**	**-920.71**
	runtime	0.00	0.00	0.04	0.06	0.07	0.01	0.02	0.42	0.00	0.10	0.02	0.04	0.02	0.25	0.10	0.05	0.01	0.05	0.05	0.05	1.25	0.09	1.15	0.07	0.06	1.25	1.19	0.24	0.00	0.87	6.87	25.31	0.02	0.01	0.15	
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.04 GB	0.04 GB	0.92 GB	0.01 GB	0.01 GB	0.07 GB

scene-decomposition		n-Fusion	ngm-ICM	ngm-LBP-LP1	ngm-LBP-LP2	ngm-LP1	ngm-LP2	ngm-LP-3	ngm-TBW-LP1	ngm-TBW-LP2	BPS-TAB	ngm-BPS	ngm-LBP-0.5	ngm-LBP-0.95	ngm-TRBP-0.5	ngm-TRBP-0.95	ngm-TRBPS	ADDO	MPLP	MPLP-C	ngm-ADSAL	ngm-BUNDLE-A	ngm-BUNDLE-H	ngm-SC-A	ngm-BUNDLE-A+	ngm-SC-A+	ngm-BUNDLE-A-	ngm-SC-A-	ngm-LP-LP	TBW5-TAB	BRAOBB-1	BRAOBB-2	BRAOBB-3	ADDO-BB	ngm-CombiLP	ngm-HLP
9000933	value	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49	-907.49
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-907.49	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	4.43	0.00	0.00	0.10	0.01	0.04	0.02	0.17	0.07	0.04	0.01	0.03	0.03	0.03	1.31	0.19	1.19	0.09	0.06	1.28	1.17	0.24	0.00	0.75	6.76	25.29	0.01	0.01	0.15
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.07 GB
9000989	value	-908.40	-902.17	-908.40	-908.40	-901.77	-907.90	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40	-908.40
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-908.40	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.54	0.00	0.00	0.11	0.01	0.04	0.02	0.37	0.13	0.07	0.01	0.03	0.03	0.03	0.28	0.20	0.24	0.09	0.06	0.29	0.25	0.05	0.00	0.14	15.56	29.00	0.01	0.02	0.42
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.05 GB	0.05 GB	0.96 GB	0.01 GB	0.07 GB
9001001	value	-915.65	-914.96	-915.65	-915.65	-914.96	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-915.65	-915.65	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.65	-915.95	-916.29	-915.65	-915.65	-Inf	-Inf	-915.65	-915.65	-915.65	-915.65	-915.65
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.53	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.43	0.31	1.25	0.09	0.06	0.27	0.25	0.05	0.00	0.14	1.08	7.04	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.78 GB	0.01 GB	0.07 GB
9001034	value	-856.09	-853.64	-856.09	-856.09	-853.64	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-856.09	-856.09	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.09	-856.29	-857.14	-856.09	-856.09	-Inf	-Inf	-856.09	-856.09	-856.09	-856.09	-856.09
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.53	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.43	0.31	1.25	0.09	0.06	0.27	0.25	0.05	0.00	0.14	1.08	7.04	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.02 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.78 GB	0.01 GB	0.07 GB
9001071	value	-768.19	-764.03	-768.19	-768.19	-764.03	-765.16	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-767.67	-767.67	-768.19	-768.19	-768.19	-768.19	-768.19
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-768.19	-768.19	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19	-Inf	-Inf	-768.19	-768.19	-768.19	-768.19	-768.19	-768.19
	runtime	0.01	0.00	0.04	0.05	0.00	0.02	0.65	0.00	0.00	0.09	0.01	0.04	0.01	0.13	0.05	0.03	0.00	0.03	0.03	0.06	0.95	0.43	0.88	0.05	0.05	0.93	0.88	0.23	0.00	0.76	6.75	24.42	0.06	0.02	0.15
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.84 GB	0.01 GB	0.07 GB
9001184	value	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-997.33	-997.33	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33	-Inf	-Inf	-997.33	-997.33	-997.33	-997.33	-997.33	-997.33
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.44	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.34	0.19	1.19	0.09	0.06	0.28	0.25	0.05	0.00	0.14	6.85	26.54	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.84 GB	0.01 GB	0.07 GB
9001300	value	-923.49	-923.47	-923.49	-923.49	-923.47	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-923.49	-923.49	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.49	-923.77	-924.34	-923.49	-923.49	-Inf	-Inf	-923.49	-923.49	-923.49	-923.49	-923.49
	runtime	0.00	0.00	0.04	0.06	0.00	0.02	0.41	0.00	0.00	0.09	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.26	0.19	1.07	0.09	0.06	1.32	2.53	0.11	0.00	0.75	6.85	25.16	0.00	0.02	0.18
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.77 GB	0.01 GB	0.07 GB
9001317	value	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-717.51	-717.51	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.51	-717.81	-718.68	-717.51	-717.51	-Inf	-Inf	-717.51	-717.51	-717.51	-717.51	-717.51
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.41	0.00	0.00	0.09	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.26	0.19	1.07	0.09	0.06	1.32	2.53	0.11	0.00	0.75	6.85	25.16	0.00	0.02	0.18
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.77 GB	0.01 GB	0.07 GB
9001619	value	-861.88	-859.51	-861.39	-861.39	-859.51	-861.68	-861.88	-861.88	-861.88	-861.39	-861.39	-861.39	-861.39	-861.39	-861.39	-861.39	-861.39	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.68	-861.68	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-861.88	-861.88	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-861.88	-862.05	-861.88	-861.88	-Inf	-Inf	-861.88	-861.88	-861.88	-861.88	-861.88
	runtime	0.01	0.00	0.04	0.06	0.00	0.02	0.46	0.01	0.00	0.11	0.01	0.05	0.02	0.15	0.07	0.04	0.02	0.07	0.07	0.09	1.12	0.11	1.03	0.06	0.06	1.11	1.02	0.27	0.00	18.35	22.71	25.14	0.01	0.04	0.20
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.79 GB	0.01 GB	0.07 GB
9001713	value	-834.00	-830.99	-834.00	-834.00	-830.99	-833.82	-831.62	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-833.88	-833.88	-834.00	-834.00	-834.00	-834.00	-834.00
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-834.00	-834.00	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.00	-834.12	-834.99	-834.00	-834.00	-Inf	-Inf	-834.00	-834.00	-834.00	-834.00	-834.00
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.43	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.34	0.19	1.19	0.09	0.06	1.28	1.17	0.24	0.00	0.75	6.85	25.16	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.78 GB	0.01 GB	0.07 GB
9001991	value	-842.91	-842.18	-843.18	-843.18	-842.91	-841.45	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-843.18	-843.18	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.18	-843.41	-844.00	-843.18	-843.18	-Inf	-Inf	-843.18	-843.18	-843.18	-843.18	-843.18
	runtime	0.00	0.00	0.04	0.06	0.00	0.02	0.42	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.33	0.19	1.24	0.09	0.06	1.28	1.17	0.24	0.00	0.75	6.85	25.16	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.77 GB	0.01 GB	0.07 GB
9002004	value	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-972.40	-972.40	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.40	-972.42	-972.48	-972.40	-972.40	-Inf	-Inf	-972.40	-972.40	-972.40	-972.40	-972.40
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.04	0.02	0.33	0.13	0.07	0.01	0.03	0.03	0.03	1.34	0.19	1.19	0.09	0.06	1.28	1.17	0.24	0.00	0.75	6.85	25.16	0.01	0.02	0.17
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.77 GB	0.01 GB	0.07 GB
9002021	value	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-857.82	-857.82	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-857.82	-Inf	-Inf	-857.82	-857.82	-857.82	-857.82	-857.82
	runtime	0.01	0.01	0.03	0.05	0.01	0.02	0.41	0.00	0.00	0.10	0.01	0.04	0.02	0.20	0.09	0.05	0.01	0.02	0.02	0.02	1.27	0.05	1.13	0.03	0.04	1.25	1.13	0.22	0.00	0.77	6.78	27.88	0.01	0.01	0.18
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.96 GB	0.01 GB	0.07 GB
9002090	value	-933.27	-931.73	-933.27	-933.27	-931.73	-932.59	-932.59	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-932.95	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-933.27	-933.27	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.27	-933.56	-934.27	-933.27	-933.27	-Inf	-Inf	-933.27	-933.27	-933.27	-933.27	-933.27
	runtime	0.00	0.00	0.04	0.06	0.00	0.02	0.44	0.00	0.00	0.11	0.01	0.05	0.02	0.34	0.14	0.06	0.01	0.03	0.03	0.04	1.35	0.07	1.27	0.07	0.07	1.28	1.17	0.24	0.00	0.77	6.78	28.92	0.01	0.01	0.19
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.96 GB	0.01 GB	0.07 GB
9002114	value	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-824.19	-824.19	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.19	-824.20	-824.28	-824.19	-824.19	-Inf	-Inf	-824.19	-824.19	-824.19	-824.19	-824.19
	runtime	0.01	0.01	0.04	0.05	0.01	0.02	0.40	0.00	0.00	0.09	0.01	0.04	0.02	0.33	0.13	0.06	0.01	0.03	0.03	0.03	1.34	0.19	1.05	0.09	0.06	1.22	1.10	0.22	0.00	0.71	6.71	25.44	0.22	0.01	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.78 GB	0.01 GB	0.07 GB
9002474	value	-801.67	-800.38	-801.67	-801.67	-800.38	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.74	-802.28	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-801.67	-801.67	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-801.67	-Inf	-Inf	-801.67	-801.67	-801.67	-801.67	-801.67
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.40	0.00	0.00	0.09	0.01	0.04	0.02	0.33	0.13	0.06	0.01	0.03	0.03	0.03	1.34	0.19	1.05	0.09	0.06	1.22	1.10	0.22	0.00	0.74	6.74	25.34	0.01	0.01	0.16
	memory (GB)	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.01 GB	0.04 GB	0.03 GB	0.04 GB	0.04 GB	0.78 GB	0.01 GB	0.07 GB
9002577	value	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71
	bound	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-1231.71	-1231.71	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-Inf	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71	-Inf	-Inf	-1231.71	-1231.71	-1231.71	-1231.71	-1231.71
	runtime	0.00	0.00	0.04	0.05	0.00	0.02	0.40	0.00	0.00	0.10	0.01	0.04	0.01	0.22	0.10	0.04	0.01	0.02	0.02	0.04	1.16	0.28	1.11	0.05	0.02	1.22	1.06	0.21